UNIT 8 MODEL SPECIFICATION ISSUES*

Structure

8.0 OBJECTIVES

After going through this unit you should be in a position to:

- explain the nature of problems associated with model specification;
- identify the consequences of model misspecification;
- find out the consequences of including an irrelevant variable;
- find out the consequences of excluding a relevant variable;
- detect a mis-specified modelandsuggest remedial measures; and
- explain various tests for model selection.

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8.1 INTRODUCTION

Regression analysis, as you know, is based on certain assumptions. These assumptions pertain to specification of the functional form and behaviour of the error variable. The inferences drawn from OLS estimation, therefore, are based on fulfilment of these assumptions.

While estimating a regression model, correct specification of the model is very important for obtaining unbiased and efficient estimates. However, several questions arise: How do we detect whether the OLS assumptions are met for a given dataset? Is the selected functional form appropriate for analysis? Are the variables included in the model relevant? In the presence of any such specification issue, what is the impact on the estimators? Are the inferences drawn still valid?

In this unit we will look out for the major specification issues and the impact it will have on prediction.

8.2 POSSIBLE PROBLEMS IN SPECIFICATION

Although extremely significant, specification is least understood aspect of regression analysis. In general, model specification should have theoretical basis rather than empirical or methodological one.

8.2.1 Importance of Model Specification

The causal relationship between regressand (explained variable) and regressor (explanatory variable) in a multiple regression model involves three steps: (i) specification of the model, (ii) estimation of the parameters of this model, and (iii) interpretation of these parameters.

Since specification forms the first and most critical step, any error in it will lead to error in estimation and interpretation. If the regressors in multiple regression analysis are independent and uncorrelated, their partial regression coefficients will be equal to their simple regression coefficients. Consequently, addition or deletion of orthogonal independent variables will not affect the partial regression coefficient of other independent variables in a multiple regression model. But the standard error and hence error variance of dependent variables are significantly affected by inclusion or omission of orthogonal independent variables. Inclusion of independent variables in regression equation increases the standard error of the partial regression coefficient by decreasing their degrees of freedom. Simultaneously, however, the addition of an independent variable may decrease these same standard errors by decreasing the error variance in the dependent variable.

8.2.2 Sources of Specification Error

There could be numerous ways in which a bias or an error can enter a regression model. Some of these are (i) inclusion of redundant/ superfluous/ irrelevant variables, (ii) omission of relevant variables, (iii) incorrect functional form, (iv) incorrect assumption regarding the error term, and (v) error in measurement of variables.

Let us consider the multiple regression model

$$Y = \beta_1 X_1 + \beta_2 X_2 + \cdots .. + \beta_k X_k + u \qquad \ldots(8.1)$$

We have two sets of assumptions: (i) specification of the model, and (ii) specification of the error variable. The assumptions regarding specification of the model are as follows:

a) The regression model is specified correctly – all relevant variables are included, and no relevant variable is excluded from the model.

b) The model is linear in parameters.

c) The variables are measured without errors.

d) The explanatory variables are independent of one another – there is no perfect correlation among them.

The assumptions regarding specification of the error variable are as follows:

(i) $E(u_i) = 0$

(ii) $E(u_i)^2 = \sigma^2$

(iii) $E(u_i u_j) = 0$ for $i \neq j$

(iv) $E(X_i u_i) = 0$ (regressor is non-stochastic; the error term is not correlated with any of the explanatory variables)

Violation of the above assumptions results in several types of problems in the desirable properties of the estimators.

8.2.3 Implications of Specification Error

Let us begin with assumption (a) given above. A model refers to a simplified version of reality. It allows us toexplain,analyse and predict economic behavior. An economic model can be for a microeconomic agent such as household or firm. In macroeconomics, it represents the behavior of the economy. In economic models we identify relevant economic variables (such as income, output, expenditure, investment, savings, exports, etc.) and establish relationship among them. The relationships among these variables may be expressed through diagrams or mathematical equations. There could be economic models without mathematical expressions, but suchmodelsmay not be precise.

While building an econometric model we first consider the logic or economic theory behind the model. The empirical or methodological considerations come later. The accuracy of the estimated parameters and the inferences drawn from the model depend upon the correct specification of the model.

An econometric model comprises a dependant variable, independent variable(s),andanerror term. As you know, regression signifies a cause-and-effect relationship – variation in the dependent variable is explained by the variation in the independent variables. Thus, while specifying a regression model, we should see to it that the dependant variable islogically explained by the independent variables. If an independent variable does not explain the dependent variable, it should not be included in the model.

Next is the functional form of the regression model, which should be specified correctly. Let me illustrate the point through an example. In the case of a firm, we assume that there are two factors of production, viz., capital and labour. We aggregate all types of labour into a homogeneous category – we do not distinguish between a manager and a worker in the field! Thus, we ignore the details and concentrate on the major issues in a model. Secondly, we assume that the production function takes a particular form, say Cobb-Douglas. But it is just an assumption! The production function, in reality, could be of some other formalso such as CES or translog. Thus, we need to logically explain the functional form (regression equation) of the model.

Assumption (b) says that the regression model is linear in parameters. Equation (8.1) is linear in parameters (there are no such terms as β_i^2, for example). Suppose our regression model is

$$Y_i = \beta_1 + \beta_2 X_i + \beta_3 X_i^2 + u_i \qquad \dots (8.2)$$

Equation (8.2) is a quadratic function, but it is still linear in parameters. Examples of non-linear regression models are logarithmic functions, logistic functions, trigonometric functions, exponential functions, etc. Some of these functions can be transformed to linear functions (see Unit 3).If we cannot transform a non-linear function into a linear function, we cannot estimate the non-linear function by OLS method.

Many a time we specify the functional form correctly, but the variables are not measured correctly. There could be errors in measurement of the independent variable(s) or in the dependent variable. Such issues are called error-in-variables; and we will deal with them in Unit 12. Remember that error in variable is different from the error term (u_i) included in a regression model. The error term reflects the variables that could not be included in the regression equation (excluded variables). Measurement error in a variable, on the other hand, reflects the situation where the variable is included in the model, but it is not correctly measured. There could be several sources of measurement error: (i) it may not be possible to measure certain variables (e.g. taste, preference, business environment, intelligence, etc.), or (ii) the variable can be defined conceptually but it may not be possible to measure the variable correctly (e.g., education – it can be measured by a proxy, that is, years of schooling). There are certain other factors that lead to errors in variables such as: (i) non-response error, (ii) reporting error, and (iii) computing error.

In case the assumption (d) is violated we have the problem of multicollinearity. It is a problem observed in multiple regression models, when we include too many variables in a regression equation. It is basically a data problem. Independence of the explanatory variables is ensured when the matrix X has full rank.We will discuss problems arising out of multicollinearity in Unit 10.

8.3 INCLUSION OF VARIABLES IN A MODEL

Regression analysis derives its robustness from the assumption that the econometric model under study is correctly specified. We discuss below the consequences of (i) inclusion of irrelevant variables, and (ii) omission of relevant variables in a regression model.

Each of the above problem results in a different kind of bias.

8.3.1 Inclusion of Irrelevant Variables

Let us consider the case where certain irrelevant variable is included in the regression model. Suppose the true model is

$$Y_i = \beta_0 + \beta_1 X_{1i} + u_i \qquad \text{… (8.3)}$$

But we somehow include a redundant variable, i.e., we estimate the following equation:

$$Y_i = \beta_{0s} + \beta_{1s} X_{1i} + \beta_{2s} X_{2i} + v_i \qquad \text{… (8.4)}$$

For the true model (8.3), the slope coefficient is expressed as

$$\hat{\beta}_1 = \frac{\sum y x_1}{\sum x_1^2} \qquad \text{… (8.5)}$$

which is unbiased.

For the model (8.4) that we have taken, we obtain

$$\tilde{\beta}_1 = \hat{\beta}_{1s} = \frac{(\sum y x_1)(\sum x_2^2) - (\sum y x_2)(\sum x_1 x_2)}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2} \qquad \text{… (8.6)}$$

Now the true model in deviation form is

$$y_i = \beta_1 x_1 + (u_i - \bar{u}) \qquad \text{… (8.7)}$$

In equation (8.6) let us substitute the value ofy_i from (8.7) and simplify. If we take expected value of (8.6), after substitution and simplification, we obtain

$$\mathrm{E}(\tilde{\beta}_1) = \mathrm{E}(\hat{\beta}_{1s}) = \beta_1 \frac{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2} \qquad \text{… (8.8)}$$

From equation (8.8) we find that

$$E(\tilde{\beta}_1) = \beta_1$$

Thus, inclusion of an irrelevant variable provides us with unbiased estimator of β_1.

The estimator of the redundant variable $\hat{\beta}_{2s}$is given by

$$\hat{\beta}_{2s} = \frac{(\sum y x_2)(\sum x_1^2) - (\sum y x_1)(\sum x_1 x_2)}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2} \qquad \text{… (8.9)}$$

If we substitute for y_i from (8.7) in (8.9) and re-arrange terms, we obtain

$$E(\tilde{\beta}_2) = E(\hat{\beta}_{2s}) = \beta_2 \frac{(\sum x_1 x_2)(\sum x_1^2) - (\sum x_1 x_2)(\sum x_1^2)}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2} \quad \ldots (8.10)$$

Thus, $E(\tilde{\beta}_2) = E(\hat{\beta}_{2s}) = 0$

So, we find that $\hat{\beta}_{2s}$, which is absent from the true model, has its coefficient '0'.

Thus, we obtain unbiased estimators for both the parameters. This leads us to conclude that inclusion of irrelevant variables is not that harmful as omission of relevant variables. As an extra variable is added to the model, we observe that there is an increase in R-squared. The variance of the parameters will not be efficient.

Therefore, the specification error in the case of inclusion of irrelevant variables in the model will produce unbiased but inefficient least squares estimators of the parameters. The larger variance reduces the precision of the estimates resulting in wider confidence intervals. This may lead to type II error (the error of not rejecting a null hypothesis when the null hypothesis is actually not true).

8.3.2 Omission of Relevant Variables

Now let us look into the other side of the spectrum – excluding a relevant variable. Since a relevant variable is not included in the model (although it influences the dependent variable) its impact will be included in the residuals. As a result, the residuals will show a systematic pattern rather than being white noise as required by Gauss-Markov theorem. Also, the coefficient of the included variable will be biased.

Suppose the true equation (in deviation form) is

$$y = \beta_1 x_1 + \beta_2 x_2 + u \quad \ldots (8.11)$$

Instead of estimating equation (8.11) suppose we omitted x_2. The following equation is estimated,

$$y = \beta_1^* x_1 + e \quad \ldots (8.12)$$

Equation (8.12) is a case of omitted variable, and hence incorrect model specification. In the model with omitted variable (incorrect model) the estimate of β_1^* is

$$\hat{\beta}_1^* = \frac{\sum x_1 y}{\sum x_1^2} \quad \ldots (8.13)$$

In order to calculate the bias in the estimated value of β_1 in the incorrect model (equation (8.12)) as compared to the true model (equation (8.11)), we take the following steps:

We substitute the expression of y from the true model (8.11) in equation (8.13). This gives us

$$\hat{\beta}_1^* = \frac{\sum x_1(\beta_1 x_1 + \beta_2 x_2 + u)}{\sum x_1^2} = \beta_1 + \beta_2 \frac{\sum x_1 x_2}{\sum x_1^2} + \frac{\sum x_1 u}{\sum x_1^2} \quad \ldots (8.14)$$

Since E $(\sum x_1 u) = 0$ we get

$$E(\hat{\beta}_1^*) = \beta_1 + \beta_2 b_{21} \qquad \dots (8.15)$$

where $b_{21} = \frac{\sum x_1 x_2}{\sum x_1^2}$ is the regression coefficient from a regression of x_2 (omitted variable) on x_1.

Thus, $\hat{\beta}_1^*$ is a biased estimator for β_1 and the bias is given by

$$E(\hat{\beta}_1^*) - \beta_1 = \beta_2 b_{21} \qquad \dots (8.16)$$

Thus, bias = (coefficient of the excluded variable, β_2) × (regression coefficient in a regression of the excluded variable on the included variable, b_{21})

The variance of β_1^* (parameter of the incorrect model) can also be derived by using the formula for variance. As it is a bit complex, we do not present it here. You should note that the variance of β_1^* is higher than that of β_1. An implication of the above is that usual tests of significance concerning parameters are invalid, if some of the relevant variables are excluded from a model.

In order to detect omitted variables, residuals ($\hat{u}_i$) can be examined for any pattern that they exhibit (see Unit 5). If a variable is omitted, it will be reflected in the form of a systematic pattern in the residuals. If the residuals show autocorrelation which can be detected using Durbin-Watson d-statistic (see Unit 9), there exists model misspecification where some variable is omitted from the model.

Thus, we know that

(i) When an irrelevant variable is included in the model: (a) the estimators of parameters are unbiased, (b) there is a decline in the efficiency of the estimators, and (c) the estimator of the error variance is unbiased. Thus, conventional tests of hypothesis are valid. The inferences drawn could be somewhat erroneous.

(ii) When a relevant variable is dropped from the model: (a) estimators of parameters are biased, (b) decline in the efficiency of the estimators, and (c) estimator of error variance is biased (over-estimate). Thus, conventional tests of hypothesis are invalid. The inferences drawn are faulty.

Recall from Unit 3 that there two approaches to building an economic model: (i) general to specific (also called top-down approach), and (ii) specific to general (also called bottom-up approach). In the first approach we consider a general model by including many variables. Subsequently, we exclude the irrelevant variables. In the second approach, we begin with a simple model and keep on adding variables based on our logic.

Although, there is no consensus on the approach to be followed in econometric model building, researchers prefer the general-to-specific approach. Now you can understand the reason behind it – if we exclude certain relevant variable(s) from a model, we do not obtain unbiased estimators even for the variables included in the model!

When we follow the general-to-specific approach, we may retain only those variables which are statistically significant (i.e., we exclude the variables which are not statistically significant). In the case of the specific-to-general approach, i.e., expanding the model after testing for the statistical significance of the variables is criticized because the true level of significance and the nominal level of significance differ if all the variables are not included in the model.

Check Your Progress 1

1) List the assumptions of the classical regression model. What problems do you encounter if these assumptions are violated?

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2) Do you agree that correct specification of an econometric model is important? Why?

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3) List three types of specification error that we encounter in an econometric model.

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4) Explain the consequences of inclusion of an irrelevant variable.

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4) Explain the consequences of excluding a relevant variable.

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8.4 SPECIFICATION ERRORTEST

There is a tendency on the part of researchers to assume a linear relationship between variables. This however is not always true. If the true relationship is non-linear and we take a linear regression model for estimation, we will not be able to draw correct inferences. Hence, detection of the specification error can help us in obtaining efficient estimates. Once it is found that specification errors are made, finding a remedy is not a major problem.

One general test for model specification,that is very popular, is the RESET (Regression Specification Error Test). The idea underlying RESET is simple.

Let us assume thatwe consider the model (8.17) as given below for estimation.

$$y = \beta_1 x_1 + \beta_2 x_2 + u \qquad \ldots(8.17)$$

The RESET introduces quadratic and cubic functions of the estimated dependant variable in the regression model and compares it to the model without the additional regressors using F-test. If the increase in R^2 is statistically significant, there exists model misspecification. Now we will discuss the step-by-step RESET,

(i) From the chosen model $y = \beta_1 x_1 + \beta_2 x_2 + u$, obtain the estimated y, that is $\hat{y}$.

(ii) Re-run the above chosen model introducing $\hat{y}$ in some form as an additional regressor (quadratic and cubic form is quite popular and yields good results). Thus, the new model will be

$$y = \beta_1 x_1 + \beta_2 x_2 + \beta_3 \hat{y}^2 + \beta_4 \hat{y}^3 + v \qquad \ldots (8.18)$$

(iii) Let the R^2obtained from the model in the second step above be R^2_{new} and that obtained from step one be R^2_{old}. The F test can be used to find out if the increase in R^2 from using the model in step two is statistically significant using the following F-test from Unit 4:

$$F = \frac{(R^2_{new} - R^2_{old}) / \text{ number of new regressors}}{(1 - R^2_{new}) / (n - \text{number of new parameters in the model})} \qquad \ldots (8.19)$$

(iv) If the computed F-value is statistically significant, say, at the 5 percent level, wereject the null hypothesis that the model is correctly specified. It implies we accept the alternate hypothesis that the regression model is mis-specified.

Limitations of the RESET

The RESET does not test for omitted variables. It tests whether the functional form of the model is correctly specified or not. If there isnon-linearity in the functional form and we consider a linear model, the RESET test will reject the null hypothesis. In other words, we accept the alternate hypothesis. The alternative hypothesis in the RESET, however, that there is misspecification of the functional form. Thus, an important limitation of the RESET is that it does not provide further leads: What should we do if there is misspecification? What functional form should we adopt?

8.5 MODEL SELECTION CRITERIA

From the discussion so far, we observe that there could be many competing econometric models for certain economic phenomenon. For example, for quarterly data on GDP of India, there could be several autoregressive-moving average (ARMA) models before us. There could be certain seasonal variation in the GDP data along with the secular trend. Such variations are represented by ARMA models with different orders of AR and MA. Similarly, for the estimation of production function of a firm, there are quite a few functional forms such as Cobb-Douglas, CES, translog, etc. A pertinent issue is selection of regressors in a model.

There are certain criteria on the basis of which we select the most appropriate regression model for particular situation. We discuss these criteria below.

8.5.1 The $R^2$and Adjusted-R^2Criteria

We have discussed the concept of coefficient of determination (R^2) in Unit 4. As you know, the coefficient of determination indicates the explanatory power of a model. If, for example, $R^2 = 0.82$ we can infer that 82 per cent variation in the dependent variable is explained by the explanatory variable in the model. The R^2 of a model measures the goodness of fit of a regression model and lies between 0 and 1.

$$R^2 = \frac{explained\ sum\ of\ squares}{total\ sum\ of\ squares} = 1 - \frac{residual\ sum\ of\ squares}{total\ sum\ of\ squares} \qquad \dots (8.20)$$

A higher value of R^2 indicates better fit of a model and vice versa. The R^2measures only in-sample goodness of fit and it cannot forecast the out-of-sample observations, which happen to be an important function of a regression model. Second, we can compare among models having the same dependant variable.Third, with every additional regressor, even if irrelevant, theR^2-value does not decrease. Hence a researcher may go on adding more independent variables although this will increase the variance of forecast error.

Adjusted-R²

As a penalty for adding regressors to increase the R^2 value, the concept of adjusted-R^2, denoted by $\bar{R}^2$has been introduced.

$$\bar{R}^2 = 1 - \frac{Residual\ sum\ of\ squares/n-k}{total\ \ sum\ of\ squares/n-1} = 1 - (1 - \bar{R}^2)\frac{n-1}{n-k} \qquad \ldots (8.21)$$

where *n* is the sample size, and *k* is the number of explanatory variables. From (8.21) we see that $\bar{R}^2 \leq R^2$. It indicates that the adjusted-R^2penalizes for adding regressors. There is an increase in the value of Adjusted-R^2 only if the absolute *t*-value of the added regressor is greater than 1. Hence $\bar{R}^2$ ensures meaningful and valid comparisons. Remember that comparison between models is possible for the same dependent variables. If dependent variables are different, we cannot compare among models.

8.5.2 The AIC and SIC Criteria

The Akaike Information Criterion (AIC) criterion is based on the maximum likelihood estimation of regression models. We consider a small number of regression models based on economic theory. We find out the value of the log-likelihood function of each model. The AIC criterion is based on the idea of imposing a penalty for adding regressors.

$$AIC = e^{2k/n}\frac{\sum \hat{u}_i^2}{n} = e^{2k/n}\frac{residual\ sum\ of\ squares}{n} \qquad \ldots (8.22)$$

where *k* is the number of regressors (including the intercept) and *n* is the number of observations.

For mathematical convenience, we can write

$$ln\,AIC = \left(\frac{2k}{n}\right) + ln\left(\frac{residual\ sum\ of\ squares}{n}\right) \qquad \ldots (8.23)$$

where $ln\,AIC =$ natural log of AIC and $\left(\frac{2k}{n}\right) =$ penalty factor.

We find out the value of '*ln AIC*' according to equation (8.23) for all the contending models.We compare the value of '*ln AIC'* of the models, and the model with the lowest value of AIC is preferred. The advantages of AIC over the coefficient of determination (R^2) are as follows: (i) AIC criterion is useful for not only the in-sample but also the out-of-sample forecasting performance. (ii) AIC criterion can be used for both nested and non-nested models. (iii) AIC criterion can also help in determining the lag length in an AR(p) model.

Schwartz Information Criterion (SIC)

The SIC criterion is also based on the method of maximum likelihood estimation. It is also called the Bayesian Information Criterion (BIC), or the Schwartz Bayesian Criterion (SBC). The SIC criterion, as in the case of adjusted-R^2 and the AIC criterion, penalises regression models having larger number of regressors.

The SIC is defined as:

$$\text{SIC} = n^{k/n}\frac{\sum \hat{u}^2}{n} = n^{k/n}\frac{residual\ sum\ of\ squares}{n} \qquad \ldots (8.24)$$

The above can be expressed in log-form as

$$\ln \text{SIC} = \frac{k}{n}\ln n + \ln\frac{residual\ sum\ of\ squares}{n} \qquad \dots (8.25)$$

where $[(k/n)\ln n]$ is the penalty factor. A lower value of SIC implies a better model. The SIC, like AIC, can also be used for comparison of both in-sample and out-of-sample forecasting performance of the model.

8.5.3 Mallow's C_p Criterion

Mallow's $C_p$criterion for model selection is defined as

$$C_p = \frac{RSS_p}{\hat{\sigma}^2} - (n - 2p) \qquad \dots (8.26)$$

where *n* is the number of observations, total regressors are k out of which only p are included in the model and RSS_p is the residual sum of squares obtained from p (included) regressors. $E(\hat{\sigma}^2)$ is the unbiased estimator of the true σ^2. If the model with only p regressors is the true model, then $E(RSS_p) = (\text{n} - \text{p})\sigma^2$.Therefore, (8.26) is approximately equal to

$$E(C_p) \approx \frac{(n-p)\sigma^2}{\sigma^2} - (n - 2p) \approx p \qquad \dots (8.27)$$

While choosing a regression model according to the $C_p$criterion, a model with a lower$C_p$value is selected, about equal to p. To put it differently, following the principle of parsimony, we will choose a model with *p* regressors (p<k) that gives a fairly good fit to the data.

8.6 CAUTION ABOUT MODEL SELECTION CRITERIA

We highlighted earlier that econometric models should be grounded in economic theory and reasoning. Hence, when refining an econometric model, it is important to consider the theoretical justification for including or omitting a variable. To ensure a properly defined model, it is essential to go through earlier studies to grasp the theoretical principles. Additionally, the quality of the model we develop is contingent upon the quality of the data we gather. If the data collected is free from issues like multicollinearity or autocorrelation, we are more likely to end up with a stronger model.

Many a time, we observe certain relationship between two variables. Such relationships, however, may be superficial or spurious. Let us take an example. At a traffic light, cars stop when the signal is red. It does not mean that cars cannot move when there is a red light in front of them. It also does not mean that traffic lights have some damaging effect on moving cars. The reason is observance of traffic rules. Unless we look into the traffic rules and go by observation only, our reasoning will be wrong. The dependent variable and the independent variable both may be affected by another variable. In such cases the relationship is 'confounded'. Confounded relationship indicates a situation where a variable, Z, affects two other variables, X and Y, in such a way that we observe certain association between X and Y. In the example given above traffic rules affect the working of traffic lights as well as car drivers.

You should note one more issue regarding selection of econometric models. Different test criteria may suggest different models. For example, economic logic suggests that there could be two possible econometric models (say, model A and model B) for a particular issue. You may come across a situation such that $\bar{R}^2$ test suggests model A and AIC criterion suggest model B. In such situations you should carry out several tests and then only chose the best model.

Adjusted-R^2, Mallows C_p, p-values, etc. may point to different regression equations without much clarity to the econometrician. Thus, we conclude that none of the methods for model selection listed above are adequate by itself. There is no substitute to theoretical understanding of the related literature, accurately collected data, practical understanding of the problem, and common sense while specifying an econometric model.

CHECK YOUR PROGRESS 2

1) Explain how the RESET is applied.

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2) What are the limitations of RESET?

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3) What precaution one should one take while dealing with model selection criteria?

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8.7 LET US SUM UP

Correct specification of an econometric model determines the accuracy of the estimates obtained. Therefore, correct specification of an econometric model is very important. Economic theory and logic guide us in specification of econometric models. In order to correctly specify an econometric model, all relevant explanatory variables should be included in the model. No relevant explanatory variable should be excluded from the model. Further, the functional form of the model should be correct.

Selection of an appropriate econometric model is a difficult task. We have to take into account economic theory and logic behind the econometric model. There could be many competing models for a particular issue. There are certain criteria on the basis of which the best econometric model is selected. These criteria could be $\bar{R}^2$, AIC, SIC, and Mallow's C_p.We have described the formulae for these test criteria in the Unit.

8.8 KEY WORDS

Bias : The difference between the expected value of an estimator and the population parameter that the estimator is supposed to be estimating.

RESET : RESET is used to test for the presence of error in model specification, particularly its functional form. The RESET is based on F-test.

AIC Criterion : If there are k number of explanatory variables in a regression model, then $ln\, AIC = \left(\frac{2k}{n}\right) + ln\left(\frac{residual\ sum\ of\ squares}{n}\right)$. The AIC criterion is used in model selection among competing regression models.

SIC Criterion : The SIC criterion is used in model selection among competing regression models. If n is the sample size and k is the number of regressors in a model, then SIC = $n^{k/n}\frac{\sum \hat{u}^2}{n} = n^{k/n}\frac{residual\ sum\ of\ squares}{n}$

8.9 SOME USEFUL BOOKS

Dougherty, C. (2011). *Introduction to Econometrics*, Fourth Edition, Oxford University Press

Gujarati, D N and D C Porter (2009), *Basic Econometrics*, Fifth Edition, McGraw Hill

Johnston, J., Dinardo, J. (2007), "*Econometric Methods*", 4th Edition, McGraw-Hill Education (Asia), International Edition, pp. 109-125

Maddala, G. S. and K. Lahiri (2009), *Introduction to Econometrics*, Fourth Edition, John Wiley and Sons

Wooldrige, Jeffrey (2022), *Introductory Econometrics*, Seventh Edition, Cengage Learning

8.10 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) The basic assumptions of the classical regression model are as follows:

   a) The regression model is linear in parameters

   a) $E(X_i u_i) = 0$ (regressor is non-stochastic)

   b) $E(u_i) = 0$

   c) $E(u_i)^2 = \sigma^2$

   d) $E(u_i u_j) = 0$ for $i \neq j$

   e) The explanatory variables (X_i) are independent of one another.

2) Go through Section 8.2. It is important because incorrect specification has serious implications on desirable properties of the estimators. Therefore, inferences drawn from mis-specified models are not correct.

3) The major sources of specification error are: (i) inclusion of redundant/ superfluous/ irrelevant variables, (ii) omission of relevant variables, (iii) incorrect functional form, (iv) incorrect assumption regarding the error term, and (v) error in measurement of variables.

4) Due to inclusion of irrelevant variables in a model, we obtain unbiased but inefficient least squares estimators of the parameters. The larger variance reduces the precision of the estimates resulting in wider confidence intervals. This may lead to type II error.

5) When a relevant variable is dropped from the model, we obtain biased estimators of the parameters and there is a decline if the efficiency of the estimators. The estimator of error variance is higher, as a result of which tests of hypothesis are invalid. Therefore, the inferences drawn are faulty.

Check Your Progress 2

1) Go through Section 8.4 and answer.

2) The RESET leaves us with much ambiguity. It does not tell us the next course of action.

3) Go through Section 8.6 and answer.

UNIT 9 AUTOCORRELATION*

Structure

9.0 OBJECTIVES

After going through this unit, you should be in a position to

- identify the problem of autocorrelation in regression analysis,
- explain how the problem of autocorrelation affects the estimators of the regression model parameters,
- explain the various methods available to detect autocorrelationin a dataset, and
- suggest remedial measures for the problem of autocorrelation.

9.1 INTRODUCTION

An importantassumption of the classical regression model is that the error terms u_i are independent of one another. However, this may not be true always and in certain cases the error terms may be related to one another. In notations,

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$E(u_i, u_j) \neq 0; i \neq j$ …(9.1)

We find several examples in our day to day life where the successive observations appear to follow a natural order over time. The upward or downward trend in time series data generally has an in-built momentum which makes successive observations interdependent. Hence, the successive error terms may be correlated. For example, the BSE Sensex may be seen moving upwards or downwards for successive days. In this Unit we deal mainly with correlation between error terms in time series data.

9.2 CONCEPT OF AUTOCORRELATION

There are two ways in which the error terms may be correlated:

Spatial correlation: It is the correlation between error terms in cross-sectional data. For instance, recall the demonstration effect, which says that consumption behaviour of households is affected by that of their neighbours (at times termed as "keeping up with the Joneses" effect). Households try to match or keep up to the standards of their neighbours. Thus, in a household study, the error terms may be found to be correlated. In the present Unit, however, we do not discuss issues related to spatial correlation.

Serial correlation or Autocorrelation: In time series data, the correlation between error terms is known as autocorrelation or serial correlation. There are various reasons why time series data may exhibit autocorrelation. The series may be *non-stationary* (mean, variance and covariance do not remain constant over time) which causes the error terms to be correlated. Omission of a relevant variable from the model is also captured by the error term causing them to be correlated. Even specifying an incorrect functional form (for instance, fitting a linear cost function instead of a cubic function) will cause the error terms to be correlated with one another. Sometimes the explained variable depends on its lagged values. For instance, current consumption depends on past consumption as consumption habits are slow to change. In such cases we need to include lagged value of consumption as explanatory variable. If the lagged variable is not included, the error term may capture that effect resulting in correlation between the error terms. Interpolation and extrapolation of data may also cause residuals to follow a systematic pattern.

Technically, $u_{t-2}, u_{t-1}, u_t, u_{t+1}, u_{t+2}$ .... may be correlated to one another. Here, 'u' denotes the error term and the subscript denotes the time period. We have changed the subscripts to 't' keeping the view the fact that autocorrelation takes place mostly in time series data. First order autocorrelation is correlation between u_t and u_{t-1}. Second order autocorrelation is correlation between u_t and u_{t-2} and so on. Autocorrelation of order 'k' is the correlation between $u_t$and u_{t-k}. For 'n' observations, there could be $(n-1)$ autocorrelations.

9.3 CONSEQUENCES OF AUTOCORRELATION

In the presence of autocorrelation, the OLS estimators are:

- Linear
- Unbiased
- Asymptotically normally distributed

The variance of the estimators, however, gets affected. Thus in the presence of autocorrelation, the OLS estimator $\hat{\beta}$ is not the best linear unbiased estimator (BLUE). It implies that the usual tests of significance discussed in Units 3 and 4 cannot be applied. We discuss further on each of the above propositions.

9.3.1 Unbiasedness

Let $y_t = \beta x_t + u_t$ … (9.2)

Let us denote the OLS estimator of β by $\hat{\beta}$.

We know from Unit 3 that

$$\widehat{\beta} = \frac{\sum x_t y_t}{\sum {x_t}^2} = \beta + \frac{\sum x_t u_t}{\sum {x_t}^2} \quad \ldots (9.3)$$

In the presence of autocorrelation, the assumptions $E(u_t) = 0$ and $E(x_t u_t) = 0$ continue to be true.

Thus, $E\left(\frac{\sum x_t u_t}{\sum {x_t}^2}\right) = 0$

Hence, $E(\hat{\beta}) = \beta$

That is, $\hat{\beta}$ is unbiased.

9.3.2 Minimum Variance

In order to derive the variance of $\hat{\beta}$ in the presence of autocorrelation, let us consider a two-variable regression model $y_t = \beta x_t + u_t$ and assume that the error terms are auto-correlated. In notations,

$E(u_t, u_{t+s}) \neq 0 \ (s \neq 0)$

We assume that the error term follows a '**first order auto-regressive'** or **AR(1) process** such that

$u_t = \rho u_{t-1} + \varepsilon_t$ …(9.4)

In the above AR(1) process, we assume that current error (u_t) equals certain proportion of previous error (u_{t-1}) plus a random shock (ε_t). Here, 'ρ' is the *coefficient of autocovariance or first order coefficient of autocorrelation.* The random shock ε_t satisfies the usual OLS assumptions.

$E(\varepsilon_t) = 0$

$Var(\varepsilon_t) = {\sigma_\varepsilon}^2$, and

$Cov(\varepsilon_t, \varepsilon_{t+s}) = 0;\ s \neq 0$ …(9.5)

The random shock (ε_t) which satisfies the above properties is usually called the **white noise**. The error terms u_t can be correlated in many forms. For instance, an AR(2) process is given by

$u_t = \rho_1 u_{t-1} + \rho_2 u_{t-1} + \varepsilon_t$

Here u_t depends upon error terms of previous two periods. Similarly, an AR(3) process is given by

$u_t = \rho_1 u_{t-1} + \rho_2 u_{t-1} + \rho_3 u_{t-2} + \varepsilon_t$

and so on.

Sometimes, the error term u_t follows a Moving Average (MA) process. For instance, a first order moving average process is given by MA(1) where,

$u_t = \varepsilon_t + \rho_1 \varepsilon_{t-1}$ … (9.6)

Here u_t depends upon the white noise of previous period. We will discuss further on AR and MA processes in Unit 18 of this course. In order to keep matters simple, in the present Unit, we assume that the error term follows an AR(1) process.

(a) $\boldsymbol{Var(u_t)}$

For an AR(1) process the variance of the error term is given by,

$var(u_t) = var\ (\rho u_{t-1} + \varepsilon_t) = \rho^2 var\ (u_{t-1}) + var(\varepsilon_t)$

Recall from equation (9.3) that $var(\varepsilon_t) = {\sigma_\varepsilon}^2$.

Since $var\ (u_t) = var\ (u_{t-1}) = \sigma^2$, the above equation can be written as

$\sigma^2 = \rho^2\sigma^2 + {\sigma_\varepsilon}^2$

On re-arranging terms in the above equation we obtain

$\sigma^2 - \rho^2\sigma^2 = {\sigma_\varepsilon}^2$

or, $\sigma^2(1-\rho^2) = {\sigma_\varepsilon}^2$

This gives us, $var(u_t) = \sigma^2 = \frac{{\sigma_\varepsilon}^2}{(1-\rho^2)}$ …(9.7)

In equation (9.7), the variance of the error term (u_t) is homoscedastic even though the error terms are correlated.

(b) $\boldsymbol{Cov\ (u_t, u_{t-1})}$

Covariance between u_t and u_{t-1} is given by

$Cov\ (u_t, u_{t-1}) = E\{u_t - E(u_t)\}\{u_{t-1} - E(u_{t-1})\}$

Since $E(u_t) = E(u_{t-1}) = 0$, the covariance of the error term, $Cov\ (u_t, u_{t-1})$, is given by

$Cov\ (u_t, u_{t-1}) = E(u_t u_{t-1})$

On successive substitution we find that

$$E(u_t u_{t-1}) = E(\rho u_{t-1}{}^2 + \varepsilon_t u_{t-1}) = \rho E(u_{t-1}{}^2) = \rho \frac{\sigma_\varepsilon{}^2}{(1-\rho^2)} \qquad \text{... (9.8)}$$

[Notice that in an AR(1) process u_{t-1} is not influenced by ε_t as the former occurs in the earlier time period;thus $E(\varepsilon_t u_{t-1}) = 0$]

Similarly, $cov\ (u_t, u_{t-2}) = \rho^2 \frac{\sigma_\varepsilon{}^2}{(1-\rho^2)}$

In general terms we can write that

$$cov\ (u_t, u_{t-s}) = \rho^s \frac{\sigma_\varepsilon{}^2}{(1-\rho^2)} \qquad \text{… (9.9)}$$

We need to assume here that |ρ| <1, otherwise the variance and covariance given above are undefined. In case |ρ| ≥1, we encounter is the popular 'unit root problem' (see Unit 17 for further discussion on this).

(c) $\boldsymbol{Var(\hat{\beta})_{AR(1)}}$

We know from Unit 3 that in the classical regression model the OLS estimator for variance of $\hat{\beta}$, say $Var(\hat{\beta})_{OLS} = \frac{\sigma^2}{\sum x_t{}^2}$

The underlying assumption is that the error terms are not correlated, i.e.,

$E(u_t, \varepsilon_{t+s}) = 0; s \neq 0$

When this assumption is violated, and u_t follows an AR process, the variance and covariance of u_t are as follows:

$$var(u_t) = \frac{\sigma_\varepsilon{}^2}{(1-\rho^2)}$$

$$cov\ (u_t, u_{t-s}) = \rho^s \frac{\sigma_\varepsilon{}^2}{(1-\rho^2)}$$

Recall that the OLS estimator of β, that is, $\hat{\beta} = \frac{\sum x_t y_t}{\sum x_t{}^2} = \beta + \frac{\sum x_t u_t}{\sum x_t{}^2}$

Hence, $var(\hat{\beta} - \beta) = var\left(\frac{\sum x_t u_t}{\sum x_t{}^2}\right) = E(\frac{\sum x_t u_t}{\sum x_t{}^2})^2$

For AR(1) process, $var(\hat{\beta})_{AR(1)}$is given by

$$E(\frac{\sum x_t u_t}{\sum x_t{}^2})^2 = \frac{1}{(\sum x_t{}^2)^2} var(\sum x_t\ u_t) \qquad \text{... (9.10)}$$

We retain the assumption that variable x_t is non-stochastic and $E(x_t u_t) = 0$.

Thus, $E(\frac{\sum x_t u_t}{\sum x_t{}^2})^2 =$

$$\frac{1}{(\sum x_t{}^2)^2} (\sigma^2 \sum x_t{}^2 + 2Cov(u_t, u_{t-1}) \sum x_t\ x_{t-1} + 2Cov(u_t, u_{t-2}) \sum x_t\ x_{t-2} + ...)$$

$$= \frac{1}{(\sum x_t^{\,2})^{\,2}} (\sigma^2 \sum x_t^{\,2} + 2\rho\sigma^2 \sum x_t \; x_{t-1} + 2\rho^2\sigma^2 \sum x_t \; x_{t-2} + \ldots)$$

$$= \frac{\sigma^2}{(\sum x_t^{\,2})} \; (1 + 2\rho \frac{\sum x_t \, x_{t-1}}{\sum x_t^{\,2}} + 2\rho^2 \frac{\sum x_t \, x_{t-2}}{\sum x_t^{\,2}} \ldots\ldots\ldots)$$

You should notice that the problem of autocorrelation introduces an additional expression in $Var(\hat{\beta})_{OLS}$.

Now, assume that x_t also follows an AR(1) process with $var(x_t) = \sigma_x^{\,2}$ and autocorrelation coefficient of 'r'. Since x_t is **assumed to be stochastic** (see Unit 10 of this course), we take the asymptotic variance of $\hat{\beta}$ in the presence of AR(1) process. $Var(\hat{\beta})_{AR(1)}$ is given by,

$$Var(\hat{\beta})_{AR(1)} = E(\frac{\sum x_t \; u_t}{\sum x_t^{\,2}})^2$$

$$= \frac{\sigma^2}{(\sum x_t^{\,2})} \; (1 + 2\rho r + 2\rho^2 r^2 \ldots\ldots\ldots) = \frac{\sigma^2}{\mathrm{T}\sigma_x^{\,2}} \frac{(1+\rho r)}{(1-\rho r)} \qquad \ldots(9.11)$$

(T = number of observations)

Hence, we can write

$$Var(\hat{\beta})_{AR(1)} = Var(\hat{\beta})_{OLS} \frac{(1+\rho r)}{(1-\rho r)} \qquad \ldots (9.12)$$

Equation (9.12) has the following implication:

$Var(\hat{\beta})_{OLS}$ may underestimate the $Var(\hat{\beta})_{AR(1)}$ in the presence of autocorrelation. The nature of bias however depends on the values taken by ρ and r. If ρ and r are both positive, then,

$$Var(\hat{\beta})_{OLS} < Var(\hat{\beta})_{AR(1)}$$

From (9.12) we observe that in the presence of AR(1) process, the formula for calculation of variance of the OLS estimator $\hat{\beta}$is

$$Var(\hat{\beta})_{AR(1)} = \frac{\sigma^2}{\mathrm{T}\sigma_x^{\,2}} \frac{(1+\rho r)}{(1-\rho r)}. \qquad \ldots(9.13)$$

In such a case, using the usual OLS formulae for computation of variance of OLS estimators leads to biased results. Moreover, depending on the values of ρ and r, the variance of the OLS estimator $Var(\hat{\beta})_{AR(1)}$ may be found to be higher than what it would be in the absence of autocorrelation. Hence, the presence of autocorrelation has a tendency to inflate the standard errors.

Application of OLS may give misleading standard errors of estimators in the presence of autocorrelation. In this case, Generalised Least Squares (GLS) should be used as it gives estimators which have minimum variance. When we allow for autocorrelation and use $Var(\hat{\beta})_{AR(1)}$, wider confidence intervals are obtained as compared to those obtained from GLS procedure. Thus, the likelihood of accepting the null hypothesis increases and the coefficient may be declared insignificant despite being significantly different from zero.

9.3.3 Biasedness of $\mathbf{Var(\hat{\beta})_{OLS}}$

$Var\ (u_t) = \sigma^2$

and, $Var(\hat{\beta})_{OLS} = \frac{\sigma^2}{(\sum x_t{}^2)}$

For unbiasedness of $Var(\hat{\beta})_{OLS}$, we are required to prove that,

$E[Var(\hat{\beta})_{OLS}] = Var(\hat{\beta})_{OLS}$

For the above to hold true, $E(\hat{\sigma}^2)$ should equal σ^2. For a two variable regression model,

$$\hat{\sigma}^2 = \frac{\sum \hat{u}_t{}^2}{T-1} \qquad \ldots (9.14)$$

$$E(\sum \hat{u}_t{}^2) = E(\sum u_t{}^2) + E[(\hat{\beta}-\beta)^2 \sum x_t{}^2] - 2\,E[(\hat{\beta}-\beta)\sum x_t\ u_t$$

[Since $\quad \hat{u}_t = y_t - \hat{y}_t = \beta x_t + u_t - \hat{\beta} x_t = u_t + \beta x_t - \hat{\beta} x_t$

$= u_t - x_t\,(\hat{\beta}-\beta)]$

Or, $E(\sum \hat{u}_t{}^2) = T\sigma^2 + E(\frac{\sum x_t u_t}{\sum x_t{}^2})^2 \sum x_t{}^2 - 2E(\frac{\sum x_t u_t}{\sum x_t{}^2})\ \sum x_t\ u_t$

Or, $E(\sum \hat{u}_t{}^2) = T\sigma^2 + E\,\frac{\sum (x_t u_t)^2}{\sum x_t{}^2} - 2E\,\frac{(\sum x_t u_t)^2}{\sum x_t{}^2}$

Or, $E\left(\sum \hat{u}_t{}^2\right) = T\sigma^2 + \frac{\sigma^2(1+r\rho)}{(1-r\rho)} - 2\frac{\sigma^2(1+r\rho)}{(1-r\rho)}$

(Refer to equations (9.10) and (9.11))

$$\text{Or, } E\left(\sum \hat{u}_t{}^2\right) = \sigma^2[T - \frac{(1+r\rho)}{(1-r\rho)}] \qquad \ldots (9.15)$$

In classical regression model we have seen that $E(\frac{\sum \hat{u}_t{}^2}{T-1}) = E(\hat{\sigma}^2)$. Here in the AR(1) model we find that $E\left(\sum \hat{u}_t{}^2\right) = \sigma^2[T - \frac{(1+r\rho)}{(1-r\rho)}]$. Thus in the case of AR(1), we have

$$E(\hat{\sigma}^2) \neq \sigma^2 \qquad \ldots (9.16)$$

We cannot say whether the bias is upward or downward; it depends upon the sign of '*r*' and 'ρ'. If both '*r*' and 'ρ' are positive, then $E(\hat{\sigma}^2) < \sigma^2$ and the bias is downward.

Check Your Progress 1

1) "If the OLS method is applied to time series data, we get biased estimates as the errors are autocorrelated". True or False? Explain.

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2) "Autocorrelation in error terms caused biased standard errors." True or False? Explain.

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3) For an AR(2) process, show that $cov\ (u_t, u_{t-2}) = \rho^2 \frac{\sigma_\varepsilon{}^2}{(1-\rho^2)}$

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9.4 DETECTION OF AUTOCORRELATION

In the previous section we discussed the consequences of autocorrelation. In this section we discuss some of the methods to detect autocorrelation.

9.4.1 Residual Plot

The actual errors in a regression model are not known to us. What we know are the estimated values of the error terms, i.e., residuals. By plotting the residuals '$\hat{u}$' against time 't', we can observe if there is a systematic pattern. If no systematic pattern is observed, it indicates no autocorrelation. If, however, the residuals show a linear downward or upward trend, or a cyclical or quadratic pattern, then it indicates the presence of autocorrelation.

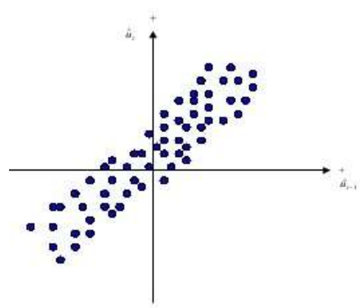

Fig.9.1: Positive Autocorrelation

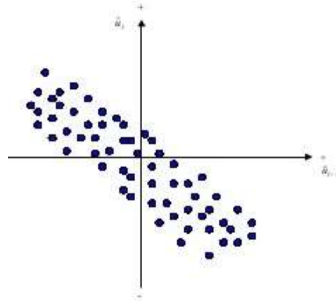

Fig. 9.2: Negative Autocorrelation

Alternatively, we can plot $\hat{u}_t$ against$\hat{u}_{t-1}$ as shown in Fig. 9.1 and Fig. 9.2. The residual plot method tests for the AR(1) process. Fig. 9.1 shows positive autocorrelation while Fig. 9.2 displays negative autocorrelation. A major limitation of the residual plot is that its interpretation is highly subjective. In view of this, we look for quantitative tests for autocorrelation.

Some of the quantitative tests are discussed below.

9.4.2 Durbin-Watson (DW) Test

A popular test for autocorrelation is the Durbin-Watson (DW) test. It is given by

$$d = \frac{\sum_{t=2}^{n}(\hat{u}_t - \hat{u}_{t-1})^2}{\sum_{t=1}^{n}\hat{u}_t^{\,2}} \qquad ...(9.17)$$

It is the ratio of the sum of squared differences in successive residuals to the overall residual sum of squares (RSS). It can be further elaborated as

$$d = \frac{\sum \hat{u}_t^{\,2} + \sum \hat{u}_{t-1}^{\,2} - 2\sum \hat{u}_t\,\hat{u}_{t-1}}{\sum \hat{u}_t^{\,2}}$$

There are $(T-1)$ observations in the numerator as taking successive differences results in loss of one observation.For large size samples, $\sum \hat{u}_t^{\,2}$ and $\sum \hat{u}_{t-1}^{\,2}$ are nearly equal.

Hence, $d = 2 - \frac{2\sum \hat{u}_t\hat{u}_{t-1}}{\sum \hat{u}_t^{\,2}} \approx 2(1-\hat{\rho})$ …(9.18)

If there is no autocorrelation in a regression model, then $\hat{\rho} = 0$. By applying this value in (9.18) we find that$d = 2$. On the other hand, If $\hat{\rho} = +1, d = 0$, and if $\hat{\rho} = -1, d = 4$. Thus, d being close to zero implies positive autocorrelation, whereas d being close to 4 implies negative autocorrelation. Thus, d lies between 0 and 4, i.e., $0 \leq d \leq 4$. We can set our null hypothesis of no autocorrelation as $H_0: d = 2$. If d is found to be around 2, we accept the H_0; if it exceeds the critical value we reject the H_0.

Somehow the *d*-statistic does not have a particular critical value for acceptance or rejection of the null hypothesis of no autocorrelation. However, it reports the lower and upper limits $d_L$and d_U respectively. These limits are reported for the number of observations in the sample and the number of independent variables (excluding the constant term) in the regression model.

If the value of d lies between 0 and d_L, there is positive autocorrelation. If $4 - d_L < d < 4$, there is negative autocorrelation. If $d_U < d < 4 - d_U$, there is no autocorrelation. If the value of d lies between $d_L$and d_U ($d_L \leq d \leq d_U$or $4 - d_U \leq d \leq 4 - d_L$), this test is inconclusive as this is the zone of indecision (see Fig. 9.3). This is an important limitation of the Durbin-Watson test. Typically, it is observed that as sample size increases, the zone of indecision shrinks.

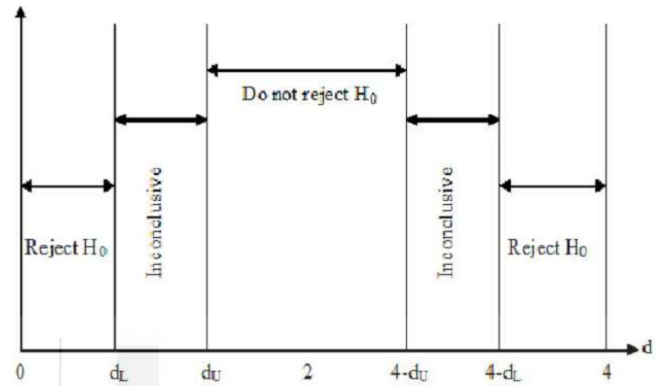

Fig. 9.3: Rejection Zones under the DW Test

The DW statistic is based on certain assumptions: i) intercept term is present in the regression model, ii) the independent variables are fixed in repeated sampling (maintaining this assumption is difficult in case of time series data), iii) the error terms are normally distributed, and iv) the error terms follow AR(1) process. Hence, the d-statistic may be unreliable if the error terms are not NIID (normally, independently and identically distributed). However, in large samples, the *d*-statistic follows a standard normal distribution. Further, this test can be used only in the case of first order autocorrelation. The DW test cannot be used if we have lagged dependent variables in the regression model.

9.4.3 Breusch-Godfrey (BG) Test

The BG test overcomes the limitations of the DW test. It is a general test for higher order autocorrelations. It can be applied in models having lagged values of dependent variable as independent variables. It can also be applied in case of moving averages of white noise error terms. This test is derived from the Lagrange Multiplier (LM) principle. Hence, it is also known as LM test.

Let the regression model be specified as follows:

$y_t = \sum_{i=1}^{k} x_{it}\, \beta_i + u_t$; t = 1,2,...,n … (9.19)

Here, lagged dependent variables can also be included as independent variables in the$x's$.

Let u_t follow an AR(p) process such that

$u_t = \rho_1 u_{t-1} + \rho_2 u_{t-2} + \ldots. + \rho_p u_{t-p} + \varepsilon_t$ … (9.20)

In the above equation,

$\varepsilon_t \sim IN(0, \sigma^2)$

Now we specify the null hypothesis as

H$_0$: $\rho_1 = \rho_2 = \cdots = \rho_p = 0$ …(9.21)

The BG Test involves the following steps:

(i) Use OLS to estimate the primary regression equation $y_t = \sum_{i=1}^{k} x_{it}\,\beta_i + u_t$ and get $\hat{u}_t$.

(ii) Regress $\hat{u}_t$ on the original values of x_{it} (as well as any other variable used as regressor) and lagged values of the residuals (for example, if p = 3, we will include three lagged values of the estimated residuals as regressors). The auxiliary regression equation is as follows:

$$\hat{u}_t = \sum_{i=1}^{k} x_{it}\,\gamma_i + \sum_{i=1}^{p} \hat{u}_{t-i}\,\rho_i + \tau_t \quad \text{…(9.22)}$$

(iii) If the sample size is large, then Breusch and Godfrey have shown that

$$(T-p)R^2 \sim \chi_p^2 \quad \text{…(9.23)}$$

In equation (9.23), T is the sample size, p is the number of autoregressive parameters in the auxiliary regression, and R^2 is the coefficient of determination of the auxiliary regression equation. BG test is asymptotically (it means, as sample size increases infinitely) distributed as χ^2 with p degrees of freedom.

(iv) Once we get the computed value of the χ^2 in (9.23) we can compare it with the standard value (tabulated value). If computed value exceeds tabulated value we reject the null hypothesis.

Check Your Progress 2

1) Explain the steps followed the Durbin Watson test.

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2) What are the limitations of the DW test? Can you suggest some alternative tests which can overcome these limitations?

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3) Consider the model $y_t = \beta_1 + \beta_2 x_{2t} + \beta_3 x_{3t} + u_t$. The DW statistic is 0.85. The number of observations is 80. Also given that $d_L = 1.41$ and $d_U = 1.50$. Is there a problem of autocorrelation in the data?

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9.5 REMEDIAL MEASURES

If the DW statistic is found to be statistically significant, it is not necessarily due to AR(1) errors. It can be due to the impact of autocorrelated omitted variables or due to AR(2) or moving average errors. In this case, the natural question which arises is: what is the appropriate course of action if DW statistic is significant? First, we should check if the correlation in the error terms is pure autocorrelation or it is due to some omitted variables/ model mis-specification. If there is pure autocorrelation, then we can appropriately transform the original model. The method of Generalised Least Squares (to be discussed in Unit 11) can be employed to correct pure autocorrelation. For large samples, the Newey West Method can be employed. The various remedial measures are described below.

1. Quasi-first Differencing

Given, $y_t = \alpha + \beta x_t + u_t$; t = 1, 2, ...,T

We can obtain $y_{t-1} = \alpha + \beta x_{t-1} + u_{t-1}$

Now, $\rho y_{t-1} = \alpha\rho + \beta\rho x_{t-1} + \rho u_{t-1}$

Taking the difference with respect to $y_t = \alpha + \beta x_t + u_t$, we get,

$$y_t - \rho y_{t-1} = \alpha(1-\rho) + \beta(x_t - \rho x_{t-1}) + \varepsilon_t \qquad \text{…(9.24)}$$

[Errors follow AR(1) process such that $u_t = \rho u_{t-1} + \varepsilon_t$]

Equation (9.24) is known as Generalised or Quasi-Difference equation. The error terms ε_t satisfies the usual classical assumptions (independent having constant variance $\sigma_e{}^2$). Now, the OLS procedure can be applied by regressing the transformed variable $y_t{}^* = (y_t - \rho y_{t-1})$ on $x_t{}^* = (x_t - \rho x_{t-1})$. The resulting estimators are best, linear and unbiased (BLUE). In taking differences, one observation is lost; hence, $(T-1)$ observations are used.In order to compensate for loss of one observation, **Prais-Winsten transformation** is used wherein the first observation on y and x are transformed such that $y_1{}^* = y_1\sqrt{1-\rho^2}$ and $x_1{}^* = x_1\sqrt{1-\rho^2}$. Effectively, this procedure is the GLS procedure. The GLS procedure has T observations which includes$y_1{}^*$ and $x_1{}^*$.

Applying OLS to the transformed model which satisfies the classical assumptions is GLS. In this method, the variables are transformed by incorporating the autocorrelation parameter 'ρ'. The GLS procedure produces efficient estimates since all available information (about autocorrelation or heteroscedasticity) is used. In the presence of autocorrelation, the GLS estimator is the best linear unbiased estimator (BLUE).

$$\hat{\beta}_{GLS} = \frac{\sum_{t=2}^{n}(x_t - \rho x_{t-1})(y_t - \rho y_{t-1})}{\sum_{t=2}^{n}(x_t - \rho x_{t-1})^2} + \mathrm{C} \qquad \text{… (9.25)}$$

$$Var\hat{\beta}_{GLS} = \frac{\sigma^2}{\sum_{t=2}^{n}(x_t - \rho x_{t-1})^2} + \mathrm{D} \qquad \text{…(9.26)}$$

(C and D are correction factors)

One shortcoming of the quasi-differencing method is that ρ should be known. However, ρ is seldom known. There are various methods of estimating ρ which are discussed in Section 9.6. A limitation of using $\hat{\rho}$ is that it can be subject to sampling error. An alternative to quasi-differencing is the first-differencing method.

2. First-Differencing

Recall the quasi-difference equation given at (9.24)

$$y_t - \rho y_{t-1} = \alpha(1-\rho) + \beta(x_t - \rho x_{t-1}) + \varepsilon_t$$

If ρ= +1, then the above equation reduces to a first difference equation,

$$y_t - y_{t-1} = \beta(x_t - x_{t-1}) + (u_t - u_{t-1})$$

Or, $y_t - y_{t-1} = \beta(x_t - x_{t-1}) + \varepsilon_t$ …(9.27)

As mentioned earlier, the error term ε_t follows classical assumptions of the stochastic error term. Although u_t and $u_{t-1}$are correlated, the first difference of the error terms $(u_t - u_{t-1})$ are not correlated.Now, we can regress $(y_t - y_{t-1})$ on $(x_t - x_{t-1})$ by OLS method *without including the constant term* in the model. If there is a linear trend term in the regression equation (for example, $y_t = \alpha + \delta t + \beta x_t + u_t$), then the first difference equation has a constant term:

$(y_t - y_{t-1}) = \delta + \beta(x_t - x_{t-1}) + (u_t - u_{t-1})$ …(9.28)

Essentially, the first difference equation may be estimated if the Durbin-Watson statistic is small (rule of thumb: $d < R^2$). In a strict sense, first difference transformation is valid only if ρ= +1. In order to test whether ρ= +1 we use the Berenblut-Webb test.

The first difference transformation induces stationarity in the time series. If ρ= +1, then the variance $Var(u_t)$= $\frac{\sigma_\varepsilon{}^2}{(1-\rho^2)}$ and $Cov\,(u_t, u_{t-s})$ = $\rho^s \frac{\sigma_\varepsilon{}^2}{(1-\rho^2)}$ become infinite. Hence, the series u_t becomes non-stationary. With first difference transformation, the error term is ε_t, which does not have such problems.

Note that the R^2 of the levels equation (say, $y_t = \alpha + \beta x_t + u_t$) and the first difference equation ($(y_t - y_{t-1}) = \delta + \beta(x_t - x_{t-1}) + (u_t - u_{t-1})$) cannot be compared because they have different dependent variables. However, the Residual Sum of Squares (RSS) can be compared after making some adjustments.

The steps to be followed are as follows:

(i) Estimate the RSS from the first difference equation as well as the levels equation.

(ii) Multiply the RSS from the levels equation by$\frac{2(n-k-1)(1-\rho)}{(n-k)}$ (or, $\frac{(n-k-1)d}{(n-k)}$, which can be derived from the above)

(iii) Use the estimated ρ from the levels equation, that is, $\hat{\rho}$, [or, the DW statistic, $d \approx 2(1-\hat{\rho})$]). Here, k is the number of regressors.

(iv) Now, the RSS from the first difference equation and the levels equation can be compared.

Recall that the variance of the error term $var\ (u_t)$ = RSS/degrees of freedom. Therefore, variance of the error term in first difference equation

$$= Var\ (u_t - u_{t-1})$$

$$= Var\ (u_t) + Var\ (u_{t-1}) - 2\ Cov\ (u_t, u_{t-1})$$

$$= \sigma^2 + \sigma^2 - 2\sigma^2\rho = 2(1-\rho)\sigma^2$$

You should note that the above arguments do not hold in the presence of lagged dependent variables among the explanatory variables. Hence, this discussion excludes them by assumption. Having obtained comparable RSS, comparable R^2 can be obtained as well. Using $RSS = S_{yy}\ (1 - R^2)$, we get,

$$\frac{(1-R_D^2)}{(1-R_1^2)} = \left(\frac{RSS_0}{RSS_1}\right)\frac{(n-k-1)d}{(n-k)} \qquad \ldots(9.29)$$

Here, $S_{yy} = \sum {y_i}^2 - \bar{y}^2$

${R_1}^2 = R^2$ from the first difference equation

${R_D}^2 =$ comparable R^2 from the levels equation (i.e., without differencing)

RSS_0 = RSS from the levels equation

RSS_1 = RSS from the first difference equation

Sometimes, we get high R^2 in levels form equation and low R^2 in first difference equation. But the DW statistic may be found to be low in the levels form equation. This often suggests that the equation is mis-specified. In this case, we should use the first difference equation. Low R^2 in first difference equation may indicate that the dependent and independent variables are unrelated.

It should also be noted that the above discussion on first differencing and quasi-first differencing assumes a particular correlation structure between u_t and u_{t-1}. Sometimes, u_t can be found to be correlated with u_{t-4}. The DW statistic will not be able to detect this, and it has to be modified as follows:

$$d_4 = \frac{\sum(\hat{u}_t - \hat{u}_{t-4})^2}{\sum {\hat{u}_t}^2} \qquad \ldots (9.30)$$

In this case, fourth differencing or quasi-fourth differencing will have to be used, that is, regress $y_t - \hat{\rho} y_{t-4}$ on $x_t - \hat{\rho} x_{t-4}$, where, $\hat{\rho}$ is the correlation coefficient between $\hat{u}_t$ and $\hat{u}_{t-4}$.

3. The Newey-West Method

Under this method, the OLS method is used but the standard errors are corrected for autocorrelation. For large samples, the Newey-West method gives autocorrelation corrected standard errors of estimators. These are called Newey-West standard errors. In fact, these standard errors are corrected for both autocorrelation and heteroscedasticity and are also known as HAC (Heteroscedasticity-and-Autocorrelation-Corrected Standard Errors). This is an extension of White's Heteroscedasticity-Corrected standard errors. The Newey-West method is valid for large samples only.

9.6 METHODS OF ESTIMATING ρ

If 'ρ' is not close to 1 as required for First-Differencing method, it can be estimated using some of the methods described below.

1. From Durbin-Watson Statistic

The Durbin-Watson statistic $d \approx 2(1-\hat{\rho})$. Hence,

$$\hat{\rho} \approx 1 - \frac{d}{2} \qquad \ldots(9.31)$$

However, this method can be used in large samples only.

2. From the Residuals

Errors follow AR(1) process such that $u_t = \rho u_{t-1} + \varepsilon_t$. Estimate ρ by regressing $\hat{u}_t$ on $\hat{u}_{t-1}$ from the regression,

$$\hat{u}_t = \rho \hat{u}_{t-1} + v_t \qquad \ldots(9.32)$$

(v_t is the white noise).

3. Iterative Procedure

Taking some starting value of ρ , this method estimates $\hat{\rho}$ by successive approximation. Some iterative methods are discussed below:

3a. ***Cochrane–Orcutt Procedure:*** This method estimates $y_t = \alpha + \beta x_t + u_t$ by OLS and the residuals $\hat{u}_t$ are obtained. Hence, $\hat{\rho}$ which is the first-order coefficient of autocorrelation is obtained as follows:

$$\hat{\rho} = \sum \hat{u}_t\, \hat{u}_{t-1} / \sum \hat{u}_t^{\,2} \qquad \ldots(9.33)$$

This method can also be applied in the case of higher order autoregressive process.

3b. ***Durbin's Procedure:***Rewrite the equation,

$$y_t - \rho y_{t-1} = \alpha(1-\rho) + \beta(x_t - \rho x_{t-1}) + \varepsilon_t$$

as $y_t = \alpha(1-\rho) + \rho y_{t-1} + \beta x_t - \rho\beta x_{t-1} + \varepsilon_t \qquad \ldots(9.34)$

Now, regressing y_t on y_{t-1}, x_t and x_{t-1}, the coefficient of y_{t-1}gives an estimate of ρ. Using this estimate of ρ, find $(y_t - \rho y_{t-1})$ and $(x_t - \rho x_{t-1})$ and regress $(y_t - \rho y_{t-1})$ on $(x_t - \rho x_{t-1})$ to get an estimate of β. The constant term in this regression corresponds to $\alpha(1-\rho)$. Since the estimate of ρ is known, α can be obtained. Hence we get α and β, the estimates of the parameters of the original equation $y_t = \alpha + \beta x_t + u_t$.

A disadvantage of using this method is that it involves a number of variables in the regression. This is more of a problem if there are several independent variables.

4. Grid-Search Procedures

4a. *Hidreth and Lu Procedure:* Let $y_t^* = y_t - \rho y_{t-1}$ and $x_t^* = x_t - \rho x_{t-1}$. Assuming different values of ρ in the range +1 to -1 at intervals of 0.1, obtain y_t^* on x_t^* for different values of ρ and regress y_t^* on x_t^*. From each such regressions, obtain the RSS. Now, choose the regression having minimum RSS and choose the value of ρ corresponding to it. Smaller intervals say, 0.01 can be taken around the chosen value of ρ and this procedure can be repeated.

4b. *Maximum Likelihood Procedure:* If the error terms ε_t are normally distributed, the log likelihood function can be written as follows:

$$\log L = constant - \frac{T}{2}\log \sigma_e^2 + \frac{1}{2}\log(1-\rho^2) - \frac{Q}{2\sigma_e^2} \qquad \ldots(9.35)$$

Here, $Q = \sum[y_t - \rho y_{t-1} - \alpha(1-\rho) - \beta(x_t - \rho x_{t-1})]^2 \qquad \ldots(9.36)$

To get the ML estimates, we use the grid-search procedure. For each value of ρ, we compute the RSS (ρ) and choose that value of ρ for which $[\frac{T}{2}\log RSS(\rho) - \frac{1}{2}\log(1-\rho^2]$ is the minimum.

In conclusion, we can say that estimating 'ρ' by these different methods does not make much difference in large samples, because the OLS estimators are consistent even in the presence of autocorrelation. Using any of these methods, we get consistent estimates of the true 'ρ'. Having obtained an estimate of 'ρ', it is used to estimate GLS. Since '$\hat{\rho}$' is used, it is often known as *estimated GLS* (EGLS) or *feasible GLS* (FGLS)).

In small samples, however, the estimated coefficients (parameters of the transformed model) may not be BLUE because '$\hat{\rho}$' is used instead of true 'ρ'. The estimated OLS coefficients display usual optimum properties asymptotically (when sample size increases infinitely). Moreover, in small samples, it is sometimes better to continue using OLS. As a rule of thumb, if sample size is 20 and $\hat{\rho} < 0.3$, OLS is better than FGLS ($\hat{\rho}$ is the estimated first order coefficient of autocorrelation). Additionally, it must be noted that this discussion essentially assumes that the error terms are AR(1). If they are higher order autoregressive, it is sometimes advised to continue assuming AR(1) process rather than ignoring autocorrelation altogether and using usual OLS. The issue is debatable, and some researchers contradict this by saying that doing so may worsen the problem.

Check Your Progress 3

1) Consider the model $y_t = \alpha + \beta x_t + u_t$

$u_t = \rho u_{t-1} + \varepsilon_t;\ 0 \leq \rho \leq 1$

$\varepsilon_t \sim IN(0, \sigma^2)$

By regressing Δy_t on Δx_t, is it possible to get more efficient estimates of β than by regressing $y_t$onx_t?

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2) Given $y_t = 2.8 + 0.5\ x_t + 0.8\ \ y_{t-1}$

$\qquad\qquad\qquad(0.38)\qquad(0.06)$

The p-values are given in parenthesis. $R^2 = 0.95$. $DW = 1.89$. Here, R^2 is high and DW is also closer to 2. Identify the problem in this equation.

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9.7 LET US SUM UP

Autocorrelation is a typical problem in time series data. Even with autocorrelated errors, it is seen that the OLS estimators are unbiased. However, they no longer have minimum variances and the variances are also biased. Thus, the usual tests of significance invalid in the presence of autocorrelation among error terms. Of the various tests to detect autocorrelations, the DW test is the most popular. But, even a statistically significant DW statistic does not necessarily mean that the errors are autocorrelated. Sometimes incorrect specification of the model or omission of relevant variables may cause the DW statistic to be statistically significant. Thus, one should first verify if there is pure autocorrelation or not before proceeding for correcting it. There are other problems with the DW statistic. It cannot detect higher order autocorrelations and it cannot be applied to models having lagged explained variables as independent variables. In view of these shortcomings, the BG test scores over the DW test. Having detected the presence of pure autocorrelation, the next step is to correct for it. The method of GLS can be used but one has to estimate ρ for it. If ρ is close to 1, then first-difference transformation can be applied. There are various methods of estimating 'ρ'. In large samples, any method of estimating ρ provides consistent estimates of true ρ. This is because despite autocorrelation, the OLS estimators are consistent, that is, they tend towards their true parameter values with increase in sample size. However, in small samples, the results have to be interpreted cautiously. The estimated OLS coefficients (parameters of the transformed model) have the optimum properties but only in large samples.

9.8 KEY WORDS

Generalised Least Squares : GLS is basically application of OLS to the transformed model which satisfies the classical assumption.In this method, the variables are transformed by incorporating the autocorrelation parameter 'ρ'. The GLS procedure produces efficient estimates since all available information (about autocorrelation or heteroscedasticity) is used. In the presence of autocorrelation, the GLS estimator is BLUE.

Serial correlation or Autocorrelation : In time series data, the correlation between error terms is known as autocorrelation or serial correlation.

White Noise : An error term 'ε_t' which satisfies the following properties:

$E(\varepsilon_t) = 0$

$Var(\varepsilon_t) = \sigma_\varepsilon{}^2$

$Cov(\varepsilon_t, \varepsilon_{t+s}) = 0;\ s \neq 0$

9.9 SOME USEFUL BOOKS/REFERENCES

Brooks, Chris (2014), '*Introductory Econometrics for Finance*,' Third Edition, Cambridge University Press

Gujarati, Damodar N. and Sangeetha (2007), '*Basic Econometrics*,' Fourth Edition, Tata McGraw Publishing Company Limited, New Delhi, Edition 2007

Maddala, G. S. and K. Lahiri (2009), *Introduction to Econometrics*, Fourth Edition, John Wiley and Sons

Wooldrige, Jeffrey (2022), *Introductory Econometrics*, Seventh Edition, Cengage Learning

9.10 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) False. The consequences of autocorrelation are given in Section 9.3.
2) True. Refer to Section 9.3.
3) You should represent u_t in terms of $u_{t-2}$and find the covariance between the two.

Check Your Progress 2

1) The procedure of DW test is given in Section 9.4.
2) The limitations of DW test are given in Section 9.4.

3) Yes, there is positive autocorrelation. Go through Section 9.4 for details.

Check Your Progress 3

1) Refer to Section 9.5 and Section 9.3.
2) In models having lagged explained variables, the DW test cannot be used.

UNIT 10 MULTICOLLINEARITY*

Structure

10.0 OBJECTIVES

After reading this unit, you should be in a position to

- explain the problem of multicollinearity in regression analysis;
- identify how the problem of multicollinearity affects the results of regression analysis;
- explain various methods used to detect multicollinearity in a regression model; and
- suggest remedial measures in case multicollinearity is present in a dataset.

10.1 INTRODUCTION

Multicollinearity is basically the presence of high correlation between explanatory variables in a multiple regression model. An assumption of the OLS method, as you may recall from Unit 3, is that the explanatory variables are not correlated with one another (in technical terms, they are orthogonal). A non-zero correlation between explanatory variables is typically observed. Low correlation between explanatory variables may not be a problem, but if the correlation *between explanatory variables is found to be very high, it causes serious*

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problems in parameter estimation by inflating the standard errors and making t-ratios statistically insignificant.

10.2 CONCEPT OF MULTICOLLINEARITY

As pointed out above, multicollinearity is high intercorrelation between the explanatory variables. Two types of multicollinearity need to be discussed: *perfect multicollinearity* and *near multicollinearity.*

Perfect Multicollinearity: In this case two or more explanatory variables bear an exact linear relationship. In the presence of perfect multicollinearity, estimating all the coefficients of the model is not possible. For example, two variables 'x_2' and 'x_3' used in a regression model are such that:

$$x_2 = 2\, x_3 \qquad \dots (10.1)$$

Here, x_2 and x_3 are perfectly related to each other. In this case if we regress x_2 on x_3 we find that the coefficient of determination r_{23}^2 is equal to 1. A scatter plot of x_2 against x_3 will show all points to remain on a straight line. To cite another example, suppose in a regression model (in deviation form)

$$y = \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u \qquad \dots (10.2)$$

the independent variables are related to each other by the equation $x_3 = 2x_1 + 3x_2$

If we substitute $x_3 = 2x_1 + 3x_2$ in equation (10.2), we obtain

$$y = \beta_1 x_1 + \beta_2 x_2 + \beta_3 (2x_1 + 3x_2) + u, \quad \text{or}$$

$$y = (\beta_1 + 2\beta_3)\, x_1 + (\beta_2 + 3\beta_3) x_2 + u \qquad \dots (10.3)$$

Note that equation (10.3) is regression model where y is regressed on x_1 and x_2 with parameters $(\beta_1 + 2\beta_3)$ and $(\beta_2 + 3\beta_3)$. Thus it is possible to estimate the linear combinations of the parameters $\beta_1 + 2\beta_3$ and $\beta_2 + 3\beta_3$, but it is not possible to estimate β_1, β_2 and β_3 separately. That is, when multicollinearity is present in a regression model, disentangling the separate effect of each explanatory variable on the explained or dependent variable is difficult. In a *m*-variable regression model, there is an exact linear relationship between the variables if,

$$\varphi_1 x_1 + \varphi_2 x_2 + \dots + \varphi_m x_m \qquad \dots (10.4)$$

The parameters $\varphi_1, \varphi_2 \dots \varphi_m$ are constants, not all zero simultaneously.

Near Multicollinearity: In this case, correlation between the explanatory variables is high but not perfect (r_{23}^2 is not exactly, but nearly equal to1). In this case, the scatter plot of x_2 against x_3 will show all points lying close to the straight line, but not exactly on it. In a *m*-variable regression model, multicollinearity is less than perfect if,

$$\varphi_1 x_1 + \varphi_2 x_2 + \dots + \varphi_m x_m + v \qquad \dots (10.5)$$

where v is a stochastic error term.

You should note that in multicollinearity problem we consider only linear relationships between the explanatory variables. It does not apply to non-linear relationships between variables.

10.3 CONSEQUENCES OF MULTICOLLINEARITY

Even with near multicollinearity, the estimators obtained by OLS are:

- Unbiased
- Consistent
- Best Linear Unbiased Estimate (BLUE)

Some of the consequences of multicollinearity are discussed below.

1. Regression coefficients are indeterminate or not precisely estimable

With perfect multicollinearity, the parameters of the explanatory variables cannot be determined. For example, let us consider the regression model

$$y_i = \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{u}_i \qquad \ldots(10.6)$$

Here, y, x_1 and x_2 are taken in deviation form from the respective means, $\hat{\beta}_1$ and $\hat{\beta}_2$ are estimated regression coefficients and *'i'* range from 1 to *n*. Suppose there is perfect multicollinearity problem such that $x_{2i} = \tau x_{1i}$ (where τ is a constant). Thus the above regression model can be written as

$$y_i = \hat{\beta}_1 x_{1i} + \hat{\beta}_2(\tau\, x_{1i}) + \hat{u}_i \,, \qquad \text{or}$$

$$y_i = (\hat{\beta}_1 + \hat{\beta}_2 \tau) x_{1i} + \hat{u}_i \qquad \ldots(10.7)$$

$$\text{Let} \hat{\beta}_1 + \hat{\beta}_2 \tau = \alpha \qquad \ldots(10.8)$$

In the above formulation it is not possible get the estimators for $\hat{\beta}_1$ and $\hat{\beta}_2$ separately. However, α $\left(= \hat{\beta}_1 + \hat{\beta}_2\right)$ can be estimated which is a linear function of the two regression coefficients, $\hat{\beta}_1$ and $\hat{\beta}_2$. It is worth mentioning that we can obtain a unique value for α but there will be infinite solutions for $\hat{\beta}_1$ and $\hat{\beta}_2$, given the values of α and τ. Thus obtaining a unique set of values for $\hat{\beta}_1$ and $\hat{\beta}_2$ is not possible. With less than perfect multicollinearity, the regression coefficients can be estimated but not precisely, because their standard errors turn out to be very high (see discussion below).

2. Standard errors of the coefficients are very high or infinite

The variance and covariance of the estimators are very large in the presence of multicollinearity. With perfect multicollinearity, the standard errors of the estimators reach a value of infinity. With less than perfect multicollinearity, the standard errors of the regression coefficients are very large. In the regression model $y_i = \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{u}_i$, the variances of the estimated regression coefficients $\hat{\beta}_1$ and $\hat{\beta}_2$ are given by

$$Var\left(\hat{\beta}_1\right) = \frac{\sigma^2}{\sum x_{1i}^2 (1 - r_{12}^2)}, \text{ and}$$

$$Var\ (\hat{\beta}_2) = \frac{\sigma^2}{\sum x_{2i}^2(1-r_{12}^2)} \quad \ldots(10.9)$$

The covariance of $\hat{\beta}_1$ and $\hat{\beta}_2$ is given by

$$Cov\ (\hat{\beta}_1, \hat{\beta}_2) = \frac{-r_{12}^2\sigma^2}{(1-r_{12}^2)\sqrt{\sum x_{1i}^2 x_{2i}^2}} \quad \ldots(10.10)$$

Here, σ^2 is the variance of the error term u_i and r_{12} is the correlation coefficient between explanatory variables x_1 and x_2. From (10.9) and (10.10), we find that as r_{12} tends to unity (implying perfect multicollinearity), the variance and covariance of the estimators tend to infinity.

3. Confidence intervals are wider

Recall from Unit 3 that confidence interval depends on the size of the standard error. If the standard error for an estimator is high, its confidence interval is wide and the null hypothesis is readily accepted. There is a greater likelihood of Type II error (the hypothesis is false but the test statistic does not reject it).

4. R^2 is high but few t-ratios are significant

In the presence of multicollinearity, isolating the explanatory power of each independent variable becomes difficult. The R^2 is found to be high, but the individual variable's regression coefficients are found to exhibit low levels of statistical significance or they may be statistically insignificant.

You may recall that $t = \frac{\hat{\beta}}{se\ (\hat{\beta})}$. If there is an increase in the standard error of $\hat{\beta}$, there is a decrease in its t-ratio. In the presence of multicollinearity, as individual regression coefficients exhibit high standard errors, the t-ratios become smaller and statistically not significant. The null hypothesis of all slope coefficients being jointly zero is rejected on the basis of the F-test (reflected in the high R-squared value), but individual regression coefficients are declared insignificant on the basis of the *t*-test.

5. Regression results become very sensitive to data or specification changes

In the presence of multicollinearity, addition or removal of any independent variable causes large changes in the values of regression coefficients and their standard errors.

Check Your Progress 1

1) "In multiple regression, a high correlation in the sample among the regressors implies that the least square estimators of the coefficients are biased". True or False? Justify your answer.

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2) You are given a regression model $Y = \alpha + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 (X_{2i} + X_{3i}) + \beta_5 X_{2i}{}^2 + \beta_6 X_{3i}{}^2 + u_i$. Which of the parameters in this model can be estimated by OLS?

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10.4 DETECTION OF MULTICOLLINEARITY

There are a few rules of thumb for detecting multicollinearity. Here, the idea is not to see if multicollinearity is present or absent, but to measure its degree or strength in the sample at hand.

10.4.1 Various Methods to Detect Multicollinearity

There are several methods to detect the presence of multicollinearity.

1. Correlation Matrix

Let us assume that a regression model has three explanatory variables, viz. 'x_1', 'x_2' and 'x_3'. A matrix of pair-wise correlations among the above variables is given as follows:

correlation	x_1	x_2	x_3
x_1	-	0.1	0.85
x_2	0.1	-	0.3
x_3	0.85	0.3	-

Here, a high correlation between 'x_1' and 'x_3' shows the presence of multicollinearity. However, such a simple method of detection of multicollinearity will not prove useful if three or more variables are collinear. For example, suppose, $x_1 + x_2 = x_3$. In this case, the coefficient of determination (R^2) obtained from regressing x_3 on x_1 and x_2 equals unity. However, the pair-wise correlations between x_3 and x_1, between x_1 and x_2 and between x_1 and x_2 may not be high. Thus, pair-wise correlations are not a useful indicator of multicollinearity when three or more variables are collinear.

2. High Standard Errors of the Regression Coefficients

Due to high inter-correlation among independent variables, the estimated regression coefficients have high standard errors. But this is not necessarily true as demonstrated below. Even low inter-correlation among independent variables can cause high standard errors. In the regression model $y_i = \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{u}_i$ given above, the variance of the estimated regression coefficients $\hat{\beta}_1$ and $\hat{\beta}_2$ are given by the following:

$Var\,(\hat{\beta}_1) = \frac{\sigma^2}{\sum x_{1i}^2(1-r_{12}^2)}$

$Var\,(\hat{\beta}_2) = \frac{\sigma^2}{\sum x_{2i}^2(1-r_{12}^2)}$

We can see that the variance of $\hat{\beta}_1$ will be high if i) σ^2 is high, ii) $\sum x_{1i}^2$ is low, and iii) r_{12}^2 is high. Thus, just knowing that r_{12}^2 is high, does not tell us whether multicollinearity exists or not. In fact, standard errors can be high if σ^2 is high and $\sum x_{1i}^2$ is low. Hence, the proposition that high pair-wise correlation between explanatory variables necessarily cause high standard errors is not true. Extending the above formula to a *k*-variable regression model, the variance of the *j*th regression coefficient is given by:

$$Var\,(\hat{\beta}_j) = \frac{\sigma^2}{\sum x_j^2(1-R_j^2)} \qquad \dots (10.11)$$

Here, $\sum x_j^2 = \sum (X_j - \bar{X})^2$

$\hat{\beta}_j$ is the estimator of $\hat{\beta}_j$ the regression coefficient of explanatory variable X_j. R_j^2 is the R^2 obtained from the regression of X_j on the remaining explanatory variables. Alternatively, R_j^2 is the squared multiple correlation coefficient between X_j and other regressors. From (10.11) we can see that "high R_j^2 is neither necessary nor sufficient condition of getting high standard errors. Thus multicollinearity by itself need not cause high standard errors" (Maddala, 2001).

3. Variance Inflation Factor (VIF)

Another indicator of multicollinearity is the VIF. In the regression model $y_i = \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{u}_i$ given above, the VIF is given by:

$$VIF = \frac{1}{(1-r_{12}^2)} \qquad \dots (10.12)$$

In the case of a multiple regression model, the VIF is given by:

$$VIF_j = \frac{1}{(1-R_j^2)} \qquad \dots (10.13)$$

You can observe from (10.12) that VIF tends to ∞ as r_{12}^2 tends to 1. With zero correlation between the independent variables, the VIF is unity. That is, higher intercorrelation between the explanatory variables scales up the VIF and the variances and standard errors of regression coefficients. As seen from the formulas below, $Var\,(\hat{\beta}_1)$ and $Var\,(\hat{\beta}_2)$ increases as VIF increases.

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum x_{1i}^2} VIF \qquad \dots (10.14)$$

$$Var\,(\hat{\beta}_2) = \frac{\sigma^2}{\sum x_{2i}^2} VIF \qquad \dots (10.15)$$

However, as already discussed, high VIF need not necessarily cause high standard error for $\hat{\beta}_1$. If σ^2 is low and $\sum x_{1i}^2$ is high (or, $\sum x_{2i}^2$ is high) we obtain high variance and standard error. Whether multicollinearity is posing a problem or not can be best understood by looking at the standard errors and t-ratios. For $\hat{\beta}_{2,}$ the high intercorrelation among the independent variables may not cause high standard errors. Thus, VIF can be a misleading indicator of the extent to which multicollinearity is posing a problem. Moreover, VIF has to be computed separately for each explanatory variable. In this sense, VIF is rather useful in identifying the variables that are highly correlated with other explanatory variables and hence can be dropped.

4. Tolerance

The inverse of VIF is known as Tolerance (TOL) and it is another indicator of collinearity between variables.

$$TOL_j = \frac{1}{VIF_j} = (1 - R_j^{\,2}) \qquad \ldots (10.16)$$

TOL_j ranges from 0 (perfect collinearity) to 1 (no collinearity).

5. Klien's Rule of Thumb

Klein argued that multicollinearity is a problem only if $R_j^{\,2}$ exceeds the overall R^2. This condition can be stated as:

$$R_j^{\,2} > R^2 \qquad \ldots (10.17)$$

However, this rule has limitations as it is possible to obtain significant regression coefficients even if $R_j^{\,2} > R^2$.

6. Condition Number (CN)

Condition number is an overall measure of multicollinearity. Remember that if the explanatory variables are exactly linearly dependent, the matrix of explanatory variables *X'X* is singular. *X'X* is close to being singular if the variables are almost linearly dependent; even if not exactly linearly dependent. The 'condition number' measures multicollinearity from this perspective. In a k-variable regression model, the matrix *X'X* of explanatory variables is used to find the maximum and minimum eigen value. Eigen values or characteristic roots of any $n \times n$ matrix A are obtained by solving for γ from the characteristic equation $|A - \gamma I| = 0$. This is an n^{th} degree equation in γ having 'n' characteristic roots. The word 'eigen' is borrowed from German, which means 'proper'.

$$\text{Condition Number} = \sqrt{\frac{maximum\ eigen\ value}{minimum\ eigen\ value}} \qquad \ldots (10.18)$$

Condition number can be calculated by taking the maximum eigen value and any other eigen value. In relation to the maximum eigen value, if the other eigen value is low, it shows the presence of linearly-dependent explanatory variables in the data or near-linear dependencies. The condition number (CN) is equal to 1 if there is no multicollinearity. The condition number increases with increase in collinearity. There are some informal rules of thumb with respect to CN. If the CN is 15, multicollinearity is moderate. If the CN exceeds 30, multicollinearity is severe. By suitably transforming the variables, the CN can be made close to 1.

7. Theil's Measure of Multicollinearity

VIF and CN do not look at the intercorrelation between dependent and independent variables. They only examine the intercorrelation between independent variables. Theil's Measure of Multicollinearity (m') examines the correlation between the explained variable and the explanatory variables. It is indicated by:

$$m' = R^2 - \sum_{p=1}^{k}(R^2 - R_{-p}{}^2) \quad \ldots (10.19)$$

Here, R^2 is the squared multiple correlation obtained by regression of 'y' on 'k' explanatory variables. $R_{-p}{}^2$ is the squared multiple correlation obtained by regression of 'y' on 'k' explanatory variables omitting x_p. $(R^2 - R_{-p}{}^2)$ is the incremental contribution of the p^{th} variable to R^2. The incremental contribution of 'k' explanatory variables $(x_1, x_2, \ldots, x_k)$ add up to R^2 if these variables are mutually uncorrelated. In that case, $m' = 0$. But with multicollinearity, 'm'' can be highly positive or even negative, making it a not so useful indicator of the problem of multicollinearity.

10.4.2 Problems in Measurement of Multicollinearity

In order to understand the problems in measurement of multicollinearity, consider the following example where the independent variables can be defined in various ways. Defining C (per capita real consumption), Y (per capita real income), Y_P (per capita real permanent income) and Y_T (per capita real transitory income) we have the relationship:

$$Y = Y_P + Y_T \quad \ldots (10.20)$$

Let us assume that Y_T and Y_P are not correlated, $i.e.$, E $(y_T, y_P) = 0$

The consumption function is given by any of the following three ways:

$$C = \alpha Y + \beta Y_P + u \quad \ldots (10.21)$$

By substituting (10.20) in (10.21) we obtain

$$C = \alpha Y_T + (\alpha+\beta) Y_P + u \quad \ldots (10.22)$$

The above can be expressed as

$$C = (\alpha+\beta) Y - \beta Y_T + u \quad \ldots (10.23)$$

The above three ways (equations 10.21, 10.22 and 10.23) in which the consumption function is expressed are essentially the same. But multicollinearity may be reported to be high in the first expression and there may be no mulicollinearity in the second expression since it is given that Y_T and Y_P are not correlated. What matters is how precisely we can estimate α, β or $(\alpha+\beta)$. In fact, in some cases, the linear combination $(\alpha+\beta)$ is more precisely estimable than α or β. Such linear combinations often make economic sense based on theory.

Thus, when examining whether multicollinearity is a serious issue, we should first examine the pair-wise correlation between various explanatory variables. High correlation may be suggestive of a problem although not necessarily. If the *t*-ratios are found to be statistically significant, then multicollinearity is not a serious issue. One can also check whether the estimated coefficients are stable by deleting some observations (i. e., by changing sample size).

Check Your Progress 2

1) Examine whether the following statements are 'true' or 'false'. Give an explanation for your answer.

   a) Multicollinearity is a problem cannot be decided by just looking at the pair-wise correlation between the explanatory variables.

   b) High standard error of the estimate or of a variable in a regression equation has in an equation have, evidence of high multicollinearity.

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10.5 REMEDIAL MEASURES FOR MULTICOLLINEARITY

Some of the solutions to multicollinearity are suggested below:

1. Dropping one of the correlated variables

This method may be resorted to but it can cause omission bias. Moreover, if there is strong theoretical reason for including a variable, then exclusion of the variable is unacceptable. In the model $y = \beta_1 x_1 + \beta_2 x_2 + u$, suppose variables$x_1$ and x_2 are highly correlated. The OLS estimator of β_1 is denoted by $\hat{\beta}_1$. The mean and variance of $\hat{\beta}_1$ are such that:

$$E(\hat{\beta}_1) = \beta_1 \qquad \ldots (10.24)$$

and

$$Var\,(\hat{\beta}_1) = \frac{\sigma^2}{\sum x_1{}^2(1-r_{12}{}^2)} \qquad \ldots (10.25)$$

Now, we drop the variable x_2 such that the model looks like $y = \beta_1 x_1 + v$. The Omitted variable estimator of β_1 is denoted by $\widetilde{\beta_1}$.

$$\widetilde{\beta_1} = \frac{\sum x_1 y}{\sum x_1{}^2} = \frac{\sum x_1(\beta_1 x_1 + \beta_2 x_2 + \mu)}{\sum x_1{}^2}$$

$$\widetilde{\beta_1} = \beta_1 + \frac{S_{12}}{S_{11}}\beta_2 + \frac{\sum x_1 u}{S_{11}} \qquad \ldots (10.26)$$

Here, $S_{12} = \sum x_1\, x_2, S_{11} = \sum x_1{}^2$

$$E(\widetilde{\beta_1}) = \beta_1 + \beta_2 \frac{S_{12}}{S_{11}} \qquad \ldots (10.27)$$

Hence, $\widetilde{\beta_1}$ is biased.

$$Var(\beta_1{}^*) = Var(\frac{\sum x_1 u}{S_{11}}) = \frac{\sigma^2 S_{11}}{S_{11}{}^2} = \frac{\sigma^2}{S_{11}} \quad \dots (10.28)$$

Comparing with the variance of $\hat{\beta}_1$, we see that $\widetilde{\beta_1}$ has a smaller variance. Greater the correlation coefficient r_{12}, smaller the variance of $\widetilde{\beta_1}$ as compared to $\hat{\beta}_1$.

$$\frac{Var(\widetilde{\beta_1})}{Var(\hat{\beta}_1)} = 1 - r_{12}{}^2 \quad \dots (10.29)$$

Although $\widetilde{\beta_1}$ has lesser variance, it is biased. Omitting a variable worsens the problem of multicollinearity because it causes the estimators to be biased whereas with multicollinearity, the estimators are unbiased and continue to have the minimum variance among the linear unbiased estimators (BLUE).

2. Ignoring it

Even with multicollinearity, the OLS estimators are unbiased, efficient and consistent, that is, they continue to be BLUE. The problem of multicollinearity can be ignored if the model is nevertheless statistically adequate with the coefficients having appropriate sign. Sometimes, the presence of multicollinearity may not reduce t-ratios to such an extent as to render the variables insignificant. Moreover, multicollinearity is a sample or data problem over which the researcher has little control. Even if all the coefficients are not estimable, a linear combination of them may be estimable which serves well in many cases.

3. Ratio transformations of the correlated variables

The variables can be suitably transformed by expressing them in ratios, for instance, as a percentage of GDP or in per capita terms wherever applicable. However, ratio transformation of variables should be supported by theoretical logic. First difference transformation of variables can also be done. It is mainly done as a remedy for autocorrelation. These transformations reduce the collinearity between the variables. However, such transformations are not without problems. In taking first differences, one observation is lost, hence reducing degrees of freedom by one. This may affect the analysis if the sample size is small. Also, in ratio transformations, the error terms in the transformed model may become heteroscedastic if the original model has homoscedastic errors.

4. Expanding the sample size

You might have noticed by now that multicollinearity is more of a data problem. As we mentioned earlier, standard error can be increased by reducing the error variance or increasing $\sum x_{1i}{}^2$. Thus, collecting more data if possible or pooling the data across cross-sections and over time is another solution. The independent variables may not be collinear in the population although they may be found to be collinear in the given sample. For instance, theoretically, consumption is a function of both income and wealth. But, in a sample of observations, income and wealth may be found to be collinear as wealthy people may also be having higher incomes. This is especially true in time series data. In a cross-section of observations, it is possible to obtain a sample in which the relationship between wealth and income is random. Hence, if we have the 'right' sample, multicollinearity may not be reflected in the variables.

5. Using extraneous estimates

Let us consider the following equation:

$$\log Q = \rho + \beta logP + \theta logM + \varepsilon \quad \ldots (10.30)$$

where $Q =$ quantity, P = Price, $and\ M$ = income) in (10.30) estimating both price and income elasticity precisely is difficult in time series data as income and price are generally found to be correlated. In that case, estimate of income elasticity $\hat{\theta}$ can be obtained from budget studies and by regressing ($\log Q - \hat{\theta} logM$) on log P, estimates of ρ and β can be obtained. Here, $\hat{\theta}$ is called the extraneous estimate. However, the problem is that the cross-section estimate of θ as obtained from budget studies may not be the same as its time series estimate. Hence, this process is advised only when data is lacking. When both cross-section and time series data are available, pooling different data sets will enable getting more precise estimates.

6. Getting a priori information

Sometimes, the underlying economic theory may give relevant information. For example, let us consider the Cobb-Douglas production function,

$$Y = AK^{\alpha}L^{\beta} \quad \ldots (10.31)$$

Where Y = output, K = capital, L = labour, A = efficiency parameter, α and β are share parameters of the production function. Here, K and L may be found to be collinear as the assumption of constant returns to (ERS)scale gives $\alpha + \beta = 1 implies\ \beta = 1 - \propto$). Now, we can express the above function in per labour terms as,

$$y = Ak^{\alpha} \qquad \frac{Y}{L} = \frac{AK^{\propto}L^{1-\propto}}{L}\ or\ y = A\,(K/L)^{\propto} = AK^{\propto} \quad \ldots (10.32)$$

Regressing output per labour $'y'$ on capital per labour $'k'$ gives an estimate of α. But such a priori restriction(such as CRS) are generally supposed to be tested rather than imposed on the data.

7. Ridge Regression

Here, a constant **λ** is added to the variance of explanatory variables. In matrix notation, $\boldsymbol{X'X}$ is the variance-covariance matrix of explanatory variables. By adding **λ** to the diagonal elements of $\boldsymbol{X'X}$ matrix, the ridge estimator $\widehat{\boldsymbol{\beta}}_{\boldsymbol{R}}$ can be found. Doing so decreases the intercorrelation between explanatory variables.

$$\widehat{\boldsymbol{\beta}}_{\boldsymbol{R}} = (\boldsymbol{X'X} + \boldsymbol{\lambda}\mathbf{I})^{-1}\boldsymbol{X'y} \quad \ldots (10.33)$$

Here, **X** is the matrix of explanatory variables and $\boldsymbol{y}$ is the vector of explained variables. The procedure of Ridge Regression is explained below.

(i) Constrained Least Squares

The residual sum of squares ($\boldsymbol{u'u}$) is minimised subject to the constraint that $\sum_{i=1}^{k} \beta_i{}^2$ *or* $\boldsymbol{\beta'\beta = c}$

Our objective function is which needs to be minimised,

$$(\boldsymbol{y - X\beta})'(\boldsymbol{y - X\beta}) + \boldsymbol{\lambda}(\boldsymbol{\beta'\beta - c}) \quad \ldots (10.34)$$

(Here, $\boldsymbol{\lambda}$ is the Lagrangian multiplier)

Setting the first order derivatives with respect to $\boldsymbol{\beta}$ equal to zero, we obtain

$$\boldsymbol{-2X'y + 2X'X\beta + 2\beta\lambda = 0}$$

Hence,

$$\boldsymbol{\hat{\beta}_R = (X'X + \lambda I)^{-1}X'y} \quad \ldots (10.35)$$

In the case of two explanatory variables, the constrained least squares minimisation problem is give as follows:

$$\text{Minimize } \sum (y - \beta_1 x_1 - \beta_2 x_2)^2 + \lambda\,(\beta_1{}^2 + \beta_2{}^2 - c) \quad \ldots (10.36)$$

Setting the first order partial derivatives with respect to β_1 and β_2 equal to zero, we get,

$$2\sum (y - \beta_1 x_1 - \beta_2 x_2)(-x_1) + 2\lambda\,\beta_1 = 0 \quad \ldots (10.37)$$

$$2\sum (y - \beta_1 x_1 - \beta_2 x_2)(-x_2) + 2\lambda\,\beta_2 = 0 \quad \ldots (10.38)$$

Simplifying the above two equations, we get,

$$(S_{11} + \lambda)\beta_1 + S_{12}\beta_2 = S_{1y} \quad \ldots (10.39)$$

$$S_{12}\beta_1 + (S_{22} + \lambda)\beta_2 = S_{2y} \quad \ldots (10.40)$$

Here, $S_{11} = \sum x_1{}^2$; $S_{22} = \sum x_2{}^2$; $S_{12} = \sum x_1\, x_2\; s_{1y} = \sum x_1\, y$ *and* $s_{2y} = \sum x_2\, y$.

In (10.39) and (10.40) we have two equations in two unknowns (since date on x and y are available which β_1 *and* β_2 *are unknown*) Thus, we get the ridge regressionestimators for two explanatory variables and from these equations, λ can be estimated using the criterion $\beta_1{}^2 + \beta_2{}^2 = c$.

In the case described above, applying ridge regression is natural when prior information is given. But having prior information (in the above example that $\beta_1{}^2 + \beta_2{}^2 = c$) is not always possible. Hence, choosing λ is not always easy. You should note that λ is a function of β_i and σ^2 (variance of the error term), both of which are unknown. Some authors suggest that different values of λ may be tried and that value of λ may be chosen such that the coefficients do not have "unreasonable values". Others have suggested that in order to estimate λ, initial estimates of β_i and σ^2 can be used. Iterative procedure can be applied to get the iterated ridge estimator.

One shortcoming of ridge regression is that addition of λ to the variances of explanatory variables results in biased estimators but there is a decline in the mean squared error (MSE) as well. As pointed out by some authors, there always exists $\lambda > 0$ such that,

$$\sum_{i=1}^{k} MSE\ (\tilde{\beta}_i) < \sum_{i=1}^{k} MSE\ (\hat{\beta}_i) \qquad \ldots (10.41)$$

Here, k is the number of explanatory variables, $\tilde{\beta}_i$ is the ridge regression estimator of β_i and $\hat{\beta}_i$ is the OLS estimator of β_i. Another shortcoming is that ridge regression estimators differ when different linear transformations of independent variables are used. Also, ridge regression depends on units of measurement of the independent variables. If the explanatory variables are measured in different units, adding the same value of λ to the variance of different explanatory variables does not make sense. Normalization of variables however can solve this problem.

(ii) Bayesian Interpretation

Here, some prior information on regression parameters is combined with sample information. Assuming the prior information $\beta_i \sim IN\ (0, \sigma_\beta^2)$, the ridge regression estimates of β_i's are obtained. Here, the ridge constant $\lambda = \sigma^2 / \sigma_\beta^2$ where σ^2 has to be estimated as it is unknown. However, the mean of β_i's is equal to zero is not a reasonable assumption. Hence, the ridge estimator with Bayesian interpretation does not make much sense in econometrics. More complicated generalized ridge estimators are obtained by relaxing this assumption.

(iii) Measurement Error Interpretation

In this case, random errors with mean zero and variance 'λ' are added to the explanatory variables. Because the errors are random, the covariance between the explanatory variables is unaffected but the variance of explanatory variables increases by 'λ'. Hence, ridge regression estimator is obtained. In this method, the data is not precise as random errors are added but the estimates obtained by adopting such a technique are precise.

In conclusion, ridge regression is justified in some situations but involves subjective judgement or 'vague prior information' in others. Due to these shortcomings, it is not recommended to use ridge regression as a general solution to the problem of multicollinearity.

8. Principal Components Regression

Principal Component Analysis (PCA) is a multivariate statistical technique in which the 'k' correlated explanatory variables are used to derive uncorrelated indices or components such that each of these components is a linear function of the 'k' explanatory variables. Assuming 'k' explanatory variables ($x_1, x_2, ..., x_k$), the linear functions of these variables are as follows:

$$z_1 = a_1x_1 + a_2x_2 + \cdots + a_kx_k \qquad \ldots (10.42)$$

$$z_2 = b_1x_1 + b_2x_2 + \cdots + b_kx_k \qquad \ldots (10.43)$$

Where $z_1, z_2,...,z_k$ are the principal components. Hence, there are 'k' principal components. The first principal component 'z_1' is found out by choosing a's such that the variance of z_1 is maximised subject to the condition that:

$$a_1^2 + a_2^2 + ... + a_k^2 = 1 \text{(normalization condition)} \qquad \ldots (10.44)$$

Similarly, the second, third and subsequent principal components are derived. The first principal component 'z_1' has the highest variance.

$$Var(z_1) > Var(z_2) > Var(z_3) > .... > Var(z_k) \quad ... (10.45)$$

The various principal components are orthogonal or perpendicular to each other, that is, they are not correlated to each other. Another property of the principal components is that:

$$Var(z_1) + Var(z_2) + Var(z_3) + ... + Var(z_k) = Var(x_1) + Var(x_2) + Var(x_3) + ... + Var(x_k) \quad ... (10.46)$$

Now, how does the PCA help in multicollinearity? We can regress 'y' on the z's but regressing 'y' on z's and then substituting the value of z's in terms of x's amounts to the same thing as regressing 'y' on x's. Alternatively, one can regress 'y' on a few z's but there is no reason to believe that the first principal component having maximum variance would have the highest correlation with 'y' or that the order of principal components affects its correlation with 'y'. Another problem is that the principal components often do not have any economic logic. For instance, 2 (income) + 3 (price) makes no economic sense. Also, the units of measurement of x's affect the z's. Standardization of variables can solve this problem.

In many cases which involve the construction of indices, the PCA makes economic sense. The use of PCA can also give some prior information about restrictions on the parameters. Individual regression coefficients are sometimes not estimable but their linear combinations are. Having prior knowledge about parameter restrictions can help us in estimation of individual parameters.

9. Generalised Inverse Solution

There is a generalised inverse for every matrix **A** irrespective of its dimension. If **A** is a square matrix and it is invertible, that is, the determinant is non-zero, then $\mathbf{A}^{-1}$ can be easily found. But if **A** is a non-square matrix, say of order (m × n), or it is singular (determinant is zero), then the generalised inverse of **A** is indicated by **G** (n × m matrix) such that **AGA = A**. The generalised inverse of **A** is denoted by $\boldsymbol{A'}$.

$$\mathbf{AGA} = \mathbf{A}\,\boldsymbol{A'}\mathbf{A} = \mathbf{A} \quad ... (10.47)$$

There are various generalised inverses G of an (m X n) matrix **A**. In the case of an invertible square matrix **A,** the generalised inverse G is the same as $\mathbf{A}^{-1}$. Hence $\mathbf{AA^{-1}A = A}$. For a non-singular matrix, the generalised inverse is unique.

For an (m X n) matrix having rank r(A) = r, if the matrix **A** can be partitioned such that $\mathbf{A} = \begin{pmatrix} \boldsymbol{C} & \boldsymbol{D} \\ \boldsymbol{E} & \boldsymbol{F} \end{pmatrix}$, where **C** is ($r$ X r) non-singular matrix and r(A) = r(C) = r, then the generalised inverse of **A** is:

$$\mathbf{G} = \begin{pmatrix} \boldsymbol{C^{-1}} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \quad ... (10.48)$$

The procedure to compute **G** is as follows: First, take any (r X r) non-singular sub-matrix, that is, **C**. It is not necessary to take elements in **C** from adjacent rows and columns in A. Now, find $(C^{-1})'$ and replace C by $(C^{-1})'$. Remaining elements of **A** should be replaced by 0. Finally, transpose the resulting matrix.

Example 10.1

Let $\mathbf{A} = \begin{pmatrix} 4 & 1 & 2 \\ 1 & 1 & 5 \\ 3 & 1 & 3 \end{pmatrix}$

You can find out that the determinant of the above matrix vanishes. Let us take (**A**) = 2. Accordingly, $\mathbf{C} = \begin{pmatrix} 4 & 1 \\ 1 & 1 \end{pmatrix}$.

Hence $\mathbf{C}^{-1} = \begin{pmatrix} 1/3 & -1/3 \\ -1/3 & 4/3 \end{pmatrix}$.

Thus, the generalised inverse, $\mathbf{G} = \begin{pmatrix} 1/3 & -1/3 & 0 \\ -1/3 & 4/3 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

Using the generalised inverse, the regression parameter can be estimated as $\widehat{\boldsymbol{\beta}}_{\boldsymbol{G}} = (\boldsymbol{X}'\boldsymbol{X})^{-}\boldsymbol{X}'\boldsymbol{y}$ for the model. Different solutions are obtained in parameter estimation by using different generalised inverses. Hence, a class of generalised inverse regression estimators can be obtained.

Check Your Progress 3

1) Explain the following methods:

   a) Ridge Regression

   b) Principal Component Regression

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2) A researcher estimates the following model for stock market returns, but thinks that there may be a problem with it. By calculating the *t*-ratios, and considering their significance (p-values given in brackets underneath) and by examining the value of R^2 or otherwise, suggest what the problem might be.

$y_t = 0.638 + 0.402\ x_{2t} - 0.891\ x_{3t}$ $\quad (R^2 = 0.96, \bar{R}^2 = 0.89)$

(0.436) (0.291) (0.763)

Figures in parentheses are the standard errors of the coefficients. Explain how you will proceed to solve the problem.

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3) "The relevant question to ask (if there is high multicollinearity) is not what variables to drop but what other information will help." True or False? Substantiate your answer.

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10.6 LET US SUM UP

Interpretation of individual regression coefficients becomes difficult due to high inter-correlation between explanatory variables. This is known as the multicollinearity problem. However, it must be noted that high inter-correlation between independent variables does not necessarily cause problems in parameter estimation and inference. Even with high inter-correlation, standard errors can be low if error variance is low and variation in independent variables is high. Measuring multicollinearity simply on the basis of inter-correlations is not advisable as these correlations can change by suitably transforming the independent variables. This, however, does not imply that the multicollinearity problem has been resolved. What is important is how precisely the coefficients or a linear combination of them are estimable. Solutions to multicollinearity such as ridge regression are ad hoc measures requiring a priori information. Essentially, the problem arises due to paucity of information. It is advisable to get more data (if possible) and to examine the prior information available.

10.7 KEY WORDS

Multicollinearity : It is basically, high inter-correlation between explanatory variables in a regression model. If the correlation between explanatory variables is found to be very high, it causes problems in parameter estimation by inflating the standard errors and making t-ratios statistically insignificant.

Tolerance (TOL) : The inverse of VIF is known as Tolerance (TOL) and it is another indicator of collinearity between variables.

Variance Inflation Factor (VIF) : It is an indicator of multicollinearity. In the regression model $y_i = \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{\mu}_i$, the VIF is given by $VIF = \frac{1}{(1-r_{12}{}^2)}$.

10.8 SOME USEFUL BOOKS/ REFERENCES

Baltagi, Badi H. (2008), '*Econometrics*,' Fourth Edition, Springer-Verlag Berlin, Heidelberg, ISBN 978-3-540-76515-8

Brooks, Chris (2014), '*Introductory Econometrics for Finance*,' Third Edition, Cambridge University Press

Gujarati, D N and D C Porter (2009), *Basic Econometrics*, Fifth Edition, McGraw Hill

Maddala, G. S. and K. Lahiri (2009), *Introduction to Econometrics*, Fourth Edition, John Wiley and Sons

10.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) False. Refer to Section 10.3

2) $(\beta_2 + \beta_4), (\beta_3 + \beta_4), \beta_5, \beta_6$. Refer to Section 10.2

Check Your Progress 2

1) (a) True. Refer to Sub-Section 10.4.1, points 2 and 3.

   (b) False. Refer to Sub-Section 10.4.1, points 2 and 3.

Check Your Progress 3

1) (a) Refer to Section 10.5, point 7.

   (b) Refer to Section 10.5, point 8.

2) R^2 is high but the t-ratios are insignificant which is suggestive of multicollinearity.

3) True. Refer to Section 10.5, point 1 and 6. Some hints can also be obtained from Section 10.6.

UNIT 11 HETEROSCEDASTICITY *

Structure

11.0 OBJECTIVES

After going through this unit, you should be in a position to

- explain the concept of heteroscedasticity in regression analysis;
- explain the consequences of heteroscedasticity;
- apply various methods used to detect heteroscedasticity; and
- provide solutions to the problem of heteroscedasticity.

11.1 INTRODUCTION

In the classical regression model that we discussed in Block 1 we assumed that the error terms are identically independently distributed (IID) with mean 0 and variance σ^2. An implication of the above is that the error variance (σ^2) remains constant for all observations. This assumption of constant variance, however, does not remain valid always. It may so happen that the errors are mutually uncorrelated (that is, no serial correlation) but have different variances. Thus, when the variance of the error term increases or decreases with the dependent variable, we have the case of heteroscedasticity. In other words, we say that the problem of heteroscedasticity arises when the assumption of homoscedasticity – that the variances of the stochastic disturbance term are finite and constant over the sample – is not met.

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Heteroscedasticity usually occurs in cross-section data. For example, consider the savings behaviour of households. We know that savings is relatively higher among households with higher income. It is also likely that the variance of savings among high-income families is larger than the variance among low-income families. Low-income families do not have much to save, so they cannot differ appreciably in their levels of savings. High-income families, by contrast, exhibit wide ranges, from the miser who saves virtually all his income to the spendthrift who saves virtually nothing. Secondly, there is more freedom of choice for high-income families while low-income families have to spend a major part of their income on consumption goods. Clearly, a cross-section sample of data on savings and income behaviour to estimate a regression model may not satisfy the homoscedasticity assumption.

There could be several other reasons for the presence of heteroscedasticity: There may be certain outliers in the data which would increase or decrease error variance. Further, heteroscedasticity often arises because the scale of a variable varies enormously within the sample. For example, in India the large states have relatively large state domestic product (SDP) compared small states. If we take SDP as a regressor in a model, then the variability is likely to increase enormously as we move from small to large states. However, if we take per capita SDP then the variation would not be so prominent. While Maharashtra's SDP is 175 times that of Manipur, Maharashtra's per capita SDP is about 2.5 times that of Manipur. The point we try to make is that heteroscedasticity in many cases could be due to incorrect data transformation. Thus, variables should be appropriately transformed before running a regression.

Heteroscedasticity, or unequal error variance, is more likely to be present in cross section data than in time series data. In the case of time series data, the same entity or unit is surveyed over a period of time which reduces the chances of variability in errors.

11.2 CONCEPT OF HETEROSCEDASTICITY

As pointed out above, error terms are said to be homoscedastic if the variance of the error term u_i conditional upon x_i is equal to σ^2.

$$E({u_i}^2) = Var(u_i) = \sigma^2 \qquad \text{… (11.1)}$$

(i = 1, 2, …., n)

In Fig.11.1 below, we show the homoscedasticity condition through a three-dimensional diagram. We observe that the conditional variance of the dependent variable the same for all the values of x_i (that is, x_1, x_2 and x_3).

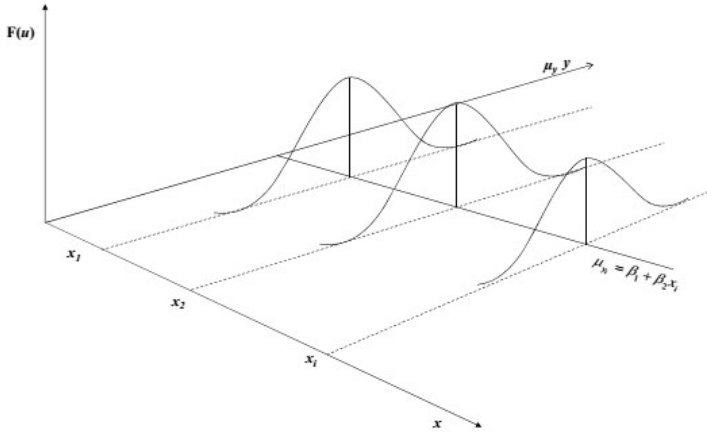

Fig. 11.1: Homoscedasticity

In case of heteroscedasticity, the variance of $u_i$conditional upon x_i is as follows:

$$E(u_i^2) = Var(u_i) = \sigma_i^2 \qquad \dots (11.2)$$

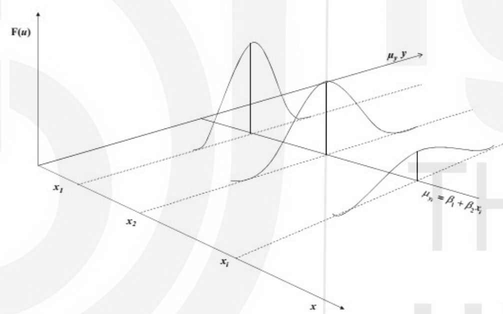

Fig. 11.2: Heteroscedasticity

11.3 CONSEQUENCES OF HETEROSCEDASTICITY

Heteroscedasticity has two important implications for estimation. First, the OLS estimators, while still being linear and unbiased, are no longer efficient. They no longer provide minimum variance estimators among the class of linear unbiased estimators (that is, they are not BLUE). Second, estimated variance of the OLS estimator is biased; so, the usual tests of statistical significance, such as *t* and *F* tests are no longer valid.

We prove these propositions below.

1. Unbiasedness of $\hat{\beta}$

Assume $y_i = \beta x_i + u_i$

Now, $\hat{\beta} = \frac{\sum x_i y_i}{\sum x_i^2} = \beta + \frac{\sum x_i u_i}{\sum x_i^2}$ $\qquad \dots (11.3)$

Here, $E(u_i) = 0$

u_i are independent of x_i.

Thus, $E(\frac{\sum x_i u_i}{\sum x_i^2}) = 0$

Hence, $E(\hat{\beta}) = \beta$

That is, $\hat{\beta}$ is unbiased.

2. Efficiency of $\hat{\beta}$

To prove that the OLS estimator $\hat{\beta}$ is inefficient in the presence of heteroscedasticity, we show that the $Var(\beta^*)$ is less than $Var(\hat{\beta})$. β^* is the Weighted Least Square (WLS) estimator of β.

Now, $Var(\hat{\beta}) = E(\hat{\beta} - \beta)^2 = E(\frac{\sum x_i u_i}{\sum x_i^2})^2 = \frac{\sum x_i^2 \sigma_i^2}{(\sum x_i^2)^2}$ … (11.4)

To find the WLS estimator of β, assume that, $\sigma_i^2 = \sigma^2 z_i^2$ and divide

$y_i = \beta x_i + u_i$ by z_i (here, z_i are known),

we get,

$$\frac{y_i}{z_i} = \beta \frac{x_i}{z_i} + \frac{u_i}{z_i} \quad \text{… (11.5)}$$

Let $\frac{u_i}{z_i} = v_i$

Now, the error term v_i has a constant variance σ^2. This method is called the Weighted Least Squares (WLS) wherein the i^th term is weighed by $\frac{1}{z_i}$.

The WLS estimator $\beta^* = \frac{\sum\left(\frac{x_i}{z_i}\right)(y_i/z_i)}{\sum(\frac{x_i}{z_i})^2} = \beta + \frac{\sum\left(\frac{x_i}{z_i}\right)v_i}{\sum(\frac{x_i}{z_i})^2}$ … (11.6)

Since the expected value of the last term is zero, $E(\beta^*) = \beta$. Thus, the WLS estimator β^* is unbiased.

$$Var(\beta^*) = \frac{\sigma^2}{\sum\left(\frac{x_i}{z_i}\right)^2} \quad \text{…(11.7)}$$

and $Var(\hat{\beta}) = \frac{\sum x_i^2 \sigma_i^2}{(\sum x_i^2)^2}$

Substituting $\sigma_i^2 = \sigma^2 z_i^2$, we get,

$$Var(\hat{\beta}_{HET}) = \frac{\sigma^2 \sum x_i^2 z_i^2}{(\sum x_i^2)^2} \quad \text{… (11.8)}$$

Thus, $\frac{Var(\beta^*)}{Var(\hat{\beta})} = \frac{\sigma^2/\sum(\frac{x_i}{z_i})^2}{\sigma^2 \sum x_i^2 z_i^2/(\sum x_i^2)^2}$

On further elaboration,

$$\frac{Var(\beta^*)}{Var(\hat{\beta})} = \frac{\left(\sum x_i^2\right)^2}{\sum\left(\frac{x_i^2}{z_i^2}\right)\sum x_i^2 z_i^2} = \frac{\left(\sum\left(x_i z_i \frac{x_i}{z_i}\right)\right)^2}{\sum(x_i z_i)^2 \sum\left(\frac{x_i}{z_i}\right)^2} = \frac{\left(\sum a_i b_i\right)^2}{\sum a_i^2 \sum b_i^2} \qquad \ldots (11.9)$$

where, $a_i = x_i z_i$ and $b_i = \frac{x_i}{z_i}$

The above expression is less than 1 implying that $Var(\beta^*)$ is less than $Var(\hat{\beta})$. Thus, the OLS estimator is inefficient in the presence of heteroscedasticity. The expression at (11.9) equals 1only if a_i and b_i are proportional to each other or ifz_i^2 is a constant in which case $\sigma_i^2 = \sigma^2 z_i^2$ shows that the error variances are equal.

Using $Var(\hat{\beta})$ gives wider confidence intervals and standard errors are high. Hence, the *t*-ratios are smaller thus rendering a coefficient statistically insignificant even though it may found to be significant using WLS.

3. Var ($\hat{\beta}$) is biased

We cannot ignore the problem of heteroscedasticity because in the presence of heteroscedasticity, the variance of $\hat{\beta}$ is biased. Whether the bias is positive or negative depends on the relationship between σ_i^2 and x_i^2.

In the presence of heteroscedasticity,

$$Var(\hat{\beta}) = \frac{\sum x_i^2 \sigma_i^2}{\left(\sum x_i^2\right)^2}$$

We know from Unit 4 that, in the case of one explanatory variable,

$$Var(\hat{\beta}) = \frac{RSS}{n-1}\left(\frac{1}{\sum x_i^2}\right)$$

where RSS = Residual Sum of Squares

$$E(RSS) = E\left[\sum(y_i - \hat{y}_i)^2\right] = E\left[\sum(y_i - \hat{\beta}x_i)^2\right] = E\left[\sum\left(\beta x_i + u_i - \hat{\beta}x_i\right)^2\right]$$

Or, $E(RSS) = \sum \sigma_i^2 - \frac{\sum x_i^2 \sigma_i^2}{\sum x_i^2}$ … (11.10)

Hence, $E(Var(\hat{\beta})) = \frac{1}{(n-1)\sum x_i^2}\left[\sum \sigma_i^2 - \frac{\sum x_i^2 \sigma_i^2}{\sum x_i^2}\right]$

and, $Var(\hat{\beta}_{HET}) = \frac{\sum x_i^2 \sigma_i^2}{\left(\sum x_i^2\right)^2}$

Thus, $E(Var(\hat{\beta})) \neq Var(\hat{\beta}_{HET})$ … (11.11)

Therefore, the variance of $\hat{\beta}$ is biased. Since the standard errors of least square estimators are found to be biased, it invalidates the inferences drawn on the basis of the tests of significance.

Note that if $\sigma_i^2 = \sigma^2$ for all 'i', then $E(RSS) = (n-1)\sigma^2$. In that case, $E\left(Var(\hat{\beta})\right) = \frac{\sigma^2}{\sum x_i^2}$. Therefore, $Var(\hat{\beta}_{HET}) = \frac{\sigma^2}{\sum x_i^2}$. Thus, the bias disappears with homoscedastic error variances.

Check Your Progress 1

1) Consider the following equation where consumption of good X depends on the following: variables: $X = \beta_0 + \beta_1 income + \beta_2 price + \beta_3 male + \beta_4 education + u$

   $E(u \vdots income, price, male, education) = 0$

   $Var(u \vdots income, price, male, education) = \sigma^2 income^2$

   Transform this equation such that the error term is homoscedastic.

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2) "Heteroscedastic errors cause biased regression coefficients and biased standard errors". State whether True or False? Justify your answer.

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11.4 DETECTION OF HETEROSCEDASTICITY

In the case of heteroscedasticity, the OLS estimators are unbiased and consistent, but they are no longer efficient. This lack of efficiency makes the usual hypothesis testing procedure faulty and calls for remedial measures. Before we make any corrections for heteroscedasticity, we should first try to detect the presence of heteroscedasticity in the error terms. Two common methods are used to detect the presence of heteroscedasticity.

(i) We can plot the residuals against the predicted values of the dependent variable and examine the patterns of the residuals. If the variance of the residuals is not constant but increases or decreases with the independent variable, then there is a possibility of heteroscedasticity in the data set.

(ii) A more careful method is to break the sample into two or more sub-groups, on the basis of one of the independent variables (say, X), and then compute the error variance of each group. Our null hypothesis is that there is no difference among the variances of these groups. To test this hypothesis, we can use the chi-square test assuming that the error terms are normally and independently distributed. A problem with this method, however, is that the breakdown of these groups is rather arbitrary in terms of different ranges of X. Therefore, the test is viewed to be merely an indication of the presence of heteroscedasticity.

In practice, our data obtained from a sample survey, gives a single value of$y$corresponding to a value of X. Getting the entire range of Y population for chosen value of x is difficult. Hence, knowing ${\sigma_i}^2$ is also difficult. There are, however, other ways of detecting the presence of heteroscedasticity in the data.

1. Residual Plot

By plotting the estimated residuals$\hat{u}_i$ againstx_i, we can analyse whether there is a systematic pattern in the residuals which suggests heteroscedasticity in the errors. If the heteroscedastic error variance is found to be proportional to the explanatory variable (for example), it gives some prior knowledge which can be used to transform the data and correct for heteroscedasticity. Alternatively, ${\hat{u}_i}^2$ can be plotted against $\hat{y}_i$ and the residual pattern can be analysed. For instance, if a linear or quadratic pattern is observed, it indicates heteroscedasticity in the errors.

2. Auxiliary Regression

We can test for heteroscedasticity by (i) regressing $\hat{u}_i$ on ${x_i}^2$, ${x_i}^3$,... oron x_i, ${x_i}^2$, ${x_i}^3$...or $\hat{u}_i$ on ${\hat{y}_i}^2$, ${\hat{y}_i}^3$...., and (ii) testing if the regression coefficients obtained are statistically significant.

3. Park Test

In heteroscedasticity, the variance is different across the observations.

Let ${\sigma_i}^2 = \sigma^2 X_i^{\beta} e^{v_i}$ … (11.12)

By taking 'natural logarithm', we obtain, $ln\, {\sigma_i}^2 = ln\sigma^2 + \beta X_i + v_i$

If we take${\hat{u}_i}^2$as a proxy for${\sigma_i}^2$, we obtain

$ln\, {\hat{u}_i}^2 = \alpha + \beta X_i + v_i$ … (11.13)

Now, Park Test for detecting heteroscedasticity involves the following steps:

a) Run the original regression model despite the heteroscedasticity problem.

b) From the regression, obtain $\hat{u}_i$and square them, that is, ${\hat{u}_i}^2$. Then, find out $ln\, {\hat{u}_i}^2$

c) Run the auxiliary regression equation indicated in equation (11.13) using an explanatory variable present in the original regression model. Then run the regression against each X variable. In other words, we run the regression against $\hat{Y}_i$, the estimated value of Y_i.

d) Test the null hypothesis$\beta = 0$, i.e., there is no heteroscedasticity.

e) If the null hypothesis could not be rejected, we infer that there is no heteroscedasticity. In this case the value of the intercept can be accepted as the common, homoscedastic variance σ^2.

f) If the null hypothesis is rejected, then we infer that there is heteroscedasticity problem in the data set.

4. Glejser Test

The steps followed in the Glejser Test are similar to those of the Park Test. An important difference between both the tests is that in place of equation (11.13) we take different functional forms for the auxiliary regression equation.

The steps followed are:

a) Obtain the residual $\hat{u}_i$ from the original regression model.

b) Take absolute value $|\hat{u}_i|$of the residuals.

c) Regress the absolute values of $|\hat{u}_i|$on the X variable that is expected to be closely associated with heteroscedastic variance σ_i^2.

d) You can take various functional forms of X_i. Some of the functional forms suggested by Glejser are

$$|\hat{u}_i| = \alpha + \frac{\beta}{x_i} + v_i \qquad \ldots (11.14)$$

$$|\hat{u}_i| = \alpha + \beta x_i + v_i \qquad \ldots (11.15)$$

$$|\hat{u}_i| = \alpha + \beta\sqrt{x_i} + v_i \qquad \ldots (11.16)$$

The above means that the Glejser test suggests various plausible (linear as well as non-linear) relationships between the residual term and the explanatory variable to investigate the presence of heteroscedasticity.

e) For each of the cases given, test the null hypothesis that there is no heteroscedasticity, i.e., H_0: $\beta = 0$ (no heteroscedasticity).

f) If H_0 is rejected, we conclude that there is evidence of heteroscedasticity.

You should note that the error term v_i can itself be heteroscedastic as well as serially correlated. For both the Park Test and the Glejser Test, the stochastic disturbance term v_i should satisfy the standard classical assumptions (see Unit 3).

5. Goldfeld-Quandt Test

The Goldfeld and Quandt (G-Q) Test is used if the sample size is not large. Firstl, the observations are ordered starting with the lowest value of 'X'. The observations are divided into two groups such that the first group contains smaller values of the independent variable 'X' whereas the second group contains larger values of 'x'. While segregating the two groups, some middle observations are omitted. Assuming that there are 'n' observations and 'c' observations are omitted, there remains $\frac{(n-c)}{2}$ observations in each group. Separate OLS regressions are fitted in each group and the RSS are obtained corresponding to each group.

Each group's RSS has $\left[\frac{(n-c)}{2}-k\right]$ degrees of freedom. Here, k is the number of parameters (including the intercept term) that has to be estimated. Let RSS_2 be the RSS from the group containing large values of 'x' while RSS_1 be the RSS from the group containing smaller values of 'x'.

Now, assume $\lambda = \frac{RSS_2/df}{RSS_1/df}$ … (11.17)

If the error terms are assumed to be normally distributed and they are homoscedastic, then λ follows an F-distribution with $\left[\frac{(n-c)}{2}-k\right]$ degrees of freedom in both numerator and denominator. The null hypothesis (H_0) is the presence of homoscedasticity. The H_0 is rejected if the computed value of λ exceeds the critical F-value at the significance level assumed.

If the model has more than one independent variable, then any of the $x's$ can be employed and the observations can be ranked according to it. Alternatively, the test can be conducted on each of the $x's$. There are two limitations of the G-Q Test: (i) identification of the appropriate x variable to be used for ordering the observations. (ii) identifying the number of central observations that has to be omitted. Omitting a few central observations allows us to differentiate between the smaller variance group RSS_1 and the larger variance group RSS_2.

6. Breusch-Pagan Test

Two limitations of the Goldfeld and Quandt test mentioned above can be avoided by adopting the Breuschand Pagan (B-P) Test, which is an asymptotic test.

Let us assume that the variance of the error term is a function of certain variables such that,

$$\sigma_i^2 = f(\alpha_1 + \alpha_2 z_{2i} + \alpha_3 z_{3i} + \ ... + \alpha_m z_{mi}) \qquad ... (11.18)$$

The steps to be followed are:

(i) Use the following regression model:

$$\sigma_i^2 = \alpha_1 + \alpha_2 z_{2i} + \alpha_3 z_{3i} + \ ... + \alpha_m z_{mi}$$

(iii) Test the null hypothesis that $\alpha_2 = \alpha_3 = \cdots = \alpha_m = 0$. If this null hypothesis is accepted, then σ_i^2 is homoscedastic. The function f(.) can take any form and the test is not affected by the functional form.

(iii) Let $\hat{\sigma}^2 = \sum \frac{\hat{u}_i^2}{n}$ ... (11.19)

(iv) Obtain S_0 = explained sum of squares from a regression of $\hat{u}_i^2$ on $z_2, z_3, ..., z_m$.

Then $\lambda = S_0/2\hat{\sigma}^4$ follows a χ^2 distribution with '*m*' degrees of freedom assuming normally distributed error terms and homoscedasticity.

7. White's Test

An assumption under the Breusch-Pagan Testis the normal distribution of the error terms. The Goldfeld-Quandt test, on the other hand, requires ordering of the observations. The White's test does not require the assumption that the error terms are normally distributed. Nor does it require ordering of observations. This test suggests regression of$\hat{u}_i^2$ on all the independent variables in the model as well as their squares and their cross products. In the case of three independent variables, for example, the White's test suggests regressing $\hat{u}_i^2$ on $x_1,\ x_2,\ x_3,\ x_1^2,\ x_2^2,\ x_3^2,\ x_1x_2,\ x_1x_3$ and x_2x_3 .

Let us take the case of two independent variables, x_2 and x_3. The steps to be followed are:

(i) Run the following auxiliary regression including the constant term:

$$\hat{u}_i^2 = \alpha_1 + \alpha_2 x_{2i} + \alpha_3 x_{3i} + \alpha_4 x_{2i}^2 + \alpha_5 x_{3i}^2 + \alpha_6 x_{2i} x_{3i} + v_i \quad ... (11.20)$$

(ii) Find the R^2 of the auxiliary regression equation given at equation (11.20).

(iii) For homoscedasticity, the null hypothesis is H_0: $\alpha_2 = \alpha_3 = \alpha_4 = \alpha_5 = \alpha_6 = 0$

(iv) Consider that nR^2 asymptotically follows a χ^2 distribution with 5 degrees of freedom (n is the sample size). Here, the number of regressors without including intercept term is the degrees of freedom. In notations, $nR^2 \sim \chi^2{}_{df}$.

(v) If the computed χ^2 is less than the critical value at the assumed level of significance, it implies that heteroscedasticity is not present.

(vi) If the computed χ^2 is greater than the critical value at the assumed level of significance, there is heteroscedasticity in the dataset.

One limitation of the White's Test is that using the squares and cross products of the independent variables as regressors leads to a decrease in the degrees of freedom. Further, the White's test statistic may be found to be statistically significant due to specification error and not necessarily due to heteroscedasticity. Thus, it is a test of both pure heteroscedasticity and/or specification error.

Check Your Progress 2

1) What are the various tests for heteroscedasticity?

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2) Describe how the White's Test can be carried out to test for the presence of heteroscedasticity.

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11.5 REMEDIAL MEASURES

The tests discussed above help us in identification of the presence of heteroscedasticity. There are various methods to eliminate heteroscedasticity from a dataset. Some of these methods are discussed below.

1. Weighted Least Squares

There are situations where we have some ideas about the pattern of heteroscedasticity in a dataset. For instance, suppose $\sigma_i{}^2 = \sigma^2 z_i{}^2$ and z_i are known. In this case we can transform the regression equation $y_i = \alpha + \beta x_i + u_i$. We can divide the above regression model throughout by z_i such that

$$\frac{y_i}{z_i} = \alpha \frac{1}{z_i} + \beta \frac{x_i}{z_i} + \frac{u_i}{z_i.} \qquad \dots (11.21)$$

In this case, let the error term be transformed into $\frac{u_i}{z_i.} = v_i$.

We find that the error term v_i has a constant variance σ^2.

$$Var\ (v_i) = E(\frac{u_i}{z_i.})^2 = \frac{\sigma_i{}^2}{z_i{}^2} = \sigma^2 \qquad \dots (11.22)$$

This method is called the Weighted Least Squares (WLS) method wherein the ith term is weighed by $\frac{1}{z_i}$.Here, the rule is to regress $\frac{y_i}{z_i}$ on $\frac{1}{z_i}$ and $\frac{x_i}{z_i}$ without the constant term.

Let us take another example: $Var(u_i) = \sigma^2 x_i$ and $\alpha = 0$. The equation $y_i = \beta x_i + u_i$ can be transformed by dividing throughout by $\sqrt{x_i}$ such that,

$$\frac{y_i}{\sqrt{x_i}} = \beta \frac{x_i}{\sqrt{x_i}} + \frac{u_i}{\sqrt{x_i}} \qquad \dots (11.23)$$

In this case, the WLS estimator $\beta^* = \frac{\sum(\sqrt{x_i})(y_i/\sqrt{x_i})}{\sum \sqrt{x_i}^2} = \frac{\bar{y}}{\bar{x}}$ … (11.24)

That is, β^* is the ratio of the means. Similarly, if $\sigma_i{}^2 = \sigma^2 x_i{}^2$, then we can transform the equation as follows:

$$\frac{y_i}{x_i} = \alpha \frac{1}{x_i} + \beta + v_i \qquad \dots (11.25)$$

Here, the constant term β is the slope coefficient of the original equation.

2. Two-Step Weighted Least Squares

If error variances are not known, for instance, $\sigma_i{}^2 = \sigma^2[E(y_i)]^2 = \sigma^2[\alpha + \beta x_i]^2$, a two-stage procedure may be employed. First, using OLS, the estimators $\hat{\alpha}$ and $\hat{\beta}$ are obtained. Second, regress $\frac{y_i}{\hat{\alpha}+\hat{\beta}x_i}$ on $\frac{1}{\hat{\alpha}+\hat{\beta}x_i}$ and $\frac{x_i}{\hat{\alpha}+\hat{\beta}x_i}$ without the constant term. The standard errors obtained for the estimates of α and β are only asymptotically valid because the weights are estimated.

3. Maximum Likelihood Method

In the above example, if the errors are normally distributed, maximise the likelihood function:

$$LogL = -nlog\sigma - \sum log\ (\alpha + \beta x_i) - \frac{1}{2\sigma^2} \sum \left(\frac{y_i - \alpha - \beta x_i}{\alpha + \beta x_i}\right)^2 \qquad \dots (11.26)$$

Such a method is efficient than the WLS method if the errors are normally distributed. Note that maximising the above likelihood function is not the same as minimizing $\sum\left(\frac{y_i-\alpha-\beta x_i}{\alpha+\beta x_i}\right)^2$ which is the method of Weighted Least Squares.

4. Generalised Least Squares (GLS)

The OLS procedure gives equal weight to each observation. Ideally, observations with higher variance ought to be given less weight and vice-versa. If ${\sigma_i}^2$ is known, then $y_i = \alpha + \beta x_i + u_i$ can be transformed by dividing throughout by σ_i such that,

$$\frac{y_i}{\sigma_i} = \frac{\alpha}{\sigma_i} + \beta\frac{x_i}{\sigma_i} + \frac{u_i}{\sigma_i}$$

Or, $\frac{y_i}{\sigma_i} = \frac{\alpha}{\sigma_i} + \beta\frac{x_i}{\sigma_i} + v_i$ … (11.27)

where, $v_i = \frac{u_i}{\sigma_i}$.

$$Var\,(v_i) = Var\left(\frac{u_i}{\sigma_i}\right) = \frac{E({u_i}^2)}{{\sigma_i}^2} = \frac{{\sigma_i}^2}{{\sigma_i}^2} = 1 \quad \dots (11.28)$$

Thus, the transformed model has homoscedastic error variance equal to unity. This transformed model satisfies the usual classical assumptions. Applying OLS to this transformed model is known as the Generalised Least Squares (GLS) method and it produces estimators which are best linear unbiased estimator (BLUE). Effectively, in GLS we minimise the weighted residual sum of squares (RSS) given by $\sum w_i\,{\hat{u}_i}^2$ where the weight $w_i = \frac{1}{{\sigma_i}^2}$. This shows that the observations with larger variance get lesser weight and vice-versa. Essentially, this is the Weighted Least Squares procedure which is a special case of the GLS procedure.

The GLS procedure in matrix form is discussed below. Let us take the regression model in matrix form

$$\boldsymbol{y} = \boldsymbol{X\beta} + \boldsymbol{u}$$

Let us assume that there is no autocorrelation or heteroscedasticity.

In that case,

$$\boldsymbol{E(uu') = \sigma^2 I.}$$

Now let is relax these assumptions, and assume that

$E(uu') = \varphi$

where φ is an arbitrary positive definite matrix. Pre-multiplying $y = X\beta + u$ by $\varphi^{-1/2}$, we get,

$$y^* = X^*\beta + u^* \quad \dots (11.29)$$

Here, $y^* = y\varphi^{-\frac{1}{2}}$

$X^* = X\varphi^{-\frac{1}{2}}$

$u^* = u\varphi^{-\frac{1}{2}}$

Thus, $E(u^*u^{*'}) = u\varphi^{-\frac{1}{2}}u'\varphi^{-\frac{1}{2}} = I$ … (11.30)

Hence, with this method, the error variances become homoscedastic. Such a method is called Generalised Least Squares (GLS). The GLS estimator β_{GLS} is given by:

$$\beta_{GLS} = (X^{*\prime}X^*)^{-1}(X^{*\prime}y^*) = (X'\varphi^{-\frac{1}{2}}X\varphi^{-\frac{1}{2}})^{-1}(X'\varphi^{-\frac{1}{2}}y\varphi^{-\frac{1}{2}})$$

$$= (X'\varphi^{-1}X)^{-1}(X'\varphi^{-1}y) \quad \dots (11.31)$$

β_{GLS} is the best linear unbiased estimator of β.

The variance of β_{GLS} is given by $Var(\beta_{GLS}) = (X^{*\prime}X^*)^{-1}$

$$= (X'\varphi^{-\frac{1}{2}}X\varphi^{-\frac{1}{2}})^{-1} = (X'\varphi^{-1}X)^{-1} \quad \dots (11.32)$$

φ is the variance-covariance matrix of the error term u. In the presence of autocorrelation and heteroscedasticity, there are elements in this matrix both on the main diagonal and off it. With heteroscedasticity, the main diagonal elements will reflect unequal variances of the error term. But the structure of this matrix is not known in practice. Hence, estimated GLS procedure (EGLS) is used wherein OLS is employed ignoring the problems of autocorrelation and heteroscedasticity. The residuals $\hat{u}$ are estimated and they are used to form the estimated variance-covariance matrix of the error term. This matrix is then employed to get the estimated GLS (EGLS) estimator of β. EGLS estimators are consistent estimators of GLS.

5. White's Estimator

If the error variances σ_i^2 are not known, the White's Heteroscedasticity corrected standard errors or robust standard errors can be used. In the regression model, since

$y_i = \beta x_i + u_i$, the variance of $\hat{\beta}$ is given by $Var(\hat{\beta}) = \frac{\sum x_i^2 \sigma_i^2}{(\sum x_i^2)^2}$.

Since σ_i^2 are not known, White suggests using $\hat{u}_i^2$ and estimate,

$$Var(\hat{\beta}) = \frac{\sum x_i^2 \hat{u}_i^2}{(\sum x_i^2)^2} \qquad \text{... (11.33)}$$

This variance is consistent, that is, it converges to $\frac{\sum x_i^2 \sigma_i^2}{(\sum x_i^2)^2}$ with increase in sample size. Hence, this is a large sample procedure. Empirically, White' robust standard errors can be found to exceed or fall short of the OLS standard errors.

White' robust standard errors can be estimated for k-variable regression model as well. Consider, $Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \ldots . + \beta_k X_{ki} + u_i$. In this case, the variance of $\hat{\beta}_j$ is given by

$$Var(\hat{\beta}_j) = \frac{\sum \hat{w}_{ji}^2 \hat{u}_i^2}{(\sum \hat{w}_{ji}^2)^2} \qquad \text{... (11.34)}$$

Here, $\hat{w}_j$ is the residual from regressing X_j on remaining regressors (auxillary regression). One limitation of the White's procedure is that the estimators obtained through it may not be as efficient as those obtained by variable transformation methods employed in WLS procedure illustrated above.

6. Logarithmic Transformation

Sometimes, logarithmic transformation of variables can be done to solve the problem of heteroscedasticity. Economic reasoning supports the use of log linear functional form in some cases. For instance, in case of Cobb- Douglas production function $Y = AK^{\alpha}L^{\beta}$, the log linear form given by $\log Y = logA + \alpha \log K + \beta \log L$is more suitable as it helps estimate the elasticity of output with respect to various inputs. In another example, price and income elasticity can be estimated from logarithmic transformation such as$\log Q = \alpha + \beta_1 \log P + \beta_2 \log Y$ [Q = output, P = price, Y = income].

7. Using Deflators

The variables can be deflated to solve the problem of heteroscedasticity. Sometimes, economic logic supports this procedure, for instance, in problems involving relative prices or real incomes. In other cases, deflation of variables is simply done to solve the problem of heteroscedasticity.

Assume that $Y = \alpha Z + \beta X + u$. Also, it is given that $\sigma_i^2 = \sigma^2 Z^2$.

Dividing $Y = \alpha Z + \beta X + u$ by Z, we get,

$$\frac{Y}{Z} = \alpha + \frac{\beta X}{Z} + \frac{u}{Z}$$

Or, $\frac{Y}{Z} = \alpha + \frac{\beta X}{Z} + u'$ ... (11.35)

Here $u' = \frac{u}{Z}$ and $Var\ (u') = \sigma^2$

The deflated variables equation $\frac{Y}{Z} = \alpha + \frac{\beta X}{Z} + u'$ should be used to obtain the estimates of α, β and σ^2 and not the original equation. Once the estimators are obtained, the inferences such as residual calculation, future values prediction, etc. should be made on the basis of the following rule. When deflation of variables is supported by economic logic, the deflated variables equation should be used to draw inferences such as residuals calculation, future values prediction, etc. However, the original equation should be used to draw inferences if deflation of variables is simply done to solve the problem of heteroscedasticity.

In case there are many deflated variables equation, one may be tempted to compare different procedures based on the standard errors of coefficients. After all, the purpose of deflation is to arrive at more efficient estimates. But, such a comparison should be avoided because with heteroscedasticity the standard errors of the coefficients are biased. One should be very careful in making such comparisons. Residuals examination will prove more useful.

Such a procedure of deflation of variables may enhance or reduce the correlation between variables but drawing of inferences should not be based on correlations. For instance, in the above example, the correlations between $\frac{Y}{Z}$ and $\frac{X}{Z}$ and that between Y and X should not be compared to decide which equation, the original one or the transformed one is better for making inferences such as residuals calculation, future values prediction, etc. The presence of a common denominator Z may increase the correlation between $\frac{Y}{Z}$ and $\frac{X}{Z}$, but such a correlation is 'spurious' if there is no close relationship between X and Y, a priori. However, the problem of spurious correlation does not exist if the original hypothesis is formulated in ratio terms, for example, relative prices or real output.

Check Your Progress 3

1) In the model, $y_i = \alpha + \beta x_i + u_i$, the variance of the error terms depends on a variable z_i. Which of the following three specifications is best suited?

   $Var\ (u_i) = \sigma^2$

   $Var\ (u_i) = \sigma^2 x_i$

   $Var\ (u_i) = \sigma^2 {x_i}^2$

   ....

   ....

   ....

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11.6 LET US SUM UP

Heteroscedasticity occurs when the variance of the error terms is unequal across observations. In the presence of heteroscedasticity, the OLS estimators are unbiased and consistent but they are no longer efficient. The estimated variances are also biased with heteroscedasticity. Thus, the usual tests of significance are not valid.

Various tests can be applied to detect the presence of heteroscedasticity. The White's test, Goldfeld-Quandt test and the Breush-Pagan test are widely used. Having detected heteroscedasticity, corrective measures such as Weighted Least Squares (WLS) method can be applied. But, for this, the pattern of heteroscedasticity must be known so that the original equation can be suitably transformed. The White's Heteroscedasticity Corrected Standard Errors or the robust standard errors are also reported by econometric software; but these can be obtained only for large samples. Sometimes, simple logarithmic transformation of variables can also solve the problem.

11.7 KEY WORDS

Homoscedasticity : The error terms are said to be homoscedastic if the variance of the error term u_i conditional upon x_i (x_i is the independent variable) is equal to σ^2. $E(u_i{}^2) = Var(u_i) = \sigma^2(i = 1, 2, ..., n)$.

Heteroscedasticity : Error terms have unequal variances. The variance of u_i conditional upon the given x_i is $E(u_i{}^2) = Var(u_i) = \sigma_i{}^2$.

Weighted Least Squares : Here, we minimise the weighted Residual Sum of Squares (RSS). Observations with higher variance get less weight and vice-versa.

11.8 SOME USEFUL BOOKS

Brooks, Chris (2014), '*Introductory Econometrics for Finance*,' Third Edition, Cambridge University Press

Gujarati, Damodar N. and Sangeetha (2007), '*Basic Econometrics*,' Fourth Edition, Tata McGraw Publishing Company Limited, New Delhi, Edition 2007

Maddala, G. S. and Kajal Lahiri (2012), '*Introduction to Econometrics*,' Fourth Edition, John Wiley and Sons Ltd

Verbeek, Marno (2017), 'A *Guide to Modern Econometrics*,' Fifth Edition, John Wiley and Sons Ltd.

Wooldrige, Jeffrey (2022), *Introductory Econometrics*, Seventh Edition, Cengage Learning

11.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) Hint: Use Weighted Least Squares technique.

2) False. Regression coefficients are unbiased but their standard errors are biased. Refer to Section 11.3, Points 1 and 3.

Check Your Progress 2

1) There are a number of tests to test for the presence of heteroscedasticity. Refer to Section 11.4 and answer.

2) The steps are given in detail in Section 11.4. Go through it.

Check Your Progress 3

1) The first specification is best suited because the error terms are homoscedastic in this case. If there is a choice among the remaining two specifications, proceed as follows. First estimate the residuals $\hat{u}_i$. Plot of $\hat{u}_i$ against x_i will give some indication of heteroscedasticity. Apply Glejser's test and regress $\hat{u}_i$ on x_i as follows.

$$|\hat{u}_i| = \alpha + \beta_1 x_i + v_i$$
$$|\hat{u}_i| = \alpha + \beta_2 \sqrt{x_i} + v_i$$

If $\beta's$ turn out to be insignificant, then there is no heteroscedasticity. If β_1 is significant, then use $Var\,(u_i) = \sigma^2 {x_i}^2$. If β_2 is significant, then use $Var\,(u_i) = \sigma^2 x_i$. If both β_1 and β_2 are significant, choose the specification, with high R^2.

UNIT 12 ERRORS IN VARIABLES*

Structure

12.0 OBJECTIVES

After going through this Unit you should be in a position to

- explain the concept of errors in variables;
- indentify the consequences of errors in variables;
- explain the concept of instrumental variables;
- test for the presence of measurement errors; and
- take remedial measures to solve the problem of measurement error.

12.1 INTRODUCTION

In ordinary least squares model we assume that sample observations are measurement accurately. All our formulae are based upon the presumption that variables (both explained and explanatory) are measured without error. The only form of error admitted to our model is in the form of disturbance term. Here the error term represents the influence of various explanatory variables that have not accurately been included in the model. However, the assumption may not be realistic particularly in the case of secondary data.

Variables, both dependent and independent, are measured subject to error. In particular, the available data may not refer to the variable as specified, as specified, as in the case of proxy variable, or there may be systematic biases in

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the collection or publication of data. If the measurement errors are systematic, in general, auxiliary equations can be specified to capture these errors. This unit will focus only on the impact of random measurement errors on the regression model.

12.2 CONSEQUENCES OF ERRORS IN VARIABLES

Most of the published data or summary information contain errors of summarizing or misrepresenting by informer. When these data are used, one of the assumptions of classical lease-squares method is violated. In this case, the classical least-squares estimators will be biased even when sample size increases, which is alternatively known as asymptotic biases or inconsistency.

Under the classical assumptions the ordinary least squares (OLS) estimators are best linear unbiased. One of the major underlying assumptions is the interdependence of repressors from the disturbance term. If this condition does not bold, OLS estimators are biased and inconsistent. This statement may be illustrated by simple errors in variables.

We discuss about the consequences in the following, if the error appears in the measurement of dependent variable, independent variable or both.

12.2.1 Measurement Error in Y

Let us assume that only the dependent variable contains error of measurement. Assume that the true regression model (written in deviation form) is

$$y_i = \beta x_i + \varepsilon_r \qquad \dots (12.1)$$

where ε_i represents errors associated with the specification of the model (the effects of omitted variables, etc.).

Assume that the variable y^*, instead of y, is obtained in the measurement process such that

$$y_i^* = y_i + u_i, \qquad \dots (12.2)$$

The measurement error u_i is not associated with the regressor. Thus we have

$$Cov(u_i, x_i) = 0$$

The regression model is estimated with y^* as the dependent variable, with non account being taken of the fact that y^* is not an accurate measure of y. therefore, instead 0 estimating Eq. (12.1), we estimate

$$y^* = y_i + u_i = (\beta x_i + \varepsilon_i) + u_i$$

or, $$y^* = \beta x_i + (\varepsilon_i + u_i)$$

or, $$y^* = \beta x_i + e_i \qquad \dots (12.3)$$

where e_i is a composite error term, containing the population disturbance term ε_i (which may be called the error term in the equation) and the measurement error term u_i.

For simplicity let us assume that $E(\varepsilon_i) = E(u_i) = 0$, $Cov(x_i, \varepsilon_i) = 0$ (which is the assumption of classical linear regression), and $Cov(x_i, u_i) = 0$, i.e., the errors of measurement in y^* are uncorrelated with x_i, and $Cov(\varepsilon_i, u_i) = 0$, the equation error and measurement error are uncorrelated.

With these assumptions, it can be shown that β estimated from either (12.1) or (12.3) will be an unbiased estimator of the true β. Thus, the errors of measurement in dependent variable y^*, do not destroy the unbiased property of the OLS estimators.

However, the variances and standard errors of β estimated from (12.1) and (12.3) will be different because, employing the usual formula, we obtain

For (12.1), $$Var(\hat{\beta}) = \frac{\sigma_\varepsilon^2}{\sum x_i^2} \qquad \ldots (12.4)$$

For (12.3), $$Var(\hat{\beta}) = \frac{\sigma_\varepsilon^2}{\sum x_i^2} = \frac{\sigma_\varepsilon^2 + \sigma_u^2}{\sum x_i^2} = \frac{\sigma_\varepsilon^2}{\sum x_i^2} + \frac{\sigma_u^2}{\sum x_i^2} \qquad \ldots (12.5)$$

Obviously, the variance given at (12.5) is larger than the variance given at (12.4). Therefore, although the errors of measurement in the dependent variable still give unbiased estimates of the parameters and their variables, the estimated variances are now larger than the case where there are no such errors of measurement.

12.2.2 Measurement Error in X

We have taken the true regression model deviation form to be

$$y_i = \beta x_i + \varepsilon_i \qquad \ldots (12.1)$$

Now let us assume that explanatory variable x_i is measured with error and the observed value becomes x_i^* such that

$$x_i^* = x_i + \nu_i$$

or, $$x_i = x_i^* - \nu_i$$

Substituting the value of x_i in (12.1), we have

$$\begin{aligned} y_i &= \beta(x_i^* - \nu_i) + \varepsilon_i \\ &= \beta x_i^* - \beta \nu_i + \varepsilon_i \\ &= \beta x_i^* + (\varepsilon_i - \beta \nu_i) \\ &= \beta x_i^* + (w_i) \end{aligned}$$

where $w_i = \varepsilon_i - \beta \nu_i$, We have assumed earlier that the measurement error in x is normally distributed with zero mean, it has no serial correlation, and it is uncorrelated with ε_i. However, we can no longer assume that the composite error term w_i is independent of the explanatory variable x_i^*.

$$Cov(w_i, x_i^*) = E[w_i - E(w_i)]\,[x_i^* - E(x_i^*)]$$

$$= E[w_i - 0]\left[\left(x_i^* - x_i\right)\right]$$

$$= E[\varepsilon_i - \beta\nu_i]\ \nu_i$$

$$= E\left[(\varepsilon_i - \nu_i) - \beta\nu_i^2\right]$$

$$= 0 - \beta\sigma_v^2$$

$$= -\beta\sigma_v^2 \qquad \ldots (12.6)$$

The composite error term w_i has mean zero as
$E(w_i) = E[\varepsilon_i - \beta\nu_i] = E(\varepsilon_i) - \beta E(\nu_i) = 0$.

Thus, the explanatory variable and the error term are correlated, which violates the crucial assumption of the classical linear regression model that the explanatory variable is uncorrelated with the stochastic disturbance term. If this assumption is violated, then OLS estimates are not only biased but also inconsistent; they remain biased asymptotically. Let us now show this.

Instead of being able to estimate true relationship (12.1), we are forced to replace x_i by x_i^*. Now the OLS estimator is

$$\hat{\beta} = \frac{\sum x_i^* y_i}{\sum x_i^{*2}}$$

$$= \frac{\sum (x_i + \nu_i) y_i}{\sum (x_i + \nu_i)^2}$$

$$= \frac{\sum (x_i + \nu_i)(\beta x_i + \varepsilon_i)}{\sum (x_i + \nu_i)^2}$$

$$= \frac{\beta \sum x_i^2 + \beta \sum x_i \nu_i + \sum x_i \varepsilon_i + \sum \nu_i \varepsilon_i}{\sum x_i^2 + \sum \nu_i^2 + 2\sum x_i \nu_i}$$

From above we find that $\hat{\beta}$ is a biased estimate as $E(\hat{\beta}) \neq \beta$. Let us see the asymptotic properties of $\hat{\beta}$.

Since ν_i and ε_i are stochastic and are uncorrelated with each other as well as with x_i, we can say that

$$\operatorname{plim}\hat{\beta} = \operatorname{plim}\frac{\beta \sum x_i^2}{\sum x_i^2 + \sum \nu_i^2}$$

$$= \frac{\beta \mathrm{Var}(x)}{\mathrm{Var}(x) + \sigma_v^2} \text{ (Since } x_i = X_i - \bar{x} \text{ by definition)}$$

$$= \frac{\beta\sigma_{x2}^2}{\sigma_x^2 + \sigma_v^2}$$

$$= \beta\frac{\beta\sigma_x^2}{\sigma_x^2 + \sigma_v^2}$$

$$= \beta \frac{1}{1 + \frac{\sigma_v^2}{\sigma_x^2}} < \beta \qquad \ldots (12.7)$$

Thus, $\hat{\beta}$ is biased even for an infinite sample and $\hat{\beta}$ is an inconsistent estimator of β.

12.2.3 Measurement Errors in both X and Y

Now let us assume that both X and Y have errors of measurement. The true model is as before

$$y_i = \beta x_i + \varepsilon_i$$

Since X and Y have errors of measurement, we observe x_i^* and y_i^* instead of x_i and y_i, such that

$$y_i^* = Y_i + u_i, \qquad x_i^* = x_i + \nu_i,$$

where u_i and ν_i present the errors in the values of y_i and x_i respectively. We make the following assumptions about the error terms:

(i) There is no correlation between the error term and corresponding variable, i.e.,

$$E(y_i, u_i) = 0, \; E(x_i, y_i) = 0$$

(i) There is no correlation between error of one variable and measurement of the other variable, i.e.,

$$E(u_i, x_i) = 0 \; E(\nu_i, y_i) = 0$$

(i) There is no correlation between errors in measurement of both the variables, i.e.,

$$E(u_i, \nu_i) = 0$$

On the basis of the above assumptions, our estimated regression equation will be

$$y_i = \beta x_i$$

or, $$y_i - u_i = \beta(x_i^* - \nu_i)$$

or, $$y_i^* = \beta x_i^* - \beta \nu_i + u_i$$

or, $$y_i^* = \beta x_i^* + (\mu_i - \beta \nu_i) \qquad \ldots(12.8)$$

Equation (12.8) shows that if we model y^* as a function of x^*, the transformed disturbance contains the measurement error in β. We then write the OLS estimator $'\beta'$ as

$$\hat{\beta} = \frac{\sum x_i^* y_i^*}{\sum x_i^{*2}}$$

By substituting the values of x^* and y^* in terms of x and y we find that

$$\hat{\beta} = \frac{\sum (x_i + \nu_i)(y_i + u_i)}{\sum (x_i + \nu_i)^2}$$

$$= \frac{\sum (x_i + \nu_i)(\beta x_i + u_i)}{\sum (x_i + \nu_i)^2}$$

$$= \frac{\beta \sum x_i^2 + \beta \sum x_i \nu_i + \sum x_i u_i + \sum \nu_i u_i}{\sum x_i^2 + \sum \nu_i^2 + 2\sum x_i \nu_i}$$

From above we observe that $E(\hat{\beta}) \neq \beta$

$$p\lim \hat{\beta} = p\lim \frac{\beta \sum x_i^2}{\sum x_i^2 + \sum \nu_i^2}$$

$$= \frac{\beta \mathrm{Var}(x)}{\mathrm{Var}(x) + \sigma_v^2}$$

$$= \frac{\beta \sigma_x^2}{\sigma_x^2 + \sigma_v^2}$$

$$= \beta \frac{\sigma_x^2}{\sigma_x^2 + \sigma_v^2}$$

$$= \beta \frac{1}{1 + \frac{\sigma_v^2}{\sigma_x^2}} < \beta \qquad \ldots (12.9)$$

Thus, $\hat{\beta}$ will not be a consistent estimator of β. The presence of measurement error of the type in question will lead to an underestimate of the true regression parameter if ordinary least squares are used.

Example 12.1

A Specific illustration of this problem in economics is the permanent income theory of the consumption function, which is an errors-in-variables model. According to this model measured income Y^* has a permanent (true) component. Y and a transitory component, Y_ε such that

$$Y^* = Y + Y_\varepsilon \qquad \ldots (12.10)$$

Measured consumption expenditure, C^*, similarly, has a permanent (true) component, C, and a transitory component, C_ε such that

$$C^* = C + C_\varepsilon \qquad \ldots (12.11)$$

As you know form MEC-002, Block 4, the permanent consumption (C) is proportional to permanent income (Y). Due to transitory components, however, the average propensity to consume declines as income of a household increases. Let us analyse this issue through errors in variables.

The transitory components represent accidental or chance factors, including cyclical variations. They can be treated as the errors in measuring income and consumption. It is usually assumed that errors, on the average, vanish:

$$E(Y_\varepsilon) = E(C_\varepsilon) = 0 \qquad \ldots (12.12)$$

It is also usually assumed that the errors are correlated neither with each other nor with the permanent components:

$$\mathrm{Cov}(Y, Y_\varepsilon) = \mathrm{Cov}(C, C_\varepsilon) = \mathrm{Cov}(Y_\varepsilon, C_\varepsilon) = 0 \quad \ldots (12.13)$$

According to permanent-income hypothesis,

$$C = \beta_1 Y + \beta_2 + u \quad \ldots (12.14)$$

that is, permanent consumption is a linear function of permanent income, where β_1 and β_2 are constant parameters and u is a stochastic disturbance term. Notice that the parameter β_1 in (12.14) represents marginal propensity to consume (MPC). Combining (12.14) with (12.10) and (12.11) yields

$$C^* = \beta_1 Y^* + \beta_2 + (\nu - \beta_1 Y_\varepsilon), \qquad \text{where } \nu = u + C_\varepsilon \quad \ldots (12.15)$$

Equation (12.15) is a simple linear regression model of the same form as (12.8). Then,

$$\text{p}\lim \hat{\beta}_1 < \beta_1 \qquad \text{(as we know from (12.9))}$$

Since the ratio β_1 is less than unity, we observe that the above term is less than the true parameter.

Thus, the estimated marginal propensity to consume, $\hat{\beta}_1$, even in the probability limit, systematically underestimates the true β_1. The greater the variance of transitory income, the greater will be the underestimating. Only when the variance of transitory (error) component of income is zero, the least squares estimator of β_1 is a consistent estimator. To solve this problem, it is necessary to construct a measure of permanent income Y before estimating the consumption function.

Check Your Progress 1

1) Explain the concept of errors in variable. What are its consequences?

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2) State whether the following statements are true or false?

a) Errors in variables lead to estimates of the regression coefficients that are biased.

b) The presence of measurement error in an equation leads to an underestimate of the true regression parameter if ordinary least squares are used.

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3) Explain briefly why measurement error in the explanatory variables leads to biased and inconsistent parameter estimates while measurement error in the dependent variable does not.

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12.3 INSTRUMENTAL VARIABLES METHOD

The problem of errors in the measurement of variables in regression models is quite important, yet econometricians do not have much to offer in the way of useful solutions. As a general rule, we tend to pass over the problem of measurement error, hoping that the errors are small enough which will not destroy the validity of the estimation procedure. One technique which is available and can solve the measurement error problem is the technique of instrumental variables estimation. We briefly outline the concept of instrumental variables because it is likely to be useful with measurement errors.

The method of instrumental variables involves the search for new variable Z which is highly correlated with the independent variable X and at the same time uncorrelated with the error term in the equation (as well as the errors of measurement of both variables). In practice, we are concerned with the consistency of parameter estimates and therefore concentrate on the relationship between the variable Z and the remaining variables in the model when the sample size gets large. We define the random variable Z to be an instrument if the following conditions are met.

1) The correlations between Z and $\varepsilon, u, \text{and } \nu$, respectively, in the regression approach zero, as the sample size gets large.
2) The correlation between Z and X is nonzero, as the sample size gets large.

We simply select one instrument (or combination of instruments) that has the highest correlation with the X variable.

Assuming for the moment that such a variable can be found, we can alter the least squares regression procedure to obtain estimated parameters that are consistent. Unfortunately, there is no guarantee that the estimation process will yield unbiased parameter estimates. To simplify the matter, let us consider the case of measurement errors in the independent variable such $y_i = \beta x_i + \varepsilon_i$ and only x is measured with error (as $x^* = x + \nu$). In order to solve the problem we take the regression equation $y_i = \beta z_i + \varepsilon_i$ Where Z is the instrumental variable. The instrumental variables estimator of the regression slope in the above model is

$$\hat{\beta} = \frac{\sum y_i z_i}{\sum x_i^* z_i} \qquad \text{… (12.16)}$$

The choice of the particular slope formula is made so that the resulting estimator will be consistent. To see this, we can derive the relationship between the instrumental-variables estimator and the true slope parameter as follows:

$$\hat{\beta} = \frac{\sum y_i z_i}{\sum x_i^* z_i}$$

$$= \frac{\sum \left(\beta x_i^* + \varepsilon_i^*\right) z_i}{\sum x_i^* z_i}$$

$$= \frac{\beta \sum x_i^* z_i + \sum z_i \varepsilon_i^*}{\sum x_i^* z_i}$$

$$= \beta + \frac{\sum z_i \varepsilon_i^*}{\sum x_i^* z_i}$$

Clearly, the choice of Z as an instrument guarantees that $\hat{\beta}$ will approach β as the sample size gets large $\left[\text{Cov}\left(z, \varepsilon^*\right) \text{approaches } 0\right]$ and will therefore be a consistent estimator of β. Remember that the variable x_i^* in (12.16) was not replaced by z_i in the denominator of instrumental variables estimator, as the estimator $\frac{\sum y_i z_i}{\sum z_i^2}$ does not yield a consistent estimator of β. Let us show this,

$$\hat{\beta} = \frac{\sum y_i z_i}{\sum z_i^2}$$

$$= \frac{\sum y_i z_i}{\sum z_i^2}$$

$$= \frac{\sum \left(\beta x_i^* + \varepsilon_i^*\right) z_i}{\sum z_i^2}$$

$$= \frac{\beta \sum x_i^* z_i + \sum z_i \varepsilon_i^*}{\sum z_i^2}$$

$$= \frac{\beta \sum \left(x_i + v_i\right) z_i + \sum z_i \varepsilon_i^*}{\sum z_i^2}$$

$$= \frac{\beta \sum x_i z_i + \beta \sum z_i v_i + \sum z_i \varepsilon_i^*}{\sum z_i^2}$$

$$p\lim \hat{\beta} = p\lim \frac{\beta \sum x_i z_i}{\sum z_i^2}$$

$$= \beta \frac{\text{Cov}\left(x_i z_i\right)}{\sigma_z^2} \neq \beta$$

The above shows that $\frac{\sum y_i z_i}{\sum z_i^2}$ will not yield a consistent estimator of β.

The technique of instrumental variables appears to provide a simple solution to a difficult problem. We have defined an estimation technique, which yields consistent estimates if we can find an appropriate instrument.

A few concluding comments may be instructive.

i) First, the OLS estimation technique is actually a special case of instrumental variables. This follows because in the classical regression model X is correlated with the error term and because X is perfectly correlated with itself.

ii) Second, if we generalize the measurement error problem to errors in more than one independent variable, one instrument is needed to replace each of the designated independent variables.

iii) Finally, we repeat that instrumental-variables estimation guarantees consistent estimation but does not guarantee unbiased estimation.

12.4 TEST OF MEASUREMENT ERRORS

Suppose we are estimating the two-variable regression model

$$y_i = \beta x_i + \varepsilon_i \qquad \ldots (12.1)$$

There is a possibility that x might be measured with error.

If $x_i = x_i^* - \nu_i$, then the actual least squares regression would be

$$\begin{aligned} y_i &= \beta\left(x_i^* - \nu_i\right) + \varepsilon_i \\ &= \beta x_i^* - \beta\nu_i + \varepsilon_i \\ &= \beta x_i^* + \left(\varepsilon_i - \beta\nu_i\right) \\ &= \beta x_i^* + \varepsilon_i^* \qquad \ldots (12.17) \end{aligned}$$

where $\varepsilon_i^* = \varepsilon_i - \beta\nu_i$

If x is measured with error, we have seen that consistent estimator of β can be obtained by using an instrument z which is correlated with x^* but uncorrelated with ε and ν. Suppose the relationship between z and x^* is given by

$$x_i^* = \gamma z_i + w_i \qquad \ldots (12.18)$$

When (12.18) is estimated using least squares, we obtain

$$x_i^* = \hat{\gamma} z_i$$

or

$$x_i^* = \hat{x}_i^* + \hat{w}_i \qquad \ldots (12.19)$$

where $\hat{w}_i$ are the regression residuals. Substituting the value of eq. (12.19) into eq. (12.17) we have

$$y_i = \beta\left(\hat{x}_i + \hat{w}_i\right)$$

$$y_i = \beta\hat{x}_i^* + \beta\hat{w}_i + \varepsilon_i^* \qquad \ldots (12.20)$$

Whether or not there is measurement error, the coefficient of $\hat{x}_i^*$ will be consistently estimated by ordinary least squares, since

$$p\lim \frac{\sum \hat{x}_i^* \varepsilon_i^*}{N} = p\lim \frac{\hat{\gamma}\sum z_i\left(\varepsilon_i - \beta\nu_i\right)}{N} = 0.$$

In fact, the least-squares estimator of the coefficient of $\hat{x}_i^*$ in eq. (12.20) is identically equal to the instrumental-variables estimator, which is given by

$$\hat{\beta} = \frac{\sum y_i z_i}{\sum x_i^* z_i}$$

To look at the coefficient of the variable $\hat{w}_i$, note that

$$\text{p}\lim \frac{\sum \hat{w}_i \varepsilon_i^*}{N} = \text{p}\lim \frac{\sum \left(x_i^* - \hat{\gamma} z_i\right)\left(\varepsilon_i - \beta \nu_i\right)}{N}$$

$$= \text{p}\lim \frac{-\beta \sum \nu_i \left(x_i + \nu_i\right)}{N}$$

$$= -\beta \sigma_v^2$$

When there is no measurement error, $\sigma_v^2 = 0$, so that OLS applied to eq. (12.20) will generate a consistent estimator of the coefficient of $\hat{w}_i$. However, when there is measurement error, the coefficient will be estimated inconsistently.

This suggests a relatively easy measurement error test. Let δ represent the coefficient of the variable $\hat{w}_i$ in eq. (12.20).

Substituting $\hat{x}_i^* = x_i^* - \hat{w}_i$, we get

$$y_i = \beta\left(x_i^* - \hat{w}_i\right) + \delta \hat{w}_i + \varepsilon_i^*$$

or $$y_i = \beta x_i^* + (\delta - \beta)\hat{w}_i + \varepsilon_i^* \qquad \ldots (12.21)$$

With no measurement error, $\delta = \beta$, so that the coefficient of $\hat{w}_i$ should equal zero. However, with measurement error, $\delta \neq \beta$, and the coefficient will (in general) be different from zero. We can test for measurement error by doing a simple two-stage procedure. First, we regress x^* on z to obtain the residuals $\hat{w}$. Then, we regress y on x^* and $\hat{w}$ and perform a t test on the coefficient of the $\hat{w}$ variable. If we are concerned with measurement error in more than one variable of a multiple regression model, an equivalent F test could be applied.

The test just described is a special case of a more general test for specification error proposed by Hausman. The Hausman specification test relies on the fact that under the null hypothesis the ordinary least-squares estimator of the parameters of the original eq. (12.17) is consistent and (for large samples) efficient, but is inconsistent if the alternative hypothesis is true. However, the instrumental-variables estimator [test least-squares estimator of eq. (12.20)] is consistent whether or not the null hypothesis is true, although it is inefficient if the null hypothesis is not valid.

The Hausman specification test is as follows: If there are two estimators $\hat{\beta}_1$ and $\hat{\beta}_2$ that converge to the true value β under the null but converge to different values under the alternative, the null hypothesis can be verified by testing whether the probability limit of the difference of the two estimators is zero.

Example 12.2

The expenditure of states and local governments (EXP) in US vary substantially by state and region. Among the important variables that explain difference in spending levels are federal grants-in-aid (AID), the income of states (INC), and the population of states (POP). When a model that relates the dependent variable EXP to the independent variables AID, INC and POP was estimated by ordinary least squares, using census data for 50 states, the following results were obtained (with t statistics in parentheses):

$$EXP = -46.81 + 0.00324\ AID + 0.00019\ INC - 0.597\ POP$$

$$(-0.56) \quad (13.64) \quad (8.12) \quad (-5.71)$$

$$R^2 = 0.993 \quad F = 2190$$

There is an important possible source of measurement error in the AID variable. This component can be subject to substantial error as the state aid programmes involves fixed sums of money, and the money involved are easy to measure even before state and local budgets are set.

We can test the presence of measurement error by using a Hausman specification test. To perform the test, we use the population of primary and secondary school children (PS) as an instrument. The test proceeds in two stages. In the first stage AID is regressed on PS, and the residual variable $\hat{w}_i$ is calculated as follows:

$$\hat{w}_i = AID - 77.95 + 0.845\,PS \qquad R^2 = 0.87$$

$$(-1.28) \quad (18.02)$$

In the second stage $\hat{w}_i$ is added to the original regression to correct for measurement error. The resulting equation is

$$EXP = -138.51 + 0.00174\ AID + 0.000\ 18INC - 0.275\ POP + 1.372\ \hat{w}_i$$

$$(-1.41) \quad (1.94) \quad (7.55) \quad (-1.29) \quad (1.73)$$

A two-tailed t test of the null hypothesis that there is no measurement error would be accepted at the 5 per cent level, since 1.73 < 1.96. However, measurement error would be important if we were using either a one-tail test, or a two-tailed test at 10 per cent significance level. Note that correcting for the possibility of measurement error has substantially lowered the coefficient on the AID variable, suggesting that measurement error caused the effect of AID on spending to be overstated.

12.5 INVERSE REGRESSION

When we have the variables y and x both measured with errors (the observed values being y^* and x^*), we consider two regression equations:

1) Regression of y^* on x^*, which is called the "direct" regression.
2) Regression of x^* on y^* which is called the "inverse" and "reverse" regression.

Inverse regression has been frequently advocated in the case of analysis of salary discrimination. Since the problem is one of usual errors in variables, both regressions need to be computed and whether inverse regression alone gives the correct estimates depends on the assumptions we make.

The usual model, in its simplest form, is that of two explanatory variables, one of which is measured with error

$$y = \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

where y = salary

x_1 = true qualifications

x_2 = gender

What we are interested in is the coefficient of x_2. The problem is that x_1 is measured with error.

Let x_i^* = measured qualifications

$$x_i^* = x_i + \nu_i$$

Suppose we adopt the notion that

$$x_2 = \begin{cases} 1, \text{ for men} \\ 0, \text{ for women} \end{cases}$$

Then $\beta_2 > 0$ implies that men are paid more than women with the same qualification and thus there is gender discrimination. A direct least squares estimation of eq. (12.22) with x_1^* substituted for x_1 and $\hat{\beta}_2 > 0$ has been frequently used as evidence of gender discrimination.

In the inverse regression we take

$$x_1^* = \gamma_1 y + \gamma_2 x_2 + w \qquad \text{… (12.23)}$$

We are asking whether men more or less qualified than women having the same salaries. The proponents of inverse regression argue that to establish discrimination, one has to have $\hat{\gamma}_2 < 0$ in eq. (12.23); that is, among men and women receiving equal salaries, men possess lower qualifications.

The usual errors in variables results however show that we should no make inferences on the basis of $\hat{\beta}_2$ and $\hat{\gamma}_2$ but obtain bounds for β_2, from the direct regression and the inverse regression estimates.

Check Your Progress 2

1) Show that in the two-variable model $\hat{\beta} = \frac{\sum y_i z_i}{\sum z_i^2}$ (where z is an instrument) will not yield a consistent estimate of the true slope parameter.

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2) What is the Hausman specification test? How would you carry out a Hausman specification test to evaluate the presence or absence of measurement error?

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12.0 LET US SUM UP

In ordinary least squares model, we assume that sample observation are measured without error, which is always not true. When this assumption does not hold, OLS estimators are biased and inconsistent. Errors may appear in the measurement of dependent variable, independent variable or both. When there is error in dependent variable, this does not destroy the unbiased property of the OLS estimators, but the estimated variances are larger than the case where there is no such errors of measurement. On the other hand, when there is error in the independent variable, or both dependent and independent variables, then OLS estimates are not only biased but also inconsistent.

We can test for the presence of measurement error by using the Hausman specification test. One technique which is available and can solve the measurement error problem is the technique of instrumental variables estimation.

12.7 KEY WORDS

Asymptotic Bias : When the classical least-squares estimator is biased even with the increase in sample size, it is known as asymptotic bias or inconsistency.

Equation Error : It represents the influence of various explanatory variables that have not accurately been included in the model and is in the form of disturbance term.

Errors in Variables : When errors are found in the measurement of variables, we say that there are errors in variables. These errors can substantially alter the properties of the estimated regression parameters.

Explanatory Variable : It is the independent variable which explains the dependent variable.

Instrumental Variable : It is a random variable, which a highly correlated with the independent variable and at the same time uncorrelated with the error term in the equation.

Inverse Regression : It is the reverse of direct regression. If we consider regression of Y on X as direct regression, then regression of X on Y is inverse regression.

Measurement Error : When either dependent variable, or independent variable or both, are measured subject to error, we say that there is measurement error.

Parameter : It is a measure of some characteristics of the population.

Proxy Variable	:	It is a variable, which is proxy for the true variable.
Specification Error	:	Errors found in the specification of the model.
Stochastic Disturbance Term	:	It is the random error term whose value is based on an underlying probability distribution.

12.8 SOME USEFUL BOOKS

Intriligator, Michael, Ronald Bodkin and Cheng Hsiao (1996); *Econometric Models, Techniques, and Applications*, Prentice-Hall International Inc., New Jersey.

Johnston, J., Dinardo, J., 2007, "*Econometric Methods*", 4th Edition, McGraw-Hill Education (Asia), International Edition, pp. 109-125

Maddala, G. S. and K. Lahiri, 2012, *Introduction to Econometrics*, Fourth Edition, Wiley

Pindyck, Robert S. and Daneil L. Rubinfeld (1998); *Econometric Models and Economic Forecasts*, Fourth Edition, Irwin/McGraw-Hill, Singapore.

Verbeek, M. 2012, "*A Guide to Modern Econometrics*", Fourth Edition, John Wiley & Sons

12.9 ANSWERS/HINTS OF CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) Go through Sections 12.1 and 12.3.
2) a) true b) true.
3) Go through Section 12.3.

Check Your Progress 1

1) Go through Section 12.4.
2) Go through Section 12.5 and Example 12.2.

UNIT 13 STOCHASTIC REGRESSORS*

Structure

13.0 OBJECTIVES

After going through this Unit you should be in a position to

- explain the endogeneity problem due to omitted variables, measurement error, and simultaneity;
- explain the need for instrumental variables and how they can be used; and
- describe the two-stage least squares estimator.

13.1 INTRODUCTION

While discussing the classical regression model (see Unit 4), we assumed that the values of the explanatory variable, x_i, are kept fixed in repeated samples. An implication of the above is that the values assumed by x_i are not random. Consequently, there is no relationship between the stochastic error term u_i and the explanatory variable x_i. This assumption is concisely written as $E(x_i u_i) = 0$.

In real life situations x_i is a random variable, that is, x_i is a stochastic regressor. In such cases $E(x_i u_i) \neq 0$, that is, x_i and u_i are correlated. In this Unit we discuss issues related to the consequences of stochastic regressors and the remedial measures that we should undertake.

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13.2 ENDOGENEITY PROBLEM

Let us consider the following population regression model, which is linear in its parameters

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 ... + \beta_K x_K + u \qquad ...(13.1)$$

where y : dependent variable

$x_1, x_2, ..., x_K$: explanatory variables

u : unobservable random disturbance or error

$\beta_1, \beta_2, ..., \beta_K$: the parameters to be estimated

The key condition on which equation (13.1) under OLS can consistently estimate the β_j is that mean value of the stochastic error is zero and is uncorrelated with each of explanatory variables. Symbolically, we can write:

$$E(u) = 0, Cov(x_j, u) = 0, \qquad j = 1,2,...,K \qquad ... (13.2)$$

In regression analysis, assumption (13.2) is sufficiently supported by the zero conditional mean assumption, which can be written as:

$$E(u \mid x_1, x_2, ..., x_k) = E(u \mid x) = 0 \qquad ... (13.3)$$

The population regression function under equation (13.1) and assumption (13.3) can be written as:

$$E(y \mid x_1, x_2, ..., x_K) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_K x_K \qquad ... (13.4)$$

An explanatory variable x_j is said to be 'endogenous' in equation (13.1), if it is correlated with the stochastic error term u. Similarly, x_j is said to be exogenous in equation (13.1), if it is not correlated with the stochastic error u. Assumption (13.3) implies that each explanatory variable is exogenous.

An explanatory variable x_j is said to be endogenous if it is correlated with the stochastic error u. Thus, endogeneity problem arises whenever the explanatory variable is correlated with the error term. There are three ways of observing endogeneity in regression models. These are:

a. Omitted Variables

b. Measurement Error

c. Simultaneity

13.2.1 Omitted Variables

The case of omitted variables was discussed in Unit 8. It arises when we are unable to include certain additional variables in our regression model due to data unavailability.Let us suppose that $E(y|x,q)$ is the conditionalexpectation function which is linear in parametersand additive in q (suppose, q represents the excluded variables). We always estimate $E(y|x)$ for unobserved q. But such a regression model is different from $E(y|x,q)$, even if q and x are correlated. In the case of excluded variables, we are writing equation (13.1) and q gets included in the stochastic error u. Thus, variable x_j is endogenous in a regression model if q (i.e., excluded variables) and x_j are correlated.

13.2.2 Measurement Error

In regression analysis we implicitly assume that the dependent and independent variables are measured without any error. That is, the variables are exact, and error is zero. However, this is only an assumption and in real life errors creep in due to various reasons such as non-response error, reporting error, and computing error. We discussed the issue of error-in-variables in Unit 12.

Whatever be the reason, measurement error in variables lead to biased and inefficient estimators. The observed variables of'the sample regression function'on which we collect data are modelled as:

$$\tilde{x} = x + u \qquad \text{… (13.5)}$$

$$\tilde{y} = y + v \qquad \text{… (13.6)}$$

Equation (13.5) and equation (13.6) show that our observed variables $\tilde{x}$ and $\tilde{y}$ are measured with additive errors u and v respectively.The measurement error in x creates an endogeneity bias because the measurement error becomes a part of the error term $u$in the regression equation. This can be viewed from equation (13.5) as follows:

$$x = \tilde{x} - u$$

Now putting the value of x in $y = \beta x + \in$ we can write:

$$y = \beta(\tilde{x} - u) + \in$$

or, $y_i = \beta\tilde{x} + (\in - \beta u)$ … (13.7)

Because of the positive correlation between $\tilde{x}$ and u observed in equation (13.5), the OLS estimation produces certain bias in $\tilde{\beta}$. Such bias may be negative or positive depending upon the sign of the true β in equation (13.7). If the true β is positive, then the bias is negative. On the other hand, if the true β is negative, then bias is positive.

13.2.3 Simultaneity

If any of the explanatory variables is determined simultaneously along with y, then the problem of simultaneity arises. This problem is also called reverse causality, that is, the explanatory variable (x) is caused by the explained variable (y). For example, x_K and u are correlated if x_K is determined partly as a function of y.

Let us consider a two variable regression model ($y_i = \beta x_i + u_i$), where the dependent variable is 'earnings (y_i)' of a person and the independent variable is 'years of schooling (x_i)'. If we analyse the model, there are several variables such as ability of the person, type of school one attended, parental education, peer-group of the person, etc. that are excluded from the model but influence earnings. All these excluded variables are clubbed together in the stochastic error term. According to this model, if there is no association between x and u, a unit change in x will result in βx change in y. Let us consider the variable 'ability' (which cannot be measured) that is clubbed with u. The variable 'ability' influences y positively – a person with higher ability will have higher value of y. But we know that a person with high ability will have higher value of 'years of schooling', i.e., x and u are correlated. Thus, a person with high ability will influence y in two ways: directly through a higher value of u, and indirectly it will lead to a higher value of x.

Very often it is not possible to distinguish among the three possible forms of endogeneity. In fact, more than one source of endogeneity is observed in any given equation. There are two methods to resolve the endogeneity problem: (i) instrumental variables (IV) method, and (ii) two-stage least squares (2SLS) method. The IV estimator and the 2SLS estimator are consistent, i.e., they approach the value of the parameter as the sample size increases infinitely.

Check Your Progress 1

1) What do you mean by endogeneity problem in regression analysis?

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2) What are the major sources of endogeneity problems?

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3) Explain the problems of measurement error.

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4) Explain the problems of omitted variable as an endogeneity problem.

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5) Explain the problems of simultaneity as an endogeneity problem.

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13.3 INSTRUMENTAL VARIABLE ESTIMATOR

In this section we discuss about the instrumental variable method of estimation, which is extensively used in empirical research. In instrumental variable method of estimation, the unobservable error (correlated with the explanatory variables) is explicitly allowed. Let us consider the following linear population regression function:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_K x_K + u \qquad \ldots (13.8)$$

$$E(u) = 0, \qquad Cov(x_j, u) = 0, \qquad j = 1, 2, \ldots, K-1 \qquad \ldots (13.9)$$

Let us assume that x_K is correlated with u. In other words, x_K is endogenous and all other explanatory variables$x_1, x_2; \ldots; x_{K-1}$ are exogenous in equation (13.8). As mentioned above, endogeneity can come from any of the sources. For the time being let us assume that$u$ containsan omitted variable that is not correlatedwith any of the explanatory variables $x_1, x_2; \ldots x_{K-1,}$ but correlated with x_K. A conditionalexpectation function can be written as in equation (13.10).

$$E(y \mid x_1, x_2, \ldots, x_K, q) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_K x_k + \gamma q \qquad \ldots (13.10)$$

In equation (13.10), q is unobservable omitted factor and is correlated with x_K. Equation (13.8) gives us inconsistent estimators of all the β_j under OLS if $Cov(x_K, u) \neq 0$.

The instrumental variable method provides a way to the endogeneity problem. In IV method we need an instrumental variable z_1 in place of the endogenous variable x_K. In equation (13.8), z_1 should not be present and z_1 should satisfy two conditions:

(i) The instrumental variable z_1 must not be correlated with u so that z_1 along with $x_1, x_2; \ldots; x_{K-1}$ is exogenous in equation (13.8). This can be written as:

$$Cov(z_1, u) = 0 \qquad \ldots (13.11)$$

(ii) The relationship between z_1 and the endogenous variable, x_K is the next requirement. Let the linear projection of x_K onto all the exogenous variables (see Unit 2) be:

$$x_K = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \ldots + \delta_{K-1} x_{K-1} + \theta_1 z_1 + r_K \qquad \ldots (13.12)$$

where, $E(r_K) = 0$ (by definition of a linear projection error)

r_K is uncorrelatedwith $x_1, x_2; \ldots ; x_{K-1}$, and z_1.

This linear projection is based on the assumption that the coefficient of z_1 is non-zero, which can be written as:

$$\theta_1 \neq 0 \qquad \ldots (13.13)$$

This condition with the statement ''both z_1 and x_K are correlated with each other'' is not always true. The assumption (13.13) means that z_1 is partially correlated with x_K, once the other exogenous variables $x_1, x_2; \ldots\ x_{K-1}$ have been netted out. In the case of x_K being the only explanatory variable, the linear projection ofx_K can be written as:

$$x_K = \delta_0 + \theta_1 z_1 + r_K \qquad \ldots (13.14)$$

where,

$$\theta_1 = \mathrm{Cov}(z_1, x_K) / \mathrm{Var}(z_1)$$

$$\theta_1 \neq 0$$

$$\mathrm{Cov}(z_1, x_K) \neq 0$$

Depending upon our requirement, we may put no restrictions on the distribution of x_K or z_1. Though in many cases x_K and z_1 are continuous, they may be discrete at times. We can also consider one or both of x_K and z_1 as binary variables. These variables may have continuous and discrete characteristics at the same time. In equation (13.12), x_K defines a linear projection, which needs to be defined alwaysin the case of second moments of all variables being finite.

The variable z_1 is considered an instrumental variable for x_K if it satisfies two conditions: (i) variable z_1 is correlated with x_k, and (ii) variable z_1 is not correlated with u. In real life problems, it is difficult to find an instrumental variable.

We can call z_1 as instrument for x_K, because $x_1, x_2, \ldots, x_{K-1}$ being uncorrelated with u serve as their own instrumental variables in equation (13.8). For the endogenous explanatory variable x_K the linear projection in equation (13.12) is known as a 'reduced form equation'. A reduced form always involves describing an endogenous variable as a linear projection onto all exogenous variables in the context of single-equation linear models (we will discuss about reduced form of a structural equation in Unit 16). Using the structural equation (13.8) and the reduced form for x_K, a reduced form for y can be obtained by plugging equation (13.12) into equation (13.8) and re-arranging:

$$y = \alpha_0 + \alpha_1 x_1 + \ldots + \alpha_{K-1} x_{K-1} + \lambda_1 z_1 + \upsilon \qquad \ldots (13.15)$$

where, υ is the reduced form error

$$\upsilon = u + \beta_K r_K$$

$$\alpha_j = \beta_j + \beta_K \delta_j$$

$$\lambda = \beta_K \theta_1$$

Now, OLS can consistently estimate the above reduced form parameters, α_j and λ_1 because by assumption υ is uncorrelated with all explanatory variables in equation (13.15). Estimating the structural parameters is more useful although estimates of the reduced form parameters are sometimes of interest.

The assumptions made on the instrumental variable z_1 solve the identification problem for the β_j in equation (13.8). The parameters β_j in terms of population moments in observablevariables can be written to indicate identification. Let us write equation (13.8) as:

$$y = x\beta + u \qquad \dots (13.16)$$

where x = (1, $x_2, \dots, x_K$) due to the constant being absorbed into x.

Let us write the ($1 \times K$) vector of all exogenous variables as

$$z \equiv (1, x_2, \dots, x_{K-1}, z_1)$$

From assumptions (13.8) and (13.9) the K population orthogonality conditions can be written as:

$$E(z'u) = 0 \qquad \dots (13.17)$$

Multiplying both sides of equation (13.16) by z', taking expectations, and using equation(13.18), we get:

$$[E(z'x)]\beta = E(z'y) \qquad \dots (13.18)$$

where, $E(z'x)$ is $K \times K$

$E(z'y)$ is $K \times 1$

Here, we observe equation (13.18) as K linear equations with K unknowns $\beta_1, \beta_2, \dots, \beta_K$ which will have unique solution if and only if the $K \times K$ matrix $E(z'x)$ has full rank, which can be written as:

$$\text{rank } E(z'x) = K \qquad \dots (13.19)$$

The solution of equation (13.18) can be written as:

$$\beta = [E(z'x)]^{-1} E(z'y) \qquad \dots (13.20)$$

Equation (13.20) identifies the vector β where the expectations $E(z'x)$ and $E(z'y)$ can be consistently estimated. We can use random sample (x, y, z_1) to estimate both expectations $E(z'x)$ and $E(z'y)$. The solution (13.20) is valid for condition (13.11), i.e., $Cov(z_1, u) = 0$. But in order to know whether we used condition (13.13) in deriving solution (13.21), let us assume that exogenous variables are not linearly dependent to each other. Thus, full rank of $E(z'z)$ will be K, which completely rules out perfect collinearity in z in the population. In this case, equation (13.19) will be valid for $\theta_1 \neq 0$. Thus, assumption (13.13) is the major identification condition along with exogeneity condition (13.11) and rank condition (13.19).

Now, the IV estimator of β for a given random sample $\{(x_i, y_i, z_{i1}) : i = 1,2,\dots,N\}$ from the population can be written as:

$$\hat{\beta} = \left(N^{-1} \sum_{i=1}^{N} z_1' x_i \right)^{-1} \left(N^{-1} \sum_{i=1}^{N} z_i' y_i \right) = (Z'X)^{-1} Z'Y \qquad \dots (13.21)$$

where, Z : $N \times K$ data matrices

X : $N \times K$ data matrices

Y : $N \times 1$ data vector on the y_i.

The IV estimator is consistent due to the law of large numbers and equation (13.21). In order to identify β, we must give equal importance to conditions (13.11) and (13) though there is a practical difference between them. The difference is condition (13.13) can be tested, whereas condition (13.11) must be maintained because the covariance in condition (11) involves the unobservable u, and therefore we cannot test anything about $Cov(z_1, u) = 0$. Using t-test we can test condition (13.13) in its reduced form (13.12). However, here r_K may not satisfy the homoscedasticity assumption of OLS. Therefore, for $\hat{\theta}_1$, a heteroskedasticity-robust t-statistic is needed. This will be true for x_K being a binary variable or some other variable with discrete characteristics.

Instrumental variable method has a few limitations. It is difficult to find an instrumental variable (z_1) that is perfectly correlated with the endogenous explanatory variable x_k. By using z_1, we are losing some of the information contained in x_k. This loss of information makes the IV model have larger error terms. Consequently, the standard error of the IV estimator is larger. Therefore, there should be strong correlation between z_1 and x_k. If the correlation is weak, it is better to go look for other instruments, or estimate the model with the endogenous variable and accept the bias.

Check Your Progress 2

1) What is meant by instrumental variable in regression analysis?

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2) Why is the estimator consistent under instrumental variable method?

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3) What is the major difference between the following two conditions while we should give equal importance to both in identifying β?

i. $Cov(z_1, u) = 0$

ii. $\theta_1 \neq 0$

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4) Explain the first condition that the instrumental variable must satisfy in order to be considered in the model for estimation.

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13.4 TWO-STAGE LEAST SQUARES ESTIMATOR

There could be several contending instruments for an endogenous variable in a regression model. Which one to choose?

Let us again consider the following two regression models:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_K x_K + u \qquad \ldots (13.8)$$

$$E(u) = 0, \qquad Cov(x_j, u) = 0, \; j = 1, 2, \ldots, K-1 \qquad \ldots (13.9)$$

where, x_K can be correlated with u. Let us suppose that there is more than one instrumental variable for x_K. Let us also assume that $z_1, z_2, \ldots, z_M$ be variables which are not correlated with u so that:

$$Cov(z_h, u) = 0, \; h = 1, 2, \ldots, M \qquad \ldots (13.22)$$

Equation (13.22) also exhibits that each z_h is exogenous in equation (13.8). There could have M different IV estimators if each z_h has some partial correlation with x_K. Further, we can take a linear combination of these instruments! Thus, apart from the instrumental variables, there could be many more IV because any linear combination of $x_1, x_2, \ldots, x_{K-1}, z_1, z_2, \ldots, z_M$ is uncorrelated with stochastic error u. In such situations, we follow the method of 'Two-Stage Least Squares (2SLS)'. In 2SLS method we choose that combination of the instruments which is most correlated with the endogenous explanatory variable (x_K).

Under certain assumptions, the two-stage least squares (2SLS) estimator is the most efficient instrumental variable (IV) estimator. To describe the 2SLS method, let us define the vector of exogenous variables given by:

$$z \equiv (1, x_1, x_2, \dots, x_{K-1}, z_1, \dots, z_M) \quad \dots (13.23)$$

where, z is a $1 \times L$ vector and L= K+M

In the 2SLS method we choose that linear combination of z (out of many possible linear combinations) which has the maximum correlation with x_K. The linear combination of z having maximum correlation with x_K is given by the linear projection of x_K on z. Now, we can write the reduced form (see Unit 16) of x_K as:

$$x_K = \delta_0 + \delta_1 x_1 + \dots + \delta_{K-1} x_{K-1} + \theta_1 z_1 + \dots + \theta_M z_M + r_K \quad \dots (13.24)$$

where, r_K with zero mean is uncorrelated with each of the right-hand-side variables.

As any linear combination of z is uncorrelated with u, we find that x_K^* (being a linear combination of z) is also uncorrelated with u. The linear projection of x_K^* can be written as:

$$x_K^* \equiv \delta_0 + \delta_1 x_1 + \dots + \delta_{K-1} x_{K-1} + \theta_1 z_1 + \dots + \theta_M z_M \quad \dots (13.25)$$

Here, we can interpret x_K^* as a part of x_K, but x_K^* is uncorrelated with u. We can use x_K^* as an instrument for x_K in equation (13.8). However, x_K^* cannot be a usable instrument because θ and δ are population parameters. The parameters in equation (13.24) can be consistently estimated by OLS if there is no multicollinearity problem in the model. The sample analogues of the x_{iK}^* for each observation i, being simply the OLS fitted values, can be written as:

$$\hat{x}_{iK} = \hat{\delta}_0 + \hat{\delta}_1 x_{i1} + \dots + \hat{\delta}_{K-1} x_{i,K-1} + \hat{\theta}_1 z_{i1} + \dots + \hat{\theta}_M z_{iM} \quad \dots (13.26)$$

Defining the vector $\hat{x}_i$ for each observation i, we obtain:

$$\hat{x}_i \equiv (1, x_{i1}, \dots, x_{i,K-1}, \hat{x}_{iK}), \qquad i = 1, 2, \dots, N, \quad \dots (13.27)$$

Now, we can use $\hat{x}_i$ as the instruments for x_i . Thus, the IV estimator for the regression model is

$$\hat{\beta} = \left(\sum_{i=1}^{N} \hat{x}_i' x_i \right)^{-1} \left(\sum_{i=1}^{N} \hat{x}_i' y_i \right) = (\hat{X}'X)^{-1} \hat{X}'Y \quad \dots (13.28)$$

The IV estimator can be converted into OLS estimator also. We can express $N \times (K+1)$ matrix $\hat{X}$ as:

$$\hat{X} = Z(Z'Z)^{-1} Z'X = P_Z X \quad \dots (13.29)$$

where, P_Z is the projection matrix. Note that $P_Z = Z(Z'Z)^{-1} Z'$ is idempotent and symmetric.

Therefore,

$$\hat{X}'X = X'P_Z X = (P_Z X)' P_Z X = \hat{X}'\hat{X}\, X \quad \dots (13.30)$$

Plugging equation (13.30) into equation (13.28) shows that the IV estimator that uses instruments $\hat{x}_i$ can be written as:

$$\hat{\beta} = (\hat{X}'\hat{X})^{-1}\hat{X}'Y \qquad \ldots (13.31)$$

The 2SLS method involves application of the least squares method twice. These are:

Step 1: First-Stage Regression

Regress x_K on 1, $x_1, \ldots, x_{K\_1}, z_1, \ldots, z_M$ by OLS method Obtain the fitted values of endogenous explanator variable, $\hat{x}_K$.

Step 2: Second-Stage Regression

Regress y on 1, $x_1, \ldots, x_{K\_1}$, $\hat{x}_K$ by OLS method. Obtain 2SLS estimators for all the parameters.

Note that both 2SLS and IV estimators will be the same if we have only one instrument for x_K..

Check Your Progress 3

1) Write down the reduced form for x_K .

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2) Write down the linear projection of x_K^* .

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3) Write down the IV estimator using $\hat{x}_i$ as the instruments for x_i .

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4) Explain the steps followed in two-stage least squares method to obtain $\hat{\beta}$.

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5) Describe the importance of the 2SLS method.

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13.5 LET US SUM UP

An assumption of the classical regression model is that the error term is independent of the explanatory variables in the model. In other words, there is no correlation between the error term and the explanatory variables. If this assumption is violated, we cannot get best linear unbiased estimators of the parameters. If any of the explanatory variables is correlated with the error term we face the endogeneity problem.

There are three possible sources of endogeneity problem in applied econometrics, viz., omitted variables, measurement error, and simultaneity. The problem of omitted variables arises when we are unable to include some variables in a regression model due to data unavailability. Errors-in-variables can occur due to various reasons such as nonresponse error, reporting error and computing error. Measurement error in explanatory variables also creates an endogeneity bias. If any of the explanatory variables is determined simultaneously along with the dependent variable, then the problem of simultaneity arises. Sharp distinction among the three possible forms of endogeneity is not possible always. We in fact observe more than one source of endogeneity in any given equation.

For solving the problem of endogeneity, we have two methods: (i) instrumental variable method, and (ii) two-stage least squares method (2SLS). In IV estimation we replace the endogenous explanatory variable with an instrumental variable. An instrumental variable should have two properties: (i) there is a strong correlation between the endogenous explanatory variable (x_K) and instrument under consideration, and (ii) the instrumental variable is not correlated with the stochastic error term (u).

There could be more than one instrumental variable for an endogenous explanatory variable. In such situations, it is difficult to make a choice among the instrumental variables. We apply the method of two-stage least squares (2SLS) in such situations. We take all possible instruments of the endogenous variable x_k, while avoiding the problem of multicollinearity among the instruments. In the first stage, we run OLS of x_k on all instrumental variables and obtain $\hat{x}_k$. In the second stage, we run OLS of y on all the explanatory variables; while replacing x_k by $\hat{x}_k$. Both 2SLS and IV estimators will be same if we have only one instrument for x_k.

13.6 KEY WORDS

Endogeneity Problem : In an econometric model if any of the explanatory variables is correlated with the stochastic error term, the concerned explanatory variable is said to be endogenous in nature.

Measurement Error : The observed variables on which we can collect the data is modelled as $\tilde{x} = x + u$ and $\tilde{y} = y + v$, which show that our observed variables $\tilde{x}$ and $\tilde{y}$ are measured with additive errors u and v respectively. Thus u and v are measurement errors.

Instrumental Variables : A set of variables known as instrumental variables are identified with the condition that they are correlated with the independent variables but uncorrelated at least asymptotically with composite disturbances so that we can get consistent estimator.

Simultaneity : If at least one of the explanatory variables is determined simultaneously along with y then the problem of simultaneity arises. Say for example, x_K and u are correlated if x_K is determined partly as a function of y. There are always no sharp distinctions among the three possible forms of endogeneity. We in fact observe more than one source of endogeneity in any given equation.

13.7 SOME USEFUL BOOKS AND REFERENCES

Gujarati, D N and D C Porter (2009), *Basic Econometrics*, Fifth Edition, McGraw Hill

Maddala, G. S. and K. Lahiri (2009), *Introduction to Econometrics*, Fourth Edition, John Wiley and Sons

Wooldrige, Jeffrey (2022), *Introductory Econometrics*, Seventh Edition, Cengage Learning

References:

Bin Chen, 2015, Modeling and testing smooth structural changes with endogenous regressors. *Journal of Econometrics, Volume 185, Issue 1, March 2015, Pages 196-215*

Eleanor Sanderson, Frank Windmeijer, 2015, A weak instrument -test in linear IV models with multiple endogenous variables. *Journal of Econometrics.*

Xavier d'Haultfoeuille. 2010. A new instrumental method for dealing with endogenousselection. *Journal of Econometrics, Volume 154, Issue 1, January 2010, Pages 1-15*

Jefrey M. Wooldridge. 2002. Econometric Analysis of Cross Section and Panel Data. The MIT Press Cambridge, Massachusetts, London, England.

13.8 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) In econometrics if an explanatory variable x_j is correlated with the stochastic error u, then the problem of endogeneity arises.

2) The major sources of endogeneity problem in applied econometrics are: Omitted Variables, Measurement Error, and Simultaneity.

3) Measurement error in independent variable x creates an endogeneity bias because it becomes a part of the error term in the regression equation.

   We know that: $X = \tilde{x} - u$ …(1)

   Now putting the value of x in equation $y = \beta x + \in$, we get:

   $y = \beta(\tilde{x} - u) + \in$

   or, $y_i = \beta\tilde{x} + (\in - \beta u)$ … (2)

   Because of positive correlation between $\tilde{x}$ and u, observed in equation (1) the OLS estimator produces bias in $\tilde{\beta}$, which may be negative or positive depending upon the sign of true β in equation (2). If the true β is positive, then bias is negative and if it is negative then bias is positive. The coefficient $\hat{\beta}$ biases towards zero as $0 < \lambda < 1$. This is known as attenuation bias with λ being the attenuation factor. Relative to the true error the residual holds two more sources of variation. First, $\hat{\beta}$ is biased towards zero and the term $(\hat{\beta} - \beta)$ does not vanish asymptotically when measurement error is absent. Second, the presence of measurement error in the regressor leads to an increase in error variance.

4) The problem of omitted variables arises when we are unable to include some additional variables due to data unavailability. Suppose $E(y|x,q)$ is the conditional expectation function, linear in parameters and additive in q. However, we shall always estimate $E(y|x)$ for unobserved q. But it has no relationship with $E(y|x,q)$ even if q and x are allowed to be correlated. But we take equation (1) where q is included in the stochastic error u. Thus, x_j is endogenous if q and x_j are correlated.

5) If at least one of the explanatory variables is determined simultaneously along with y then simultaneity arises. Say for example, x_K and u are correlated if x_K is determined partly as a function of y. There are always no sharp distinctions among the three possible forms of endogeneity. We observe more than one source of endogeneity in any given equation.

Check Your Progress 2

1) When an explanatory variable is correlated with the error term, we face the endogeneity problem in a regression model. Consequently, the OLS estimator is not unbiased. As a solution to this problem, we use the instrumental variable method. An instrument should be strongly correlated with the endogenous variable and not correlated with the error term.

2) The IV estimator is consistent due to: the law of large numbers, and $\hat{\beta} = \left(N^{-1} \sum_{i=1}^{N} z_i' x_i \right)^{-1} \left(N^{-1} \sum_{i=1}^{N} z_i' y_i \right) = (Z'X)^{-1} Z'Y$

3) In order to identify β we must give equal importance to the following two conditions: (i) $cov(z_1, u)$, and (ii) $\theta_1 \neq 0$. The major difference between both conditions is that condition (ii) can be tested, whereas condition (i) must be maintained. The covariance in condition (i) involves the unobservable u, and therefore we cannot test anything about $\text{Cov}(z_1, u) = 0$. Using t-statistic we can test condition (ii) in its reduced form: $x_K = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + ... + \delta_{K-1} x_{K-1} + \theta_1 z_1 + r_K$

4) The instrumental variable must not be correlated with u so that it is exogenous. See Section 13.3 for details.

Check Your Progress 3

1) The reduced form for x_K is

$$x_K = \delta_0 + \delta_1 x_1 + ... + \delta_{K-1} x_{K-1} + \theta_1 z_1 + ... + \theta_M z_M + r_K$$

2) The linear projection of x_K^* can be written as

$$x_K^* \equiv \delta_0 + \delta_1 x_1 + ... + \delta_1 x_1 + ... + \delta_{K-1} x_{K-1} + \theta_1 z_1 + ... + \theta_M z_M$$

3) The IV estimator using $\hat{x}_i$ as the instruments for x_i can be written as

$$\hat{\beta} = \left(\sum_{i=1}^{N} x_i' x_i \right)^{-1} \left(\sum_{i=1}^{N} \hat{x}_i' y_i \right) = (\hat{X}'X)^{-1} \hat{X}'Y$$

4) See Section 13.5 and answer.

5) There could be M different IV estimators if each z_h has some partial correlation with x_K. In reality there are many more than we can count because any linear combination of $x_1, x_2, . . . , x_{K-1}, z_1, z_2, . . . , z_M$ is uncorrelated with stochastic error u. The two-stage least squares method provides a solution to this problem. should be used. The 2SLS estimator is the most efficient IV estimator.